

SEARS & ZEMANSKY'S

UNIVERSITY PHYSICS

FIFTEENTH EDITION

YOUNG
AND
FREEDMAN



UNIT CONVERSION FACTORS

Length

1 m = 100 cm = 1000 mm = $10^6 \mu\text{m}$ = 10^9 nm
1 km = 1000 m = 0.6214 mi
1 m = 3.281 ft = 39.37 in.
1 cm = 0.3937 in.
1 in. = 2.540 cm
1 ft = 30.48 cm
1 yd = 91.44 cm
1 mi = 5280 ft = 1.609 km
1 Å = 10^{-10} m = 10^{-8} cm = 10^{-1} nm
1 nautical mile = 6080 ft
1 light-year = $9.461 \times 10^{15} \text{ m}$

Area

1 cm² = 0.155 in.²
1 m² = 10^4 cm^2 = 10.76 ft²
1 in.² = 6.452 cm²
1 ft² = 144 in.² = 0.0929 m²

Volume

1 liter = 1000 cm³ = 10^{-3} m^3 = 0.03531 ft³ = 61.02 in.³
1 ft³ = 0.02832 m³ = 28.32 liters = 7.477 gallons
1 gallon = 3.788 liters

Time

1 min = 60 s
1 h = 3600 s
1 d = 86,400 s
1 y = 365.24 d = $3.156 \times 10^7 \text{ s}$

Angle

1 rad = 57.30° = $180^\circ/\pi$
1° = 0.01745 rad = $\pi/180 \text{ rad}$
1 revolution = 360° = $2\pi \text{ rad}$
1 rev/min (rpm) = 0.1047 rad/s

Speed

1 m/s = 3.281 ft/s
1 ft/s = 0.3048 m/s
1 mi/min = 60 mi/h = 88 ft/s
1 km/h = 0.2778 m/s = 0.6214 mi/h
1 mi/h = 1.466 ft/s = 0.4470 m/s = 1.609 km/h
1 furlong/fortnight = $1.662 \times 10^{-4} \text{ m/s}$

Acceleration

1 m/s² = 100 cm/s² = 3.281 ft/s²
1 cm/s² = 0.01 m/s² = 0.03281 ft/s²
1 ft/s² = 0.3048 m/s² = 30.48 cm/s²
1 mi/h · s = 1.467 ft/s²

Mass

1 kg = 10³ g = 0.0685 slug
1 g = $6.85 \times 10^{-5} \text{ slug}$
1 slug = 14.59 kg
1 u = $1.661 \times 10^{-27} \text{ kg}$
1 kg has a weight of 2.205 lb when $g = 9.80 \text{ m/s}^2$

Force

1 N = 10⁵ dyn = 0.2248 lb
1 lb = 4.448 N = $4.448 \times 10^5 \text{ dyn}$

Pressure

1 Pa = 1 N/m² = $1.450 \times 10^{-4} \text{ lb/in.}^2$ = 0.0209 lb/ft²
1 bar = 10⁵ Pa
1 lb/in.² = 6895 Pa
1 lb/ft² = 47.88 Pa
1 atm = $1.013 \times 10^5 \text{ Pa}$ = 1.013 bar
= 14.7 lb/in.² = 2117 lb/ft²
1 mm Hg = 1 torr = 133.3 Pa

Energy

1 J = 10⁷ ergs = 0.239 cal
1 cal = 4.186 J (based on 15° calorie)
1 ft · lb = 1.356 J
1 Btu = 1055 J = 252 cal = 778 ft · lb
1 eV = $1.602 \times 10^{-19} \text{ J}$
1 kWh = $3.600 \times 10^6 \text{ J}$

Mass–Energy Equivalence

1 kg ↔ $8.988 \times 10^{16} \text{ J}$
1 u ↔ 931.5 MeV
1 eV ↔ $1.074 \times 10^{-9} \text{ u}$

Power

1 W = 1 J/s
1 hp = 746 W = 550 ft · lb/s
1 Btu/h = 0.293 W

APPLICATIONS

CHAPTER 1					
Scalar Temperature, Vector Wind	10				
CHAPTER 2					
BIO Testing Humans at High Accelerations	46				
CHAPTER 3					
BIO Horses on a Curved Path	70				
The Moons of Jupiter	82				
Watch Out: Tight Curves Ahead!	84				
Relative Velocities near the Speed of Light	87				
CHAPTER 4					
Sledding with Newton's First Law	105				
Blame Newton's Second Law	116				
CHAPTER 5					
Static Friction and Windshield Wipers	143				
BIO Pollen and Fluid Resistance	147				
BIO Circular Motion in a Centrifuge	154				
CHAPTER 6					
BIO Work and Muscle Fibers	173				
BIO Tendons Are Nonideal Springs	184				
BIO Muscle Power	187				
CHAPTER 7					
BIO Converting Gravitational Potential Energy to Kinetic Energy	203				
BIO Elastic Potential Energy of a Cheetah	212				
Nonconservative Forces and Internal Energy in a Tire	218				
Topography and Potential Energy Gradient	221				
Acrobats in Equilibrium	223				
CHAPTER 8					
BIO Woodpecker Impulse	237				
Finding Planets Beyond Our Solar System	256				
BIO Jet Propulsion in Squids	257				
CHAPTER 9					
BIO Rotational Motion in Bacteria	277				
BIO Moment of Inertia of a Bird's Wing	283				
CHAPTER 10					
BIO Combined Translation and Rotation	309				
BIO Rolling for Reproduction	314				
CHAPTER 11					
BIO Young's Modulus of a Tendon	346				
BIO Bulk Stress on an Anglerfish	348				
CHAPTER 12					
BIO Liquid Cohesion in Trees	367				
BIO Gauge Pressure of Blood	371				
BIO Why Healthy Giraffes Have High Blood Pressure	379				
BIO Listening for Turbulent Flow	383				
CHAPTER 13					
Walking and Running on the Moon	400				
BIO Biological Hazards of Interplanetary Travel	409				
CHAPTER 14					
BIO Wing Frequencies	430				
BIO Forced Oscillations	452				
BIO Canine Resonance	453				
CHAPTER 15					
BIO Waves on a Snake's Body	465				
BIO Eating and Transverse Waves	477				
BIO Surface Waves and the Swimming Speed of Ducks	478				
CHAPTER 16					
BIO Hearing Loss from Amplified Sound	506				
BIO Resonance and the Sensitivity of the Ear	519				
CHAPTER 17					
BIO Mammalian Body Temperatures	545				
BIO Fur Versus Blubber	562				
CHAPTER 18					
BIO Respiration and the Ideal-Gas Equation	581				
BIO Activation Energy and Moth Activity	599				
CHAPTER 19					
BIO The First Law of Exercise Thermodynamics	619				
BIO Exhaling Adiabatically	630				
CHAPTER 20					
BIO Biological Efficiency	645				
BIO Entropy Changes in a Living Organism	664				
Polymers Coil in Solution	668				
CHAPTER 21					
BIO Electric Forces, Sweat, and Cystic Fibrosis	685				
BIO Sharks and the "Sixth Sense"	690				
BIO A Fish with an Electric Dipole Moment	704				
CHAPTER 22					
BIO Flux Through a Basking Shark's Mouth	723				
BIO Charge Distribution Inside a Nerve Cell	734				
Why Lightning Bolts Are Vertical	736				
CHAPTER 23					
BIO Electrocardiography	755				
BIO Electron Volts and Cancer Radiotherapy	757				
BIO Potential Gradient Across a Cell Membrane	768				
CHAPTER 24					
Touch Screens and Capacitance	787				
Capacitors in the Toolbox	795				
BIO Dielectric Cell Membrane	798				
Smartphones, Capacitors, and Dielectrics	800				
CHAPTER 25					
BIO Resistivity and Nerve Conduction	818				
BIO Danger: Electric Ray!	824				
CHAPTER 26					
BIO Electromyography	855				
BIO Pacemakers and Capacitors	860				
CHAPTER 27					
BIO Spiny Lobsters and Magnetic Compasses	880				
BIO Magnetic Fields of the Body	882				
BIO Magnetic Resonance Imaging	898				
BIO Exercise Machines and the Hall Effect	905				
CHAPTER 28					
Currents and Planetary Magnetism	922				
BIO Magnetic Fields for MRI	929				
BIO Ferro Magnetic Nanoparticles for Cancer Therapy	941				
CHAPTER 29					
BIO Exploring the Brain with Induced emfs	954				
Eddy Currents Help Power Io's Volcanoes	970				
CHAPTER 30					
Inductors, Power Transmission, and Lightning Strikes	992				
A Magnetic Eruption on the Sun	997				
CHAPTER 31					
BIO Measuring Body Fat by Bioelectric Impedance Analysis	1028				
BIO Dangers of ac Versus dc Voltages	1036				
When dc Power Transmission Is Better than ac	1038				
CHAPTER 32					
BIO Ultraviolet Vision	1052				
BIO Electromagnetic Plane Waves from Space	1058				
BIO Laser Surgery	1063				
CHAPTER 33					
BIO Transparency and Index of Refraction	1082				
Circular Polarization and 3-D Movies	1096				
Birefringence and Liquid Crystal Displays	1097				
BIO Bee Vision and Polarized Light from the Sky	1098				
CHAPTER 34					
Satellite Television Dishes	1117				
Inverting an Inverted Image	1137				
BIO Focusing in the Animal Kingdom	1139				
BIO The Telephoto Eyes of Chameleons	1142				
CHAPTER 35					
BIO Phase Difference, Path Difference, and Localization in Human Hearing	1162				
BIO Interference and Butterfly Wings	1175				
BIO Seeing Below the Surface with Interferometry	1176				
CHAPTER 36					
BIO Detecting DNA with Diffraction	1199				
Bigger Telescope, Better Resolution	1203				
BIO The Airy Disk in an Eagle's Eye	1205				
CHAPTER 37					
Which One's the Grandmother?	1227				
Relative Velocity and Reference Frames	1233				
Monitoring Mass-Energy Conversion	1241				
CHAPTER 38					
BIO Sterilizing with High-Energy Photons	1259				
BIO X-Ray Absorption and Medical Imaging	1262				
Butterfly Hunting with Heisenberg	1269				
CHAPTER 39					
Using Spectra to Analyze an Interstellar Gas Cloud	1285				
BIO Fish Fluorescence	1292				
BIO Blackbody Eyes	1304				
Star Colors and the Planck Radiation Law	1305				
CHAPTER 40					
Particles in a Polymer "Box"	1332				
BIO Electron Tunneling in Enzymes	1341				
CHAPTER 41					
BIO Electron Spins and Dating Human Origins	1380				
BIO Electron Configurations and Bone Cancer Radiotherapy	1388				
X Rays in Forensic Science	1392				
CHAPTER 42					
BIO Molecular Zipper	1408				
BIO Using Crystals to Determine Protein Structure	1413				
BIO Swallow This Semiconductor Device	1429				
CHAPTER 43					
Using Isotopes to Measure Ancient Climate	1441				
BIO Deuterium and Heavy Water Toxicity	1445				
BIO A Radioactive Building	1462				
BIO Making Radioactive Isotopes for Medicine	1467				
CHAPTER 44					
BIO Pair Annihilation in Medical Diagnosis	1483				
BIO Linear Accelerators in Medicine	1485				
BIO A Fossil Both Ancient and Recent	1508				

ABOUT THE AUTHORS

Roger A. Freedman is a Lecturer in Physics at the University of California, Santa Barbara. He was an undergraduate at the University of California campuses in San Diego and Los Angeles and did his doctoral research in nuclear theory at Stanford University under the direction of Professor J. Dirk Walecka. Dr. Freedman came to UCSB in 1981 after three years of teaching and doing research at the University of Washington.

At UCSB, Dr. Freedman has taught in both the Department of Physics and the College of Creative Studies, a branch of the university intended for highly gifted and motivated undergraduates. He has published research in nuclear physics, elementary particle physics, and laser physics. In recent years, he has worked to make physics lectures a more interactive experience through the use of classroom response systems and pre-lecture videos.

In the 1970s Dr. Freedman worked as a comic book letterer and helped organize the San Diego Comic-Con (now the world's largest popular culture convention) during its first few years. Today, when not in the classroom or slaving over a computer, Dr. Freedman can be found either flying (he holds a commercial pilot's license) or with his wife, Caroline, cheering on the rowers of UCSB Men's and Women's Crew.



IN MEMORIAM: HUGH YOUNG (1930–2013)

Hugh D. Young was Emeritus Professor of Physics at Carnegie Mellon University. He earned both his undergraduate and graduate degrees from that university. He earned his Ph.D. in fundamental particle theory under the direction of the late Richard Cutkosky. Dr. Young joined the faculty of Carnegie Mellon in 1956 and retired in 2004. He also had two visiting professorships at the University of California, Berkeley.

Dr. Young's career was centered entirely on undergraduate education. He wrote several undergraduate-level textbooks, and in 1973 he became a coauthor with Francis Sears and Mark Zemansky of their well-known introductory textbooks. In addition to his role on Sears and Zemansky's *University Physics*, he was the author of Sears and Zemansky's *College Physics*.

Dr. Young earned a bachelor's degree in organ performance from Carnegie Mellon in 1972 and spent several years as Associate Organist at St. Paul's Cathedral in Pittsburgh. He often ventured into the wilderness to hike, climb, or go caving with students in Carnegie Mellon's Explorers Club, which he founded as a graduate student and later advised. Dr. Young and his wife, Alice, hosted up to 50 students each year for Thanksgiving dinners in their home.

Always gracious, Dr. Young expressed his appreciation earnestly: "I want to extend my heartfelt thanks to my colleagues at Carnegie Mellon, especially Professors Robert Kraemer, Bruce Sherwood, Ruth Chabay, Helmut Vogel, and Brian Quinn, for many stimulating discussions about physics pedagogy and for their support and encouragement during the writing of several successive editions of this book. I am equally indebted to the many generations of Carnegie Mellon students who have helped me learn what good teaching and good writing are, by showing me what works and what doesn't. It is always a joy and a privilege to express my gratitude to my wife, Alice, and our children, Gretchen and Rebecca, for their love, support, and emotional sustenance during the writing of several successive editions of this book. May all men and women be blessed with love such as theirs." We at Pearson appreciated his professionalism, good nature, and collaboration. He will be missed.



A. Lewis Ford is Professor of Physics at Texas A&M University. He received a B.A. from Rice University in 1968 and a Ph.D. in chemical physics from the University of Texas at Austin in 1972. After a one-year postdoc at Harvard University, he joined the Texas A&M physics faculty in 1973 and has been there ever since. Professor Ford has specialized in theoretical atomic physics—in particular, atomic collisions. At Texas A&M he has taught a variety of undergraduate and graduate courses, but primarily introductory physics.

TO THE STUDENT

HOW TO SUCCEED IN PHYSICS BY REALLY TRYING

Mark Hollabaugh, Normandale Community College, Emeritus

Physics encompasses the large and the small, the old and the new. From the atom to galaxies, from electrical circuitry to aerodynamics, physics is very much a part of the world around us. You probably are taking this introductory course in calculus-based physics because it is required for subsequent courses that you plan to take in preparation for a career in science or engineering. Your professor wants you to learn physics and to enjoy the experience. He or she is very interested in helping you learn this fascinating subject. That is part of the reason your professor chose this textbook for your course. That is also the reason Drs. Young and Freedman asked me to write this introductory section. We want you to succeed!

The purpose of this section of *University Physics* is to give you some ideas that will assist your learning. Specific suggestions on how to use the textbook will follow a brief discussion of general study habits and strategies.

PREPARATION FOR THIS COURSE

If you had high school physics, you will probably learn concepts faster than those who have not because you will be familiar with the language of physics. If English is a second language for you, keep a glossary of new terms that you encounter and make sure you understand how they are used in physics. Likewise, if you are further along in your mathematics courses, you will pick up the mathematical aspects of physics faster. Even if your mathematics is adequate, you may find a book such as Edward Adelson's *Get Ready for Physics* to be a great help for sharpening your math skills as well as your study skills.

LEARNING TO LEARN

Each of us has a different learning style and a preferred means of learning. Understanding your own learning style will help you to focus on aspects of physics that may give you difficulty and to use those components of your course that will help you overcome the difficulty. Obviously you will want to spend more time on those aspects that give you the most trouble. If you learn by hearing, lectures will be very important. If you learn by explaining, then working with other students will be useful to you. If solving problems is difficult for you, spend more time learning how to solve problems. Also, it is important to understand and develop good study habits. Perhaps the most important thing you can do for yourself is set aside adequate, regularly scheduled study time in a distraction-free environment.

Answer the following questions for yourself:

- Am I able to use fundamental mathematical concepts from algebra, geometry, and trigonometry? (If not, plan a program of review with help from your professor.)
- In similar courses, what activity has given me the most trouble? (Spend more time on this.) What has been the easiest for me? (Do this first; it will build your confidence.)
- Do I understand the material better if I read the book before or after the lecture? (You may learn best by skimming the material, going to lecture, and then undertaking an in-depth reading.)
- Do I spend adequate time studying physics? (A rule of thumb for a class like this is to devote, on average, 2.5 hours out of class for each hour in class. For a course that meets 5 hours each week, that means you should spend about 10 to 15 hours per week studying physics.)
- Do I study physics every day? (Spread that 10 to 15 hours out over an entire week!) At what time of the day am I at my best for studying physics? (Pick a specific time of the day and stick to it.)
- Do I work in a quiet place where I can maintain my focus? (Distractions will break your routine and cause you to miss important points.)

WORKING WITH OTHERS

Scientists or engineers seldom work in isolation from one another but rather work cooperatively. You will learn more physics and have more fun doing it if you work with other students. Some professors may formalize the use of cooperative learning or facilitate the formation of study groups. You may wish to form your own informal study group with members of your class. Use e-mail to keep in touch with one another. Your study group is an excellent resource when you review for exams.

LECTURES AND TAKING NOTES

An important component of any college course is the lecture. In physics this is especially important, because your professor will frequently do demonstrations of physical principles, run computer simulations, or show video clips. All of these are learning activities that will help you understand the basic principles of physics. Don't miss lectures. If for some reason you do, ask a friend or member of your study group to provide you with notes and let you know what happened.

Take your class notes in outline form, and fill in the details later. It can be very difficult to take word-for-word notes, so just write down key ideas. Your professor may use a diagram from the textbook. Leave a space in your notes and add the diagram later. After class, edit your notes, filling in any gaps or omissions and noting things that you need to study further. Make references to the textbook by page, equation number, or section number.

Ask questions in class, or see your professor during office hours. Remember that the only "dumb" question is the one that is not asked. Your college may have teaching assistants or peer tutors who are available to help you with any difficulties.

EXAMINATIONS

Taking an examination is stressful. But if you feel adequately prepared and are well rested, your stress will be lessened. Preparing for an exam is a continuous process; it begins the moment the previous exam is over. You should immediately go over the exam to understand any mistakes you made. If you worked a problem and made substantial errors, try this: Take a piece of paper and divide it down the middle with a line from top to bottom. In one column, write the proper solution to the problem. In the other column, write what you did and why, if you know, and why your solution was incorrect. If you are uncertain why you made your mistake or how to avoid making it again, talk with your professor. Physics constantly builds on fundamental ideas, and it is important to correct any misunderstandings immediately. *Warning:* Although cramming at the last minute may get you through the present exam, you will not adequately retain the concepts for use on the next exam.

TO THE INSTRUCTOR

PREFACE

In the years since it was first published, *University Physics* has always embraced change, not just to include the latest developments in our understanding of the physical world, but also to address our understanding of how students learn physics and how they study.

In preparing for this new Fifteenth Edition, we listened to the thousands of students who have told us that they often struggle to see the connections between the worked examples in their textbook and problems on homework or exams. Every problem seems different because the objects, situations, numbers, and questions posed change with each problem. As a result, students experience frustration and a lack of confidence. By contrast, expert problem-solvers categorize problems by type, based on the underlying principles.

Several of the revisions we have made therefore address this particular challenge by, for example, helping students see the big picture of what each worked example is trying to illustrate and allowing them to practice sets of related problems to help them identify repeating patterns and strategies. These new features are explained in more detail below.

NEW TO THIS EDITION

- **Worked example KEYCONCEPT statements** appear at the end of every Example and Conceptual Example, providing a brief summary of the key idea used in the solution to consolidate what was most important and what can be broadly applied to other problems, to help students identify strategies that can be used in future problems.
- **KEY EXAMPLE VARIATION PROBLEMS** in the new Guided Practice section at the end of each chapter are based on selected worked examples. They build in difficulty by changing scenarios, swapping the knowns and unknowns, and adding complexity and/or steps of reasoning to provide the most helpful range of related problems that use the same basic approach to solve. These scaffolded problem sets help students see patterns and make connections between problems that can be solved using the same underlying principles and strategies so that they are more able to tackle different problem types when exam time comes.
- **Expanded Caution paragraphs** focus on typical student misconceptions and problem areas. Over a dozen more have been added to this edition based on common errors made in MasteringTM Physics.
- **Updated and expanded Application sidebars** give students engaging and relevant real-world context.
- **Based on data from Mastering Physics and feedback from instructors, changes to the homework problems include the following:**
 - **Over 500 new problems**, with scores of other problems revised to improve clarity.
 - **Expanded three-dot-difficulty and Challenge Problems** significantly stretch students by requiring sophisticated reasoning that often involves multiple steps or concepts and/or mathematical skills. Challenge Problems are the most difficult problems in each chapter and often involve calculus, multiple steps that lead students through a complex analysis, and/or the exploration of a topic or application not explicitly covered in the chapter.
 - **New estimation problems** help students learn to analyze problem scenarios, assess data, and work with orders of magnitude. This problem type engages students to more thoroughly explore the situation by requiring them to not only estimate some of the data in the problem but also decide what data need to be estimated based on real-world experience, reasoning, assumptions, and/or modeling.
 - **Expanded cumulative problems** promote more advanced problem-solving techniques by requiring knowledge and skills covered in previous chapters to be integrated with understanding and skills from the current chapter.
 - **Expanded alternative problem sets** in Mastering Physics provide textbook-specific problems from previous editions to assign for additional student practice.

Standard, Extended, and Three-Volume Editions

With Mastering Physics:

- **Standard Edition:** Chapters 1–37
(ISBN 978-0-135-64663-2)
- **Extended Edition:** Chapters 1–44
(ISBN 978-0-135-15970-5)

Without Mastering Physics:

- **Standard Edition:** Chapters 1–37
(ISBN 978-0-135-21611-8)
- **Extended Edition:** Chapters 1–44
(ISBN 978-0-135-15955-2)
- **Volume 1:** Chapters 1–20
(ISBN 978-0-135-21672-9)
- **Volume 2:** Chapters 21–37
(ISBN 978-0-135-21612-5)
- **Volume 3:** Chapters 37–44
(ISBN 978-0-135-21673-6)

KEY FEATURES OF *UNIVERSITY PHYSICS*



- A **QR code** at the beginning of the new Guided Practice section in each chapter allows students to use a mobile phone to access the Study Area of Mastering Physics, where they can watch interactive videos of a physics professor giving a relevant physics demonstration (Video Tutor Demonstrations) or showing a narrated and animated worked Example (Video Tutor Solutions). All videos also play directly through links within the Pearson eText.
- End-of-chapter **Bridging Problems** provide a transition between the single-concept Examples and the more challenging end-of-chapter problems. Each Bridging Problem poses a difficult, multiconcept problem that typically incorporates physics from earlier chapters. The **Solution Guide** that follows each problem provides questions and hints that help students approach and solve challenging problems with confidence.
- Deep and extensive **problem sets** cover a wide range of difficulty (with blue dots to indicate relative difficulty level) and exercise both physical understanding and problem-solving expertise. Many problems are based on complex real-life situations.
- This textbook offers more **Examples** and **Conceptual Examples** than most other leading calculus-based textbooks, allowing students to explore problem-solving challenges that are not addressed in other textbooks.
- A research-based **problem-solving approach (Identify, Set Up, Execute, Evaluate)** is used in every Example as well as in the Problem-Solving Strategies, in the Bridging Problems, and throughout the Instructor's Solutions Manual and the Study Guide. This consistent approach teaches students to tackle problems thoughtfully rather than cutting straight to the math.
- **Problem-Solving Strategies** coach students in how to approach specific types of problems.
- The **figures** use a simplified graphical style to focus on the physics of a situation, and they incorporate blue **explanatory annotations**. Both techniques have been demonstrated to have a strong positive effect on learning.
- Many figures that illustrate Example solutions take the form of black-and-white **pencil sketches**, which directly represent what a student should draw in solving such problems themselves.
- The popular **Caution paragraphs** focus on typical misconceptions and student problem areas.
- End-of-section **Test Your Understanding** questions let students check their grasp of the material and use a multiple-choice or ranking-task format to probe for common misconceptions. Answers are now provided immediately after the question in order to encourage students to try them.
- **Visual Summaries** at the end of each chapter present the key ideas in words, equations, and thumbnail pictures, helping students review more effectively.

Mastering™ is the teaching and learning platform that empowers you to reach *every* student. By combining trusted author content with digital tools developed to engage students and emulate the office-hour experience, Mastering personalizes learning and improves results for each student. Now providing a fully integrated experience, the eText is linked to every problem within Mastering for seamless integration among homework problems, practice problems, the textbook, worked examples, and more.

Reach every student with Mastering

- **Teach your course your way:** Your course is unique. Whether you'd like to build your own auto-graded assignments, foster student engagement during class, or give students anytime, anywhere access, Mastering gives you the flexibility to easily create *your* course to fit *your* needs.
 - With **Learning Catalytics**, you'll hear from every student when it matters most. You pose a variety of questions that help students recall ideas, apply concepts, and develop critical-thinking skills. Your students respond using their own smartphones, tablets, or laptops. You can monitor responses with real-time analytics and find out what your students do—and don't—understand. Then you can adjust your teaching accordingly and even facilitate peer-to-peer learning, helping students stay motivated and engaged.

- **Expanded alternative problem sets**, with hundreds of vetted problems from previous editions of the book, provide additional problem-solving practice and offer instructors more options when creating assignments.
- **Empower each learner:** Each student learns at a different pace. Personalized learning, including adaptive tools and wrong-answer feedback, pinpoints the precise areas where each student needs practice and gives all students the support they need—when and where they need it—to be successful.
- **Interactive Pre-lecture Videos** provide an introduction to key topics with embedded assessment to help students prepare before lecture and to help professors identify student misconceptions.
 - **NEW! Quantitative Pre-lecture Videos** now complement the conceptual Interactive Pre-lecture Videos designed to expose students to concepts before class and help them learn how problems for a specific concept are worked.
- **NEW! Direct Measurement Videos** are short videos that show real situations of physical phenomena. Grids, rulers, and frame counters appear as overlays, helping students to make precise measurements of quantities such as position and time. Students then apply these quantities along with physics concepts to solve problems and answer questions about the motion of the objects in the video. The problems are assignable in Mastering Physics and can be used to replace or supplement traditional word problems; they can also serve as open-ended questions to help develop problem-solving skills.
- **NEW! Dynamic Study Modules** help students study effectively—and at their own pace. How? By keeping them motivated and engaged. The assignable modules rely on the latest research in cognitive science, using methods—such as adaptivity, gamification, and intermittent rewards—to stimulate learning and improve retention. Each module poses a series of questions about a course topic. These question sets adapt to each student’s performance and offer personalized, targeted feedback to help students master key concepts.
- **NEW! The Physics Primer** relies on videos, hints, and feedback to refresh students’ math skills in the context of physics and prepares them for success in the course. These tutorials can be assigned before the course begins or throughout the course as just-in-time remediation. They ensure that students practice and maintain their math skills, while tying together mathematical operations and physics analysis.
- **Deliver trusted content:** We partner with highly respected authors to develop interactive content and course-specific resources that keep students on track and engaged.
 - **Video Tutor Demonstrations and Video Tutor Solutions** tie directly to relevant content in the textbook and can be accessed through Mastering Physics, via the eText, or from QR codes in the textbook.
 - **Video Tutor Solutions (VTSs) for every worked example** in the book walk students through the problem-solving process, providing a virtual teaching assistant on a round-the-clock basis.
 - **Video Tutor Demonstrations (VTDs)** feature “pause-and-predict” demonstrations of key physics concepts and incorporate assessment to engage students in understanding key concepts. New VTDs build on the existing collection, adding new topics for a more robust set of demonstrations.
 - **NEW! Enhanced end-of-chapter questions** provide expanded remediation built into each question when and where students need it. Remediation includes scaffolded support, links to hints, links to appropriate sections of the eText, links from the eText to Mastering Physics, Video Tutor Solutions, math remediation, and wrong-answer feedback for homework assignments. Half of all end-of-chapter problems now have wrong-answer feedback and links to the eText.
 - **NEW! Key Example Variation Problems**, assignable in Mastering Physics, build in difficulty by changing scenarios, swapping the knowns and unknowns, and adding complexity and/or steps of reasoning to provide the most helpful range of related problems that use the same basic approach to find their solutions.
 - **NEW! Bridging Problems are now assignable in Mastering Physics**, thus providing students with additional practice in moving from single-concept worked examples to multi-concept homework problems.

- **Improve student results:** Usage statistics show that when you teach with Mastering, student performance improves. That's why instructors have chosen Mastering for over 15 years, touching the lives of more than 20 million students.

INSTRUCTIONAL PACKAGE

University Physics with Modern Physics, Fifteenth Edition, provides an integrated teaching and learning package of support material for students and instructors.

NOTE: For convenience, instructor supplements can be downloaded from the Instructor Resources area of Mastering Physics.

Supplement	Print	Online	Instructor or Student Supplement	Description
Mastering Physics with Pearson eText (ISBN 0135180678)		✓	Instructor and Student Supplement	This product features all of the resources of Mastering Physics in addition to Pearson eText 2.0. Available on smartphones and tablets, Pearson eText 2.0 comprises the full text, including videos and other rich media.
Instructor's Solutions Manual (ISBN 0135275881)		✓	Instructor Supplement	This comprehensive solutions manual contains complete solutions to all end-of-chapter questions and problems.
Instructor's Resource Materials		✓	Instructor Supplement	All art, photos, and tables from the book are available in JPEG format and as modifiable PowerPoints™. In addition, instructors can access lecture outlines as well as "clicker" questions in PowerPoint format; editable content for key features, including key equations and Problem-Solving Strategies; and the TestGen test bank.
Student's Study Guide/Solutions Manual Volume 1 (Chs 1–20) (ISBN 0135216958) Volume 2 (Chs 21–37) (ISBN 013521694X) Volume 3 (Chs 37–44) (ISBN 013559202X)	✓		Student Supplement	This combination study guide and solutions manual reinforces the textbook's research-based problem-solving approach (Identify, Set Up, Evaluate, Execute). The solutions manual contains solutions to most of the odd-numbered problems in the text, and the study guide provides a chapter-by-chapter review of key concepts and equations as well as additional example problems with solutions.

ACKNOWLEDGMENTS

I would like to thank the hundreds of reviewers and colleagues who have offered valuable comments and suggestions over the life of this textbook. The continuing success of *University Physics* is due in large measure to their contributions.

Miah Adel (U. of Arkansas at Pine Bluff), Edward Adelson (Ohio State U.), Julie Alexander (Camosun C.), Ralph Alexander (U. of Missouri at Rolla), J. G. Anderson, R. S. Anderson, Wayne Anderson (Sacramento City C.), Sanjeev Arora (Fort Valley State U.), Alex Azima (Lansing Comm. C.), Dilip Balamore (Nassau Comm. C.), Harold Bale (U. of North Dakota), Arun Bansil (Northeastern U.), John Barach (Vanderbilt U.), J. D. Barnett, H. H. Barschall, Albert Bartlett (U. of Colorado), Marshall Bartlett (Hollins U.), Paul Baum (CUNY, Queens C.), Frederick Becchetti (U. of Michigan), B. Bederson, David Bennum (U. of Nevada, Reno), Lev I. Berger (San Diego State U.), Angela Biselli (Fairfield U.), Robert Boeke (William Rainey Harper C.), Bram Boroson (Clayton State U.), S. Borowitz, A. C. Braden, James Brooks (Boston U.), Nicholas E. Brown (California Polytechnic State U., San Luis Obispo), Tony Buffa (California Polytechnic State U., San Luis Obispo), Shane Burns (Colorado C.), A. Capecelatro, Michael Cardamone (Pennsylvania State U.), Duane Carmony (Purdue U.), Troy Carter (UCLA), P. Catranides, John Cerne (SUNY at Buffalo), Shinil Cho (La Roche C.), Tim Chupp (U. of Michigan), Roger Clapp (U. of South Florida), William M. Cloud (Eastern Illinois U.), Leonard Cohen (Drexel U.), W. R. Coker (U. of Texas, Austin), Malcolm D. Cole (U. of Missouri at Rolla), H. Conrad, David Cook (Lawrence U.), Gayl Cook (U. of Colorado), Hans Courant (U. of Minnesota), Carl Covatto (Arizona State U.), Bruce A. Craver (U. of Dayton), Larry Curtis (U. of Toledo), Jai Dahiya (Southeast Missouri State U.), Dedra Demaree (Georgetown U.), Steve Detweiler (U. of Florida), George Dixon (Oklahoma State U.), Steve Drasco (Grinnell C.), Donald S. Duncan, Boyd Edwards (West Virginia U.), Robert Eisenstein (Carnegie Mellon U.), Amy Emerson Missouri (Virginia Institute of Technology), Olena Erhardt (Richland C.), William Faissler (Northeastern U.), Gregory Falabella (Wagner C.), William Fasnacht (U.S. Naval Academy), Paul Feldker (St. Louis Comm. C.), Carlos Figueroa (Cabrillo C.), L. H. Fisher, Neil Fletcher (Florida State U.), Allen Flora (Hood C.), Robert Folk, Peter Fong (Emory U.), A. Lewis Ford (Texas A&M U.), D. Frantszog, James R. Gaines (Ohio State U.), Solomon Gartenhaus (Purdue U.), Ron Gautreau (New Jersey Institute of Technology), J. David Gavenda (U. of Texas, Austin), Dennis Gay (U. of North Florida), Elizabeth George (Wittenberg U.), James Gerhart (U. of Washington), N. S. Gingrich, J. L. Glathart, S. Goodwin, Rich Gottfried (Frederick Comm. C.), Walter S. Gray (U. of Michigan), Paul Gresser (U. of Maryland), Benjamin Grinstein (UC, San Diego), Howard Grotch (Pennsylvania State U.), John Gruber (San Jose State U.), Graham D. Gutsche (U.S. Naval Academy), Michael J. Harrison (Michigan State U.), Harold Hart (Western Illinois U.), Howard Hayden (U. of Connecticut), Carl Helrich (Goshen C.), Andrew Hirsch (Purdue U.), Linda Hirst (UC, Merced), Laurent Hodges (Iowa State U.), C. D. Hodgman, Elizabeth Holden (U. of Wisconsin, Platteville), Michael Hones (Villanova U.), Keith Honey (West Virginia Institute of Technology), Gregory Hood (Tidewater Comm. C.), John Hubisz (North Carolina State U.), Eric Hudson (Pennsylvania State U.), M. Iona, Bob Jacobsen (UC, Berkeley), John Jaszczak (Michigan Technical U.), Alvin Jenkins (North Carolina State U.), Charles Johnson (South Georgia State C.), Robert P. Johnson (UC, Santa Cruz), Lorella Jones (U. of Illinois), Manoj Kaplinghat (UC, Irvine), John Karchek (GMI Engineering & Management Institute), Thomas Keil (Worcester Polytechnic Institute), Robert Kraemer (Carnegie Mellon U.), Jean P. Krisch (U. of Michigan), Robert A. Kromhout, Andrew Kunz (Marquette U.), Charles Lane (Berry C.), Stewart Langton (U. of Victoria), Thomas N. Lawrence (Texas State U.), Robert J. Lee, Alfred Leitner (Rensselaer Polytechnic U.), Frederic Liebrand (Walla Walla U.), Gerald P. Lietz (DePaul U.), Gordon Lind (Utah State U.), S. Livingston (U. of Wisconsin, Milwaukee), Jorge Lopez (U. of Texas, El Paso), Elihu Lubkin (U. of Wisconsin, Milwaukee), Robert Luke (Boise State U.), David Lynch (Iowa State U.), Michael Lysak (San Bernardino Valley C.), Jeffrey Mallow (Loyola U.), Robert Mania (Kentucky State U.), Robert Marchina (U. of Memphis), David Markowitz (U. of Connecticut), Philip Matheson (Utah Valley U.), R. J. Maurer, Oren Maxwell (Florida International U.), Joseph L. McCauley (U. of Houston), T. K. McCubbin, Jr. (Pennsylvania State U.), Charles McFarland (U. of Missouri at Rolla), James McGuire (Tulane U.), Lawrence McIntyre (U. of Arizona), Fredric Messing (Carnegie Mellon U.), Thomas Meyer (Texas A&M U.), Andre Mirabelli (St. Peter's C., New Jersey), Herbert Muether (SUNY, Stony Brook), Jack Munsee (California State U., Long Beach), Lorenzo Narducci (Drexel U.), Van E. Neie (Purdue U.), Forrest Newman (Sacramento City C.), David A. Nordling (U.S. Naval Academy), Benedict Oh (Pennsylvania State U.), L. O. Olsen, Michael Ottinger (Missouri Western State U.), Russell Palma (Minnesota State U., Mankato), Jim Pannell (DeVry Institute of Technology), Neeti Parashar (Purdue U., Calumet), W. F. Parks (U. of Missouri), Robert Paulson (California State U., Chico), Jerry Peacher (U. of Missouri at Rolla), Arnold Perlmutter (U. of Miami), Lennart Peterson (U. of Florida), R. J. Peterson (U. of

Colorado, Boulder), R. Pinkston, Ronald Poling (U. of Minnesota), Yuri Popov (U. of Michigan), J. G. Potter, C. W. Price (Millersville U.), Francis Prosser (U. of Kansas), Sheldon H. Radin, Roberto Ramos (Drexel U.), Michael Rapport (Anne Arundel Comm. C.), R. Resnick, James A. Richards, Jr., John S. Risley (North Carolina State U.), Francesc Roig (UC, Santa Barbara), T. L. Rokoske, Richard Roth (Eastern Michigan U.), Carl Rotter (U. of West Virginia), S. Clark Rowland (Andrews U.), Rajarshi Roy (Georgia Institute of Technology), Russell A. Roy (Santa Fe Comm. C.), Desi Saludes (Hillsborough Comm. C.), Thomas Sandin (North Carolina A&T State U.), Dhiraj Sardar (U. of Texas, San Antonio), Tumer Sayman (Eastern Michigan U.), Bruce Schumm (UC, Santa Cruz), Melvin Schwartz (St. John's U.), F. A. Scott, L. W. Seagondollar, Paul Shand (U. of Northern Iowa), Stan Shepherd (Pennsylvania State U.), Douglas Sherman (San Jose State U.), Bruce Sherwood (Carnegie Mellon U.), Hugh Siefkin (Greenville C.), Christopher Sirola (U. of Southern Mississippi), Tomasz Skwarnicki (Syracuse U.), C. P. Slichter, Jason Slinker (U. of Texas, Dallas), Charles W. Smith (U. of Maine, Orono), Malcolm Smith (U. of Lowell), Ross Spencer (Brigham Young U.), Julien Sprott (U. of Wisconsin), Victor Stanionis (Iona C.), James Stith (American Institute of Physics), Chuck Stone (North Carolina A&T State U.), Edward Strother (Florida Institute of Technology), Conley Stutz (Bradley U.), Albert Stwertka (U.S. Merchant Marine Academy), Kenneth Szpara-DeNisco (Harrisburg Area Comm. C.), Devki Talwar (Indiana U. of Pennsylvania), Fiorella Terenzi (Florida International U.), Martin Tiersten (CUNY, City C.), David Toot (Alfred U.), Greg Trayling (Rochester Institute of Technology), Somdev Tyagi (Drexel U.), Matthew Vannette (Saginaw Valley State U.), Eswara Venugopal (U. of Detroit, Mercy), F. Verbrugge, Helmut Vogel (Carnegie Mellon U.), Aaron Warren (Purdue U., North Central), Robert Webb (Texas A&M U.), Thomas Weber (Iowa State U.), M. Russell Wehr (Pennsylvania State U.), Robert Weidman (Michigan Technical U.), Dan Whalen (UC, San Diego), Lester V. Whitney, Thomas Wiggins (Pennsylvania State U.), Robyn Wilde (Oregon Institute of Technology), David Willey (U. of Pittsburgh, Johnstown), George Williams (U. of Utah), John Williams (Auburn U.), Stanley Williams (Iowa State U.), Jack Willis, Suzanne Willis (Northern Illinois U.), Robert Wilson (San Bernardino Valley C.), L. Wolfenstein, James Wood (Palm Beach Junior C.), Lowell Wood (U. of Houston), R. E. Worley, D. H. Ziebell (Manatee Comm. C.), George O. Zimmerman (Boston U.)

In addition, I would like to thank my past and present colleagues at UCSB, including Rob Geller, Carl Gwinn, Al Nash, Elisabeth Nicol, and Francesc Roig, for their wholehearted support and for many helpful discussions. I owe a special debt of gratitude to my early teachers Willa Ramsay, Peter Zimmerman, William Little, Alan Schwettman, and Dirk Walecka for showing me what clear and engaging physics teaching is all about, and to Stuart Johnson for inviting me to become a coauthor of *University Physics* beginning with the Ninth Edition. Special acknowledgments go out to Lewis Ford for the huge and essential task of creating the revised end-of-chapter problem sets, and to Wayne Anderson, who carefully reviewed all of the new problems and solved them, and updated the *Instructor's Solutions Manual*. Thanks also go to Michael Faux, Elizabeth Holden, and Tom Sandin for their numerous contributions to the end-of-chapter problems. I want to express special thanks to the editorial staff at Pearson: to Jeanne Zalesky for her editorial vision; to Alice Houston for her keen eye and careful development of this edition; to Karen Karlin, Margot Otway, Joanna Dinsmore, and Jason Harlow for their careful reading of the page proofs; and to Martha Steele for keeping the editorial and production pipelines flowing. Most of all, I want to express my gratitude and love to my wife, Caroline, to whom I dedicate my contribution to this book. Hey, Caroline, the new edition's done at last—let's go flying!

PLEASE TELL ME WHAT YOU THINK!

I welcome communications from students and professors, especially concerning errors or deficiencies that you find in this edition. The late Hugh Young and I have devoted a lot of time and effort to writing the best book we know how to write, and I hope it will help as you teach and learn physics. In turn, you can help me by letting me know what still needs to be improved! Please feel free to contact me either electronically or by ordinary mail. Your comments will be greatly appreciated.

August 2018

Roger A. Freedman
Department of Physics
University of California, Santa Barbara
Santa Barbara, CA 93106-9530
airboy@physics.ucsb.edu
<http://www.physics.ucsb.edu/~airboy/>
Twitter: @RogerFreedman

BRIEF CONTENTS

MECHANICS

1	Units, Physical Quantities, and Vectors	1
2	Motion Along a Straight Line	34
3	Motion in Two or Three Dimensions	66
4	Newton's Laws of Motion	100
5	Applying Newton's Laws	129
6	Work and Kinetic Energy	171
7	Potential Energy and Energy Conservation	201
8	Momentum, Impulse, and Collisions	235
9	Rotation of Rigid Bodies	272
10	Dynamics of Rotational Motion	302
11	Equilibrium and Elasticity	337
12	Fluid Mechanics	366
13	Gravitation	395
14	Periodic Motion	429

WAVES/ACOUSTICS

15	Mechanical Waves	464
16	Sound and Hearing	501

THERMODYNAMICS

17	Temperature and Heat	541
18	Thermal Properties of Matter	579
19	The First Law of Thermodynamics	613
20	The Second Law of Thermodynamics	642

ELECTROMAGNETISM

21	Electric Charge and Electric Field	678
22	Gauss's Law	718
23	Electric Potential	747
24	Capacitance and Dielectrics	781
25	Current, Resistance, and Electromotive Force	812
26	Direct-Current Circuits	844

27	Magnetic Field and Magnetic Forces	878
28	Sources of Magnetic Field	918
29	Electromagnetic Induction	953
30	Inductance	988
31	Alternating Current	1018
32	Electromagnetic Waves	1048

OPTICS

33	The Nature and Propagation of Light	1077
34	Geometric Optics	1110
35	Interference	1159
36	Diffraction	1185

MODERN PHYSICS

37	Relativity	1217
38	Photons: Light Waves Behaving as Particles	1253
39	Particles Behaving as Waves	1279
40	Quantum Mechanics I: Wave Functions	1321
41	Quantum Mechanics II: Atomic Structure	1360
42	Molecules and Condensed Matter	1408
43	Nuclear Physics	1442
44	Particle Physics and Cosmology	1483

APPENDICES

A	The International System of Units	A-1
B	Useful Mathematical Relations	A-3
C	The Greek Alphabet	A-4
D	Periodic Table of the Elements	A-5
E	Unit Conversion Factors	A-6
F	Numerical Constants	A-7
	Answers to Odd-Numbered Problems	A-9
	Credits	C-1
	Index	I-1

DETAILED CONTENTS

MECHANICS

1	UNITS, PHYSICAL QUANTITIES, AND VECTORS	1
1.1	The Nature of Physics	1
1.2	Solving Physics Problems	2
1.3	Standards and Units	3
1.4	Using and Converting Units	6
1.5	Uncertainty and Significant Figures	8
1.6	Estimates and Orders of Magnitude	10
1.7	Vectors and Vector Addition	10
1.8	Components of Vectors	14
1.9	Unit Vectors	18
1.10	Products of Vectors	19
	Summary	25
	Guided Practice	26
	Questions/Exercises/Problems	27
2	MOTION ALONG A STRAIGHT LINE	34
2.1	Displacement, Time, and Average Velocity	34
2.2	Instantaneous Velocity	37
2.3	Average and Instantaneous Acceleration	40
2.4	Motion with Constant Acceleration	44
2.5	Freely Falling Objects	50
2.6	Velocity and Position by Integration	53
	Summary	56
	Guided Practice	57
	Questions/Exercises/Problems	58
3	MOTION IN TWO OR THREE DIMENSIONS	66
3.1	Position and Velocity Vectors	66
3.2	The Acceleration Vector	69
3.3	Projectile Motion	74
3.4	Motion in a Circle	81
3.5	Relative Velocity	84
	Summary	90
	Guided Practice	91
	Questions/Exercises/Problems	92
4	NEWTON'S LAWS OF MOTION	100
4.1	Force and Interactions	100
4.2	Newton's First Law	103
4.3	Newton's Second Law	107
4.4	Mass and Weight	113
4.5	Newton's Third Law	115
4.6	Free-Body Diagrams	119
	Summary	121
	Guided Practice	121
	Questions/Exercises/Problems	123
5	APPLYING NEWTON'S LAWS	129
5.1	Using Newton's First Law: Particles in Equilibrium	129
5.2	Using Newton's Second Law: Dynamics of Particles	134
5.3	Friction Forces	141
5.4	Dynamics of Circular Motion	149
5.5	The Fundamental Forces of Nature	154
	Summary	156
	Guided Practice	157
	Questions/Exercises/Problems	158
6	WORK AND KINETIC ENERGY	171
6.1	Work	171
6.2	Kinetic Energy and the Work–Energy Theorem	176
6.3	Work and Energy with Varying Forces	181
6.4	Power	187
	Summary	190
	Guided Practice	191
	Questions/Exercises/Problems	192
7	POTENTIAL ENERGY AND ENERGY CONSERVATION	201
7.1	Gravitational Potential Energy	201
7.2	Elastic Potential Energy	210
7.3	Conservative and Nonconservative Forces	215
7.4	Force and Potential Energy	219



7.5	Energy Diagrams	222
	Summary	224
	Guided Practice	225
	Questions/Exercises/Problems	226

8 MOMENTUM, IMPULSE, AND COLLISIONS 235

8.1	Momentum and Impulse	235
8.2	Conservation of Momentum	241
8.3	Momentum Conservation and Collisions	245
8.4	Elastic Collisions	249
8.5	Center of Mass	253
8.6	Rocket Propulsion	256
	Summary	260
	Guided Practice	261
	Questions/Exercises/Problems	262



9 ROTATION OF RIGID BODIES 272

9.1	Angular Velocity and Acceleration	272
9.2	Rotation with Constant Angular Acceleration	277
9.3	Relating Linear and Angular Kinematics	279
9.4	Energy in Rotational Motion	282
9.5	Parallel-Axis Theorem	287
9.6	Moment-of-Inertia Calculations	289
	Summary	291
	Guided Practice	292
	Questions/Exercises/Problems	293

10 DYNAMICS OF ROTATIONAL MOTION 302

10.1	Torque	302
10.2	Torque and Angular Acceleration for a Rigid Body	305
10.3	Rigid-Body Rotation About a Moving Axis	308
10.4	Work and Power in Rotational Motion	314
10.5	Angular Momentum	316
10.6	Conservation of Angular Momentum	319

10.7	Gyroscopes and Precession	322
	Summary	325
	Guided Practice	326
	Questions/Exercises/Problems	327

11 EQUILIBRIUM AND ELASTICITY 337

11.1	Conditions for Equilibrium	337
11.2	Center of Gravity	338
11.3	Solving Rigid-Body Equilibrium Problems	341
11.4	Stress, Strain, and Elastic Moduli	345
11.5	Elasticity and Plasticity	351
	Summary	353
	Guided Practice	354
	Questions/Exercises/Problems	355

12 FLUID MECHANICS 366

12.1	Gases, Liquids, and Density	366
12.2	Pressure in a Fluid	368
12.3	Buoyancy	373
12.4	Fluid Flow	376
12.5	Bernoulli's Equation	378
12.6	Viscosity and Turbulence	382
	Summary	385
	Guided Practice	386
	Questions/Exercises/Problems	387

13 GRAVITATION 395

13.1	Newton's Law of Gravitation	395
13.2	Weight	399
13.3	Gravitational Potential Energy	402
13.4	The Motion of Satellites	404
13.5	Kepler's Laws and the Motion of Planets	407
13.6	Spherical Mass Distributions	411
13.7	Apparent Weight and the Earth's Rotation	414
13.8	Black Holes	415
	Summary	419
	Guided Practice	420
	Questions/Exercises/Problems	421

14 PERIODIC MOTION 429

14.1	Describing Oscillation	429
14.2	Simple Harmonic Motion	431
14.3	Energy in Simple Harmonic Motion	438
14.4	Applications of Simple Harmonic Motion	442
14.5	The Simple Pendulum	446
14.6	The Physical Pendulum	447
14.7	Damped Oscillations	449
14.8	Forced Oscillations and Resonance	451
	Summary	453
	Guided Practice	455
	Questions/Exercises/Problems	456

WAVES/ACOUSTICS

15	MECHANICAL WAVES	464
15.1	Types of Mechanical Waves	464
15.2	Periodic Waves	466
15.3	Mathematical Description of a Wave	469
15.4	Speed of a Transverse Wave	474
15.5	Energy in Wave Motion	478
15.6	Wave Interference, Boundary Conditions, and Superposition	481
15.7	Standing Waves on a String	483
15.8	Normal Modes of a String	486
	Summary	491
	Guided Practice	492
	Questions/Exercises/Problems	493
16	SOUND AND HEARING	501
16.1	Sound Waves	501
16.2	Speed of Sound Waves	506
16.3	Sound Intensity	510
16.4	Standing Sound Waves and Normal Modes	514
16.5	Resonance and Sound	518
16.6	Interference of Waves	520
16.7	Beats	522
16.8	The Doppler Effect	524
16.9	Shock Waves	529
	Summary	531
	Guided Practice	533
	Questions/Exercises/Problems	534

THERMODYNAMICS

17	TEMPERATURE AND HEAT	541
17.1	Temperature and Thermal Equilibrium	541
17.2	Thermometers and Temperature Scales	543
17.3	Gas Thermometers and the Kelvin Scale	545
17.4	Thermal Expansion	547
17.5	Quantity of Heat	552
17.6	Calorimetry and Phase Changes	555
17.7	Mechanisms of Heat Transfer	561
	Summary	568
	Guided Practice	569
	Questions/Exercises/Problems	570
18	THERMAL PROPERTIES OF MATTER	579
18.1	Equations of State	579
18.2	Molecular Properties of Matter	585
18.3	Kinetic-Molecular Model of an Ideal Gas	588
18.4	Heat Capacities	594
18.5	Molecular Speeds	597

18.6	Phases of Matter	599
	Summary	602
	Guided Practice	603
	Questions/Exercises/Problems	604

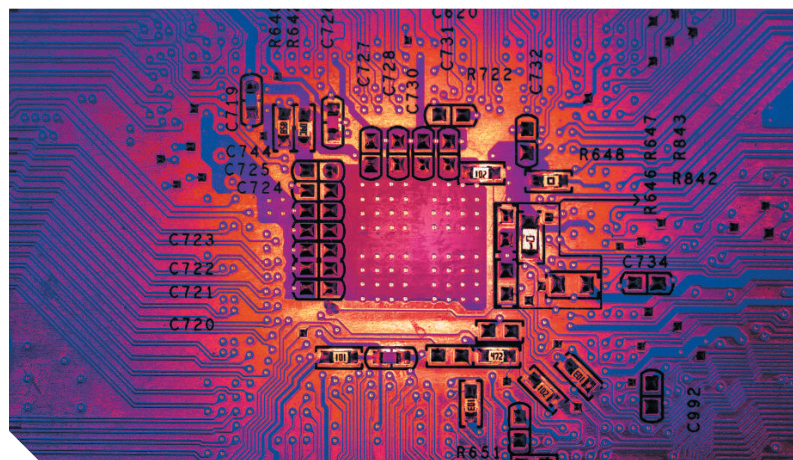
19	THE FIRST LAW OF THERMODYNAMICS	613
19.1	Thermodynamic Systems	613
19.2	Work Done During Volume Changes	615
19.3	Paths Between Thermodynamic States	617
19.4	Internal Energy and the First Law of Thermodynamics	618
19.5	Kinds of Thermodynamic Processes	623
19.6	Internal Energy of an Ideal Gas	625
19.7	Heat Capacities of an Ideal Gas	626
19.8	Adiabatic Processes for an Ideal Gas	629
	Summary	632
	Guided Practice	633
	Questions/Exercises/Problems	634



20	THE SECOND LAW OF THERMODYNAMICS	642
20.1	Directions of Thermodynamic Processes	642
20.2	Heat Engines	644
20.3	Internal-Combustion Engines	647
20.4	Refrigerators	649
20.5	The Second Law of Thermodynamics	651
20.6	The Carnot Cycle	653
20.7	Entropy	659
20.8	Microscopic Interpretation of Entropy	665
	Summary	669
	Guided Practice	670
	Questions/Exercises/Problems	671

ELECTROMAGNETISM

21	ELECTRIC CHARGE AND ELECTRIC FIELD	678			
21.1	Electric Charge	679			
21.2	Conductors, Insulators, and Induced Charges	682			
21.3	Coulomb's Law	685			
21.4	Electric Field and Electric Forces	690			
21.5	Electric-Field Calculations	694			
21.6	Electric Field Lines	700			
21.7	Electric Dipoles	701			
	Summary	706			
	Guided Practice	707			
	Questions/Exercises/Problems	708			
22	GAUSS'S LAW	718			
22.1	Charge and Electric Flux	718			
22.2	Calculating Electric Flux	721			
22.3	Gauss's Law	725			
22.4	Applications of Gauss's Law	729			
22.5	Charges on Conductors	734			
	Summary	738			
	Guided Practice	739			
	Questions/Exercises/Problems	740			
23	ELECTRIC POTENTIAL	747			
23.1	Electric Potential Energy	747			
23.2	Electric Potential	754			
23.3	Calculating Electric Potential	760			
23.4	Equipotential Surfaces	764			
23.5	Potential Gradient	767			
	Summary	770			
	Guided Practice	771			
	Questions/Exercises/Problems	772			
24	CAPACITANCE AND DIELECTRICS	781			
24.1	Capacitors and Capacitance	782			
24.2	Capacitors in Series and Parallel	786			
24.3	Energy Storage in Capacitors and Electric-Field Energy	790			
24.4	Dielectrics	793			
24.5	Molecular Model of Induced Charge	799			
24.6	Gauss's Law in Dielectrics	801			
	Summary	802			
	Guided Practice	803			
	Questions/Exercises/Problems	804			
25	CURRENT, RESISTANCE, AND ELECTROMOTIVE FORCE	812			
25.1	Current	813			
25.2	Resistivity	816			
25.3	Resistance	819			
			25.4	Electromotive Force and Circuits	822
			25.5	Energy and Power in Electric Circuits	828
			25.6	Theory of Metallic Conduction	832
				Summary	835
				Guided Practice	836
				Questions/Exercises/Problems	837
			26	DIRECT-CURRENT CIRCUITS	844
			26.1	Resistors in Series and Parallel	844
			26.2	Kirchhoff's Rules	849
			26.3	Electrical Measuring Instruments	854
			26.4	<i>R-C</i> Circuits	858
			26.5	Power Distribution Systems	863
				Summary	867
				Guided Practice	868
				Questions/Exercises/Problems	869
			27	MAGNETIC FIELD AND MAGNETIC FORCES	878
			27.1	Magnetism	878
			27.2	Magnetic Field	880
			27.3	Magnetic Field Lines and Magnetic Flux	884
			27.4	Motion of Charged Particles in a Magnetic Field	888
			27.5	Applications of Motion of Charged Particles	891
			27.6	Magnetic Force on a Current-Carrying Conductor	893
			27.7	Force and Torque on a Current Loop	897
			27.8	The Direct-Current Motor	902
			27.9	The Hall Effect	904
				Summary	906
				Guided Practice	907
				Questions/Exercises/Problems	908
			28	SOURCES OF MAGNETIC FIELD	918
			28.1	Magnetic Field of a Moving Charge	918
			28.2	Magnetic Field of a Current Element	921



28.3	Magnetic Field of a Straight Current-Carrying Conductor	923
28.4	Force Between Parallel Conductors	926
28.5	Magnetic Field of a Circular Current Loop	927
28.6	Ampere's Law	930
28.7	Applications of Ampere's Law	933
28.8	Magnetic Materials	937
	Summary	942
	Guided Practice	943
	Questions/Exercises/Problems	944

29 ELECTROMAGNETIC INDUCTION 953

29.1	Induction Experiments	954
29.2	Faraday's Law	955
29.3	Lenz's Law	962
29.4	Motional emf	965
29.5	Induced Electric Fields	967
29.6	Eddy Currents	969
29.7	Displacement Current and Maxwell's Equations	970
29.8	Superconductivity	975
	Summary	977
	Guided Practice	978
	Questions/Exercises/Problems	979

30 INDUCTANCE 988

30.1	Mutual Inductance	988
30.2	Self-Inductance and Inductors	992
30.3	Magnetic-Field Energy	995
30.4	The R - L Circuit	998
30.5	The L - C Circuit	1002
30.6	The L - R - C Series Circuit	1007
	Summary	1009
	Guided Practice	1010
	Questions/Exercises/Problems	1011

31 ALTERNATING CURRENT 1018

31.1	Phasors and Alternating Currents	1018
31.2	Resistance and Reactance	1022
31.3	The L - R - C Series Circuit	1027
31.4	Power in Alternating-Current Circuits	1031
31.5	Resonance in Alternating-Current Circuits	1034
31.6	Transformers	1036
	Summary	1040
	Guided Practice	1041
	Questions/Exercises/Problems	1042

32 ELECTROMAGNETIC WAVES 1048

32.1	Maxwell's Equations and Electromagnetic Waves	1048
32.2	Plane Electromagnetic Waves and the Speed of Light	1052
32.3	Sinusoidal Electromagnetic Waves	1057

32.4	Energy and Momentum in Electromagnetic Waves	1061
32.5	Standing Electromagnetic Waves	1066
	Summary	1069
	Guided Practice	1070
	Questions/Exercises/Problems	1071

OPTICS

33 THE NATURE AND PROPAGATION OF LIGHT 1077

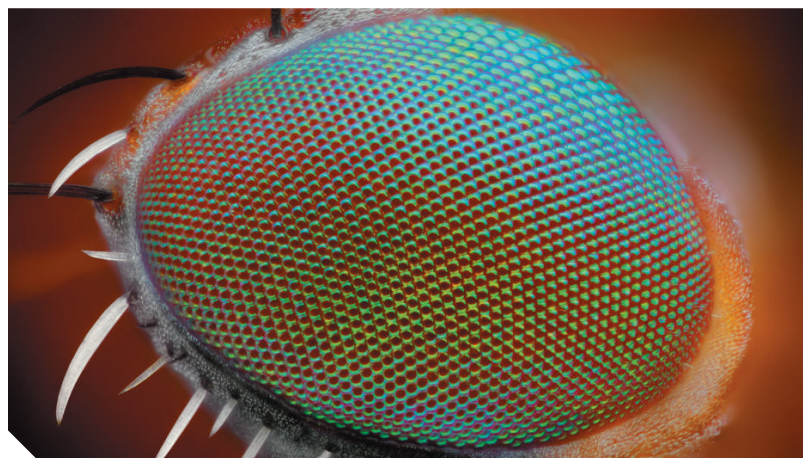
33.1	The Nature of Light	1077
33.2	Reflection and Refraction	1079
33.3	Total Internal Reflection	1085
33.4	Dispersion	1088
33.5	Polarization	1090
33.6	Scattering of Light	1097
33.7	Huygens's Principle	1099
	Summary	1101
	Guided Practice	1102
	Questions/Exercises/Problems	1103

34 GEOMETRIC OPTICS 1110

34.1	Reflection and Refraction at a Plane Surface	1110
34.2	Reflection at a Spherical Surface	1114
34.3	Refraction at a Spherical Surface	1122
34.4	Thin Lenses	1127
34.5	Cameras	1135
34.6	The Eye	1138
34.7	The Magnifier	1142
34.8	Microscopes and Telescopes	1143
	Summary	1148
	Guided Practice	1149
	Questions/Exercises/Problems	1150

35 INTERFERENCE 1159

35.1	Interference and Coherent Sources	1159
35.2	Two-Source Interference of Light	1163



35.3	Intensity in Interference Patterns	1166
35.4	Interference in Thin Films	1170
35.5	The Michelson Interferometer	1175
	Summary	1177
	Guided Practice	1178
	Questions/Exercises/Problems	1179

36 DIFFRACTION 1185

36.1	Fresnel and Fraunhofer Diffraction	1185
36.2	Diffraction from a Single Slit	1187
36.3	Intensity in the Single-Slit Pattern	1190
36.4	Multiple Slits	1194
36.5	The Diffraction Grating	1196
36.6	X-Ray Diffraction	1200
36.7	Circular Apertures and Resolving Power	1203
36.8	Holography	1206
	Summary	1208
	Guided Practice	1208
	Questions/Exercises/Problems	1209

MODERN PHYSICS

37 RELATIVITY 1217

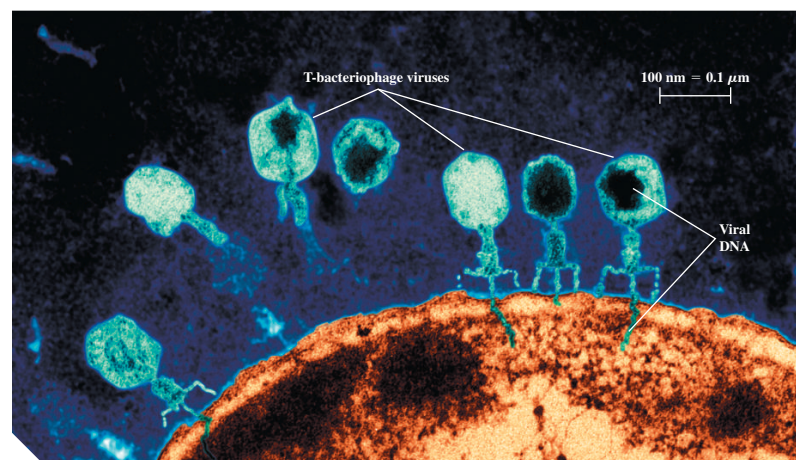
37.1	Invariance of Physical Laws	1217
37.2	Relativity of Simultaneity	1220
37.3	Relativity of Time Intervals	1222
37.4	Relativity of Length	1227
37.5	The Lorentz Transformations	1231
37.6	The Doppler Effect for Electromagnetic Waves	1235
37.7	Relativistic Momentum	1237
37.8	Relativistic Work and Energy	1239
37.9	Newtonian Mechanics and Relativity	1243
	Summary	1244
	Guided Practice	1245
	Questions/Exercises/Problems	1246

38 PHOTONS: LIGHT WAVES BEHAVING AS PARTICLES 1253

38.1	Light Absorbed as Photons: The Photoelectric Effect	1253
38.2	Light Emitted as Photons: X-Ray Production	1259
38.3	Light Scattered as Photons: Compton Scattering and Pair Production	1262
38.4	Wave-Particle Duality, Probability, and Uncertainty	1265
	Summary	1272
	Guided Practice	1273
	Questions/Exercises/Problems	1274

39 PARTICLES BEHAVING AS WAVES 1279

39.1	Electron Waves	1279
39.2	The Nuclear Atom and Atomic Spectra	1285



39.3	Energy Levels and the Bohr Model of the Atom	1290
39.4	The Laser	1300
39.5	Continuous Spectra	1303
39.6	The Uncertainty Principle Revisited	1308
	Summary	1311
	Guided Practice	1312
	Questions/Exercises/Problems	1313

40 QUANTUM MECHANICS I: WAVE FUNCTIONS 1321

40.1	Wave Functions and the One-Dimensional Schrödinger Equation	1321
40.2	Particle in a Box	1331
40.3	Potential Wells	1336
40.4	Potential Barriers and Tunneling	1340
40.5	The Harmonic Oscillator	1343
40.6	Measurement in Quantum Mechanics	1347
	Summary	1350
	Guided Practice	1351
	Questions/Exercises/Problems	1353

41 QUANTUM MECHANICS II: ATOMIC STRUCTURE 1360

41.1	The Schrödinger Equation in Three Dimensions	1360
41.2	Particle in a Three-Dimensional Box	1362
41.3	The Hydrogen Atom	1367
41.4	The Zeeman Effect	1375
41.5	Electron Spin	1378
41.6	Many-Electron Atoms and the Exclusion Principle	1385
41.7	X-Ray Spectra	1392
41.8	Quantum Entanglement	1395
	Summary	1399
	Guided Practice	1400
	Questions/Exercises/Problems	1401

42	MOLECULES AND CONDENSED MATTER	1408	44	PARTICLE PHYSICS AND COSMOLOGY	1483
42.1	Types of Molecular Bonds	1408	44.1	Fundamental Particles—A History	1483
42.2	Molecular Spectra	1411	44.2	Particle Accelerators and Detectors	1488
42.3	Structure of Solids	1415	44.3	Particles and Interactions	1492
42.4	Energy Bands	1418	44.4	Quarks and Gluons	1498
42.5	Free-Electron Model of Metals	1421	44.5	The Standard Model and Beyond	1502
42.6	Semiconductors	1425	44.6	The Expanding Universe	1504
42.7	Semiconductor Devices	1428	44.7	The Beginning of Time	1511
42.8	Superconductivity	1433		Summary	1519
	Summary	1433		Guided Practice	1520
	Guided Practice	1434		Questions/Exercises/Problems	1521
	Questions/Exercises/Problems	1435			
43	NUCLEAR PHYSICS	1442	APPENDICES		
43.1	Properties of Nuclei	1442	A	The International System of Units	A-1
43.2	Nuclear Binding and Nuclear Structure	1447	B	Useful Mathematical Relations	A-3
43.3	Nuclear Stability and Radioactivity	1452	C	The Greek Alphabet	A-4
43.4	Activities and Half-Lives	1459	D	Periodic Table of the Elements	A-5
43.5	Biological Effects of Radiation	1463	E	Unit Conversion Factors	A-6
43.6	Nuclear Reactions	1465	F	Numerical Constants	A-7
43.7	Nuclear Fission	1468		Answers to Odd-Numbered Problems	A-9
43.8	Nuclear Fusion	1472		Credits	C-1
	Summary	1475		Index	I-1
	Guided Practice	1476			
	Questions/Exercises/Problems	1477			



? Tornadoes are spawned by severe thunderstorms, so being able to predict the path of thunderstorms is essential. If a thunderstorm is moving at 15 km/h in a direction 37° north of east, how far north does the thunderstorm move in 2.0 h? (i) 30 km; (ii) 24 km; (iii) 18 km; (iv) 12 km; (v) 9 km.

1 Units, Physical Quantities, and Vectors

Physics is one of the most fundamental of the sciences. Scientists of all disciplines use the ideas of physics, including chemists who study the structure of molecules, paleontologists who try to reconstruct how dinosaurs walked, and climatologists who study how human activities affect the atmosphere and oceans. Physics is also the foundation of all engineering and technology. No engineer could design a flat-screen TV, a prosthetic leg, or even a better mousetrap without first understanding the basic laws of physics.

The study of physics is also an adventure. You'll find it challenging, sometimes frustrating, occasionally painful, and often richly rewarding. If you've ever wondered why the sky is blue, how radio waves can travel through empty space, or how a satellite stays in orbit, you can find the answers by using fundamental physics. You'll come to see physics as a towering achievement of the human intellect in its quest to understand our world and ourselves.

In this opening chapter, we'll go over some important preliminaries that we'll need throughout our study. We'll discuss the nature of physical theory and the use of idealized models to represent physical systems. We'll introduce the systems of units used to describe physical quantities and discuss ways to describe the accuracy of a number. We'll look at examples of problems for which we can't (or don't want to) find a precise answer, but for which rough estimates can be useful and interesting. Finally, we'll study several aspects of vectors and vector algebra. We'll need vectors throughout our study of physics to help us describe and analyze physical quantities, such as velocity and force, that have direction as well as magnitude.

1.1 THE NATURE OF PHYSICS

Physics is an *experimental* science. Physicists observe the phenomena of nature and try to find patterns that relate these phenomena. These patterns are called physical theories or, when they are very well established and widely used, physical laws or principles.

LEARNING OUTCOMES

In this chapter, you'll learn...

- 1.1 What a physical theory is.
- 1.2 The four steps you can use to solve any physics problem.
- 1.3 Three fundamental quantities of physics and the units physicists use to measure them.
- 1.4 How to work with units in your calculations.
- 1.5 How to keep track of significant figures in your calculations.
- 1.6 How to make rough, order-of-magnitude estimates.
- 1.7 The difference between scalars and vectors, and how to add and subtract vectors graphically.
- 1.8 What the components of a vector are and how to use them in calculations.
- 1.9 What unit vectors are and how to use them with components to describe vectors.
- 1.10 Two ways to multiply vectors: the scalar (dot) product and the vector (cross) product.

Figure 1.1 Two research laboratories.

(a) According to legend, Galileo investigated falling objects by dropping them from the Leaning Tower of Pisa, Italy, ...



... and he studied pendulum motion by observing the swinging chandelier in the adjacent cathedral.

(b) By doing experiments in apparent weightlessness on board the International Space Station, physicists have been able to make sensitive measurements that would be impossible in Earth's surface gravity.



CAUTION The meaning of “theory” A theory is *not* just a random thought or an unproven concept. Rather, a theory is an explanation of natural phenomena based on observation and accepted fundamental principles. An example is the well-established theory of biological evolution, which is the result of extensive research and observation by generations of biologists. **|**

To develop a physical theory, a physicist has to ask appropriate questions, design experiments to try to answer the questions, and draw appropriate conclusions from the results. **Figure 1.1** shows two important facilities used for physics experiments.

Legend has it that Galileo Galilei (1564–1642) dropped light and heavy objects from the top of the Leaning Tower of Pisa (Fig. 1.1a) to find out whether their rates of fall were different. From examining the results of his experiments (which were actually much more sophisticated than in the legend), he deduced the theory that the acceleration of a freely falling object is independent of its weight.

The development of physical theories such as Galileo’s often takes an indirect path, with blind alleys, wrong guesses, and the discarding of unsuccessful theories in favor of more promising ones. Physics is not simply a collection of facts and principles; it is also the *process* by which we arrive at general principles that describe how the physical universe behaves.

No theory is ever regarded as the ultimate truth. It’s always possible that new observations will require that a theory be revised or discarded. Note that we can disprove a theory by finding behavior that is inconsistent with it, but we can never prove that a theory is always correct.

Getting back to Galileo, suppose we drop a feather and a cannonball. They certainly do *not* fall at the same rate. This does not mean that Galileo was wrong; it means that his theory was incomplete. If we drop the feather and the cannonball *in a vacuum* to eliminate the effects of the air, then they do fall at the same rate. Galileo’s theory has a **range of validity**: It applies only to objects for which the force exerted by the air (due to air resistance and buoyancy) is much less than the weight. Objects like feathers or parachutes are clearly outside this range.

1.2 SOLVING PHYSICS PROBLEMS

At some point in their studies, almost all physics students find themselves thinking, “I understand the concepts, but I just can’t solve the problems.” But in physics, truly understanding a concept *means* being able to apply it to a variety of problems. Learning how to solve problems is absolutely essential; you don’t *know* physics unless you can *do* physics.

How do you learn to solve physics problems? In every chapter of this book you’ll find *Problem-Solving Strategies* that offer techniques for setting up and solving problems efficiently and accurately. Following each *Problem-Solving Strategy* are one or more worked *Examples* that show these techniques in action. (The *Problem-Solving Strategies* will also steer you away from some *incorrect* techniques that you may be tempted to use.) You’ll also find additional examples that aren’t associated with a particular *Problem-Solving Strategy*. In addition, at the end of each chapter you’ll find a *Bridging Problem* that uses more than one of the key ideas from the chapter. Study these strategies and problems carefully, and work through each example for yourself on a piece of paper.

Different techniques are useful for solving different kinds of physics problems, which is why this book offers dozens of *Problem-Solving Strategies*. No matter what kind of problem you’re dealing with, however, there are certain key steps that you’ll always follow. (These same steps are equally useful for problems in math, engineering, chemistry, and many other fields.) In this book we’ve organized these steps into four stages of solving a problem.

All of the *Problem-Solving Strategies* and *Examples* in this book will follow these four steps. (In some cases we’ll combine the first two or three steps.) We encourage you to follow these same steps when you solve problems yourself. You may find it useful to remember the acronym **I SEE**—short for *Identify*, *Set up*, *Execute*, and *Evaluate*.

PROBLEM-SOLVING STRATEGY 1.1 Solving Physics Problems

IDENTIFY the relevant concepts:

- Use the physical conditions stated in the problem to help you decide which physics concepts are relevant.
- Identify the **target variables** of the problem—that is, the quantities whose values you’re trying to find, such as the speed at which a projectile hits the ground, the intensity of a sound made by a siren, or the size of an image made by a lens.
- Identify the known quantities, as stated or implied in the problem. This step is essential whether the problem asks for an algebraic expression or a numerical answer.

SET UP the problem:

- Given the concepts, known quantities, and target variables that you found in the IDENTIFY step, choose the equations that you’ll use to solve the problem and decide how you’ll use them. Study the worked examples in this book for tips on how to select the proper equations. If this seems challenging, don’t worry—you’ll get better with practice!
- Make sure that the variables you have identified correlate exactly with those in the equations.

- If appropriate, draw a sketch of the situation described in the problem. (Graph paper and a ruler will help you make clear, useful sketches.)

EXECUTE the solution:

- Here’s where you’ll “do the math” with the equations that you selected in the SET UP step to solve for the target variables that you found in the IDENTIFY step. Study the worked examples to see what’s involved in this step.

EVALUATE your answer:

- Check your answer from the SOLVE step to see if it’s reasonable. (If you’re calculating how high a thrown baseball goes, an answer of 1.0 mm is unreasonably small and an answer of 100 km is unreasonably large.) If your answer includes an algebraic expression, confirm that it correctly represents what would happen if the variables in it had very large or very small values.
- For future reference, make note of any answer that represents a quantity of particular significance. Ask yourself how you might answer a more general or more difficult version of the problem you have just solved.

Idealized Models

In everyday conversation we use the word “model” to mean either a small-scale replica, such as a model railroad, or a person who displays articles of clothing (or the absence thereof). In physics a **model** is a simplified version of a physical system that would be too complicated to analyze in full detail.

For example, suppose we want to analyze the motion of a thrown baseball (**Fig. 1.2a**). How complicated is this problem? The ball is not a perfect sphere (it has raised seams), and it spins as it moves through the air. Air resistance and wind influence its motion, the ball’s weight varies a little as its altitude changes, and so on. If we try to include all these effects, the analysis gets hopelessly complicated. Instead, we invent a simplified version of the problem. We ignore the size, shape, and rotation of the ball by representing it as a point object, or **particle**. We ignore air resistance by making the ball move in a vacuum, and we make the weight constant. Now we have a problem that is simple enough to deal with (**Fig. 1.2b**). We’ll analyze this model in detail in Chapter 3.

We have to overlook quite a few minor effects to make an idealized model, but we must be careful not to neglect too much. If we ignore the effects of gravity completely, then our model predicts that when we throw the ball up, it will go in a straight line and disappear into space. A useful model simplifies a problem enough to make it manageable, yet keeps its essential features.

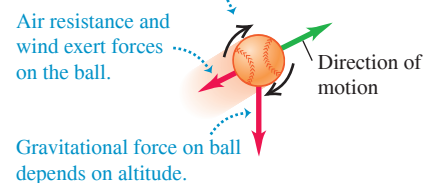
The validity of the predictions we make using a model is limited by the validity of the model. For example, Galileo’s prediction about falling objects (see Section 1.1) corresponds to an idealized model that does not include the effects of air resistance. This model works fairly well for a dropped cannonball, but not so well for a feather.

Idealized models play a crucial role throughout this book. Watch for them in discussions of physical theories and their applications to specific problems.

Figure 1.2 To simplify the analysis of (a) a baseball in flight, we use (b) an idealized model.

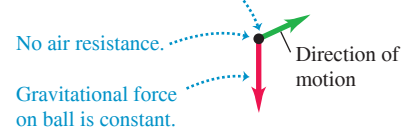
(a) A real baseball in flight

A baseball spins and has a complex shape.



(b) An idealized model of the baseball

Treat the baseball as a point object (particle).



1.3 STANDARDS AND UNITS

As we learned in Section 1.1, physics is an experimental science. Experiments require measurements, and we generally use numbers to describe the results of measurements. Any number that is used to describe a physical phenomenon quantitatively is called

a **physical quantity**. For example, two physical quantities that describe you are your weight and your height. Some physical quantities are so fundamental that we can define them only by describing how to measure them. Such a definition is called an **operational definition**. Two examples are measuring a distance by using a ruler and measuring a time interval by using a stopwatch. In other cases we define a physical quantity by describing how to calculate it from other quantities that we *can* measure. Thus we might define the average speed of a moving object as the distance traveled (measured with a ruler) divided by the time of travel (measured with a stopwatch).

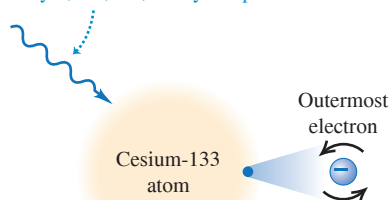
When we measure a quantity, we always compare it with some reference standard. When we say that a basketball hoop is 3.05 meters above the ground, we mean that this distance is 3.05 times as long as a meter stick, which we define to be 1 meter long. Such a standard defines a **unit** of the quantity. The meter is a unit of distance, and the second is a unit of time. When we use a number to describe a physical quantity, we must always specify the unit that we are using; to describe a distance as simply “3.05” wouldn’t mean anything.

To make accurate, reliable measurements, we need units of measurement that do not change and that can be duplicated by observers in various locations. The system of units used by scientists and engineers around the world is commonly called “the metric system,” but since 1960 it has been known officially as the **International System**, or **SI** (the abbreviation for its French name, *Système International*). Appendix A gives a list of all SI units as well as definitions of the most fundamental units.

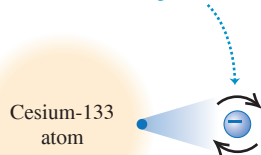
Figure 1.3 The measurements used to determine (a) the duration of a second and (b) the length of a meter. These measurements are useful for setting standards because they give the same results no matter where they are made.

(a) Measuring the second

Microwave radiation with a frequency of exactly 9,192,631,770 cycles per second ...

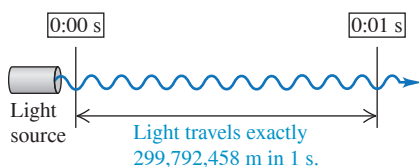


... causes the outermost electron of a cesium-133 atom to reverse its spin direction.



An atomic clock uses this phenomenon to tune microwaves to this exact frequency. It then counts 1 second for each 9,192,631,770 cycles.

(b) Measuring the meter



Time

From 1889 until 1967, the unit of time was defined as a certain fraction of the mean solar day, the average time between successive arrivals of the sun at its highest point in the sky. The present standard, adopted in 1967, is much more precise. It is based on an atomic clock, which uses the energy difference between the two lowest energy states of the cesium atom (^{133}Cs). When bombarded by microwaves of precisely the proper frequency, cesium atoms undergo a transition from one of these states to the other. One **second** (abbreviated s) is defined as the time required for 9,192,631,770 cycles of this microwave radiation (**Fig. 1.3a**).

Length

In 1960 an atomic standard for the meter was also established, using the wavelength of the orange-red light emitted by excited atoms of krypton (^{86}Kr). From this length standard, the speed of light in vacuum was measured to be 299,792,458 m/s. In November 1983, the length standard was changed again so that the speed of light in vacuum was *defined* to be precisely 299,792,458 m/s. Hence the new definition of the **meter** (abbreviated m) is the distance that light travels in vacuum in $1/299,792,458$ second (**Fig. 1.3b**). This modern definition provides a much more precise standard of length than the one based on a wavelength of light.

Mass

Until recently the unit of mass, the **kilogram** (abbreviated kg), was defined to be the mass of a metal cylinder kept at the International Bureau of Weights and Measures in France (**Fig. 1.4**). This was a very inconvenient standard to use. Since 2018 the value of the kilogram has been based on a fundamental constant of nature called *Planck's constant* (symbol h), whose defined value $h = 6.62607015 \times 10^{-34} \text{ kg} \cdot \text{m}^2/\text{s}$ is related to those of the kilogram, meter, and second. Given the values of the meter and the second, the masses of objects can be experimentally determined in terms of h . (We'll explain the meaning of h in Chapter 28.) The *gram* (which is not a fundamental unit) is 0.001 kilogram.

Other *derived units* can be formed from the fundamental units. For example, the units of speed are meters per second, or m/s; these are the units of length (m) divided by the units of time (s).

Unit Prefixes

Once we have defined the fundamental units, it is easy to introduce larger and smaller units for the same physical quantities. In the metric system these other units are related to the fundamental units (or, in the case of mass, to the gram) by multiples of 10 or $\frac{1}{10}$. Thus one kilometer (1 km) is 1000 meters, and one centimeter (1 cm) is $\frac{1}{100}$ meter. We usually express multiples of 10 or $\frac{1}{10}$ in exponential notation: $1000 = 10^3$, $\frac{1}{1000} = 10^{-3}$, and so on. With this notation, $1 \text{ km} = 10^3 \text{ m}$ and $1 \text{ cm} = 10^{-2} \text{ m}$.

The names of the additional units are derived by adding a **prefix** to the name of the fundamental unit. For example, the prefix “kilo-,” abbreviated k, always means a unit larger by a factor of 1000; thus

$$1 \text{ kilometer} = 1 \text{ km} = 10^3 \text{ meters} = 10^3 \text{ m}$$

$$1 \text{ kilogram} = 1 \text{ kg} = 10^3 \text{ grams} = 10^3 \text{ g}$$

$$1 \text{ kilowatt} = 1 \text{ kW} = 10^3 \text{ watts} = 10^3 \text{ W}$$

A table in Appendix A lists the standard SI units, with their meanings and abbreviations.

Table 1.1 gives some examples of the use of multiples of 10 and their prefixes with the units of length, mass, and time. **Figure 1.5** (next page) shows how these prefixes are used to describe both large and small distances.

The British System

Finally, we mention the British system of units. These units are used in only the United States and a few other countries, and in most of these they are being replaced by SI units. British units are now officially defined in terms of SI units, as follows:

$$\text{Length:} \quad 1 \text{ inch} = 2.54 \text{ cm (exactly)}$$

$$\text{Force:} \quad 1 \text{ pound} = 4.448221615260 \text{ newtons (exactly)}$$

The newton, abbreviated N, is the SI unit of force. The British unit of time is the second, defined the same way as in SI. In physics, British units are used in mechanics and thermodynamics only; there is no British system of electrical units.

Figure 1.4 Until 1889 a metal cylinder was used to define the value of the kilogram. (The one shown here, a copy of the one in France, was maintained by the U. S. National Institute of Standards and Technology.) Today the kilogram is defined in terms of one of the fundamental constants of nature.



TABLE 1.1 Some Units of Length, Mass, and Time

Length	Mass	Time
1 nanometer = $1 \text{ nm} = 10^{-9} \text{ m}$ (a few times the size of the largest atom)	1 microgram = $1 \mu\text{g} = 10^{-6} \text{ g} = 10^{-9} \text{ kg}$ (mass of a very small dust particle)	1 nanosecond = $1 \text{ ns} = 10^{-9} \text{ s}$ (time for light to travel 0.3 m)
1 micrometer = $1 \mu\text{m} = 10^{-6} \text{ m}$ (size of some bacteria and other cells)	1 milligram = $1 \text{ mg} = 10^{-3} \text{ g} = 10^{-6} \text{ kg}$ (mass of a grain of salt)	1 microsecond = $1 \mu\text{s} = 10^{-6} \text{ s}$ (time for space station to move 8 mm)
1 millimeter = $1 \text{ mm} = 10^{-3} \text{ m}$ (diameter of the point of a ballpoint pen)	1 gram = $1 \text{ g} = 10^{-3} \text{ kg}$ (mass of a paper clip)	1 millisecond = $1 \text{ ms} = 10^{-3} \text{ s}$ (time for a car moving at freeway speed to travel 3 cm)
1 centimeter = $1 \text{ cm} = 10^{-2} \text{ m}$ (diameter of your little finger)		
1 kilometer = $1 \text{ km} = 10^3 \text{ m}$ (distance in a 10 minute walk)		

Figure 1.5 Some typical lengths in the universe.

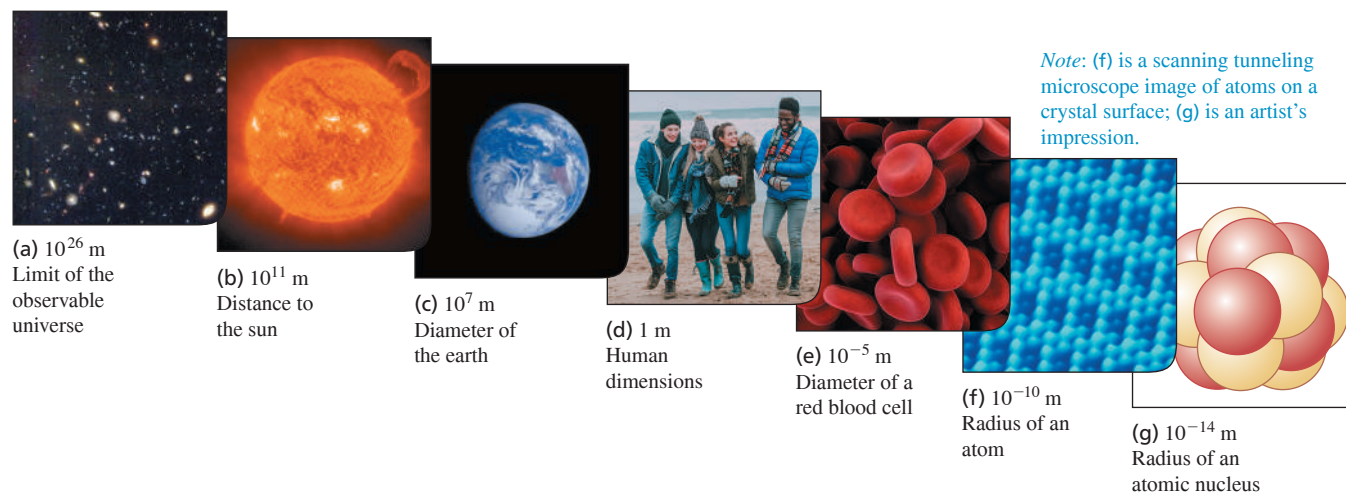
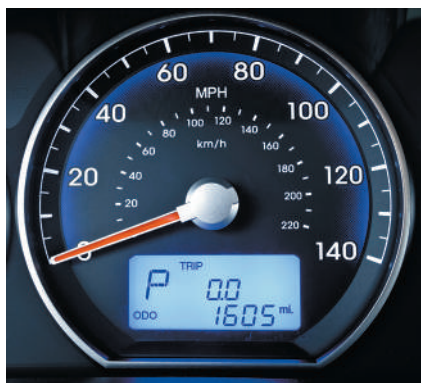


Figure 1.6 Many everyday items make use of both SI and British units. An example is this speedometer from a U.S.-built car, which shows the speed in both kilometers per hour (inner scale) and miles per hour (outer scale).



In this book we use SI units for all examples and problems, but we occasionally give approximate equivalents in British units. As you do problems using SI units, you may also wish to convert to the approximate British equivalents if they are more familiar to you (Fig. 1.6). But you should try to *think* in SI units as much as you can.

1.4 USING AND CONVERTING UNITS

We use equations to express relationships among physical quantities, represented by algebraic symbols. Each algebraic symbol always denotes both a number and a unit. For example, d might represent a distance of 10 m, t a time of 5 s, and v a speed of 2 m/s.

An equation must always be **dimensionally consistent**. You can't add apples and automobiles; two terms may be added or equated only if they have the same units. For example, if an object moving with constant speed v travels a distance d in a time t , these quantities are related by the equation

$$d = vt$$

If d is measured in meters, then the product vt must also be expressed in meters. Using the above numbers as an example, we may write

$$10 \text{ m} = \left(2 \frac{\text{m}}{\text{s}}\right)(5 \text{ s})$$

Because the unit s in the denominator of m/s cancels, the product has units of meters, as it must. In calculations, units are treated just like algebraic symbols with respect to multiplication and division.

CAUTION Always use units in calculations Make it a habit to *always* write numbers with the correct units and carry the units through the calculation as in the example above. This provides a very useful check. If at some stage in a calculation you find that an equation or an expression has inconsistent units, you know you have made an error. In this book we'll *always* carry units through all calculations, and we strongly urge you to follow this practice when you solve problems. ■

PROBLEM-SOLVING STRATEGY 1.2 Unit Conversions

IDENTIFY *the relevant concepts:* In most cases, it's best to use the fundamental SI units (lengths in meters, masses in kilograms, and times in seconds) in every problem. If you need the answer to be in a different set of units (such as kilometers, grams, or hours), wait until the end of the problem to make the conversion.

SET UP *the problem* and **EXECUTE** *the solution:* Units are multiplied and divided just like ordinary algebraic symbols. This gives us an easy way to convert a quantity from one set of units to another: Express the same physical quantity in two different units and form an equality.

For example, when we say that $1 \text{ min} = 60 \text{ s}$, we don't mean that the number 1 is equal to the number 60; rather, we mean that 1 min represents the same physical time interval as 60 s. For this reason, the ratio $(1 \text{ min})/(60 \text{ s})$ equals 1, as does its reciprocal, $(60 \text{ s})/(1 \text{ min})$. We may multiply a quantity by either of these factors

(which we call *unit multipliers*) without changing that quantity's physical meaning. For example, to find the number of seconds in 3 min, we write

$$3 \text{ min} = (3 \text{ min}) \left(\frac{60 \text{ s}}{1 \text{ min}} \right) = 180 \text{ s}$$

EVALUATE *your answer:* If you do your unit conversions correctly, unwanted units will cancel, as in the example above. If, instead, you had multiplied 3 min by $(1 \text{ min})/(60 \text{ s})$, your result would have been the nonsensical $\frac{1}{20} \text{ min}^2/\text{s}$. To be sure you convert units properly, include the units at *all* stages of the calculation.

Finally, check whether your answer is reasonable. For example, the result $3 \text{ min} = 180 \text{ s}$ is reasonable because the second is a smaller unit than the minute, so there are more seconds than minutes in the same time interval.

EXAMPLE 1.1 Converting speed units

The world land speed record of 763.0 mi/h was set on October 15, 1997, by Andy Green in the jet-engine car *Thrust SSC*. Express this speed in meters per second.

IDENTIFY, SET UP, and EXECUTE We need to convert the units of a speed from mi/h to m/s. We must therefore find unit multipliers that relate (i) miles to meters and (ii) hours to seconds. In Appendix E we find the equalities $1 \text{ mi} = 1.609 \text{ km}$, $1 \text{ km} = 1000 \text{ m}$, and $1 \text{ h} = 3600 \text{ s}$. We set up the conversion as follows, which ensures that all the desired cancellations by division take place:

$$\begin{aligned} 763.0 \text{ mi/h} &= \left(763.0 \frac{\text{mi}}{\text{h}} \right) \left(\frac{1.609 \text{ km}}{1 \text{ mi}} \right) \left(\frac{1000 \text{ m}}{1 \text{ km}} \right) \left(\frac{1 \text{ h}}{3600 \text{ s}} \right) \\ &= 341.0 \text{ m/s} \end{aligned}$$

EVALUATE This example shows a useful rule of thumb: A speed expressed in m/s is a bit less than half the value expressed in mi/h, and a bit less than one-third the value expressed in km/h. For example, a normal freeway speed is about $30 \text{ m/s} = 67 \text{ mi/h} = 108 \text{ km/h}$, and a typical walking speed is about $1.4 \text{ m/s} = 3.1 \text{ mi/h} = 5.0 \text{ km/h}$.

KEYCONCEPT To convert units, multiply by an appropriate unit multiplier.

EXAMPLE 1.2 Converting volume units

One of the world's largest cut diamonds is the First Star of Africa (mounted in the British Royal Sceptre and kept in the Tower of London). Its volume is 1.84 cubic inches. What is its volume in cubic centimeters? In cubic meters?

IDENTIFY, SET UP, and EXECUTE Here we are to convert the units of a volume from cubic inches (in.^3) to both cubic centimeters (cm^3) and cubic meters (m^3). Appendix E gives us the equality $1 \text{ in.} = 2.54 \text{ cm}$, from which we obtain $1 \text{ in.}^3 = (2.54 \text{ cm})^3$. We then have

$$\begin{aligned} 1.84 \text{ in.}^3 &= (1.84 \text{ in.}^3) \left(\frac{2.54 \text{ cm}}{1 \text{ in.}} \right)^3 \\ &= (1.84)(2.54)^3 \frac{\text{in.}^3 \text{ cm}^3}{\text{in.}^3} = 30.2 \text{ cm}^3 \end{aligned}$$

Appendix E also gives us $1 \text{ m} = 100 \text{ cm}$, so

$$\begin{aligned} 30.2 \text{ cm}^3 &= (30.2 \text{ cm}^3) \left(\frac{1 \text{ m}}{100 \text{ cm}} \right)^3 \\ &= (30.2) \left(\frac{1}{100} \right)^3 \frac{\text{cm}^3 \text{ m}^3}{\text{cm}^3} = 30.2 \times 10^{-6} \text{ m}^3 \\ &= 3.02 \times 10^{-5} \text{ m}^3 \end{aligned}$$

EVALUATE Following the pattern of these conversions, can you show that $1 \text{ in.}^3 \approx 16 \text{ cm}^3$ and that $1 \text{ m}^3 \approx 60,000 \text{ in.}^3$?

KEYCONCEPT If the units of a quantity are a product of simpler units, such as $\text{m}^3 = \text{m} \times \text{m} \times \text{m}$, use a product of unit multipliers to convert these units.

Figure 1.7 This spectacular mishap was the result of a very small percent error—traveling a few meters too far at the end of a journey of hundreds of thousands of meters.



TABLE 1.2 Using Significant Figures

Multiplication or division:

Result can have no more significant figures than the factor with the fewest significant figures:

$$\frac{0.745 \times 2.2}{3.885} = 0.42$$

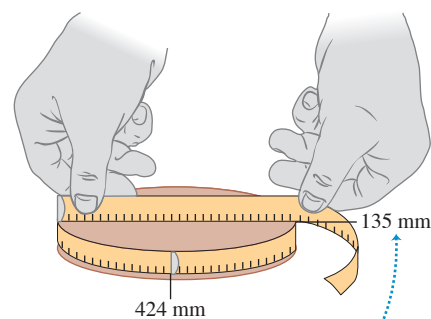
$$1.32578 \times 10^7 \times 4.11 \times 10^{-3} = 5.45 \times 10^4$$

Addition or subtraction:

Number of significant figures is determined by the term with the largest uncertainty (i.e., fewest digits to the right of the decimal point):

$$27.153 + 138.2 - 11.74 = 153.6$$

Figure 1.8 Determining the value of π from the circumference and diameter of a circle.



The measured values have only three significant figures, so their calculated ratio (π) also has only three significant figures.

1.5 UNCERTAINTY AND SIGNIFICANT FIGURES

Measurements always have uncertainties. If you measure the thickness of the cover of a hardbound version of this book using an ordinary ruler, your measurement is reliable to only the nearest millimeter, and your result will be 3 mm. It would be *wrong* to state this result as 3.00 mm; given the limitations of the measuring device, you can't tell whether the actual thickness is 3.00 mm, 2.85 mm, or 3.11 mm. But if you use a micrometer caliper, a device that measures distances reliably to the nearest 0.01 mm, the result will be 2.91 mm. The distinction between the measurements with a ruler and with a caliper is in their **uncertainty**; the measurement with a caliper has a smaller uncertainty. The uncertainty is also called the **error** because it indicates the maximum difference there is likely to be between the measured value and the true value. The uncertainty or error of a measured value depends on the measurement technique used.

We often indicate the **accuracy** of a measured value—that is, how close it is likely to be to the true value—by writing the number, the symbol $\pm$, and a second number indicating the uncertainty of the measurement. If the diameter of a steel rod is given as 56.47 ± 0.02 mm, this means that the true value is likely to be within the range from 56.45 mm to 56.49 mm. In a commonly used shorthand notation, the number 1.6454(21) means 1.6454 ± 0.0021 . The numbers in parentheses show the uncertainty in the final digits of the main number.

We can also express accuracy in terms of the maximum likely **fractional error** or **percent error** (also called *fractional uncertainty* and *percent uncertainty*). A resistor labeled “47 ohms $\pm 10\%$ ” probably has a true resistance that differs from 47 ohms by no more than 10% of 47 ohms—that is, by about 5 ohms. The resistance is probably between 42 and 52 ohms. For the diameter of the steel rod given above, the fractional error is $(0.02 \text{ mm})/(56.47 \text{ mm})$, or about 0.0004; the percent error is $(0.0004)(100\%)$, or about 0.04%. Even small percent errors can be very significant (Fig. 1.7).

In many cases the uncertainty of a number is not stated explicitly. Instead, the uncertainty is indicated by the number of meaningful digits, or **significant figures**, in the measured value. We gave the thickness of the cover of the book as 2.91 mm, which has three significant figures. By this we mean that the first two digits are known to be correct, while the third digit is uncertain. The last digit is in the hundredths place, so the uncertainty is about 0.01 mm. Two values with the *same* number of significant figures may have *different* uncertainties; a distance given as 137 km also has three significant figures, but the uncertainty is about 1 km. A distance given as 0.25 km has two significant figures (the zero to the left of the decimal point doesn't count); if given as 0.250 km, it has three significant figures.

When you use numbers that have uncertainties to compute other numbers, the computed numbers are also uncertain. When numbers are multiplied or divided, the result can have no more significant figures than the factor with the fewest significant figures has. For example, $3.1416 \times 2.34 \times 0.58 = 4.3$. When we add and subtract numbers, it's the location of the decimal point that matters, not the number of significant figures. For example, $123.62 + 8.9 = 132.5$. Although 123.62 has an uncertainty of about 0.01, 8.9 has an uncertainty of about 0.1. So their sum has an uncertainty of about 0.1 and should be written as 132.5, not 132.52. Table 1.2 summarizes these rules for significant figures.

To apply these ideas, suppose you want to verify the value of π , the ratio of the circumference of a circle to its diameter. The true value of this ratio to ten digits is 3.141592654. To test this, you draw a large circle and measure its circumference and diameter to the nearest millimeter, obtaining the values 424 mm and 135 mm (Fig. 1.8). You enter these into your calculator and obtain the quotient $(424 \text{ mm})/(135 \text{ mm}) = 3.140740741$. This may seem to disagree with the true value of π , but keep in mind that each of your measurements has three significant figures, so your measured value of π can have only three significant figures. It should be stated simply as 3.14. Within the limit of three significant figures, your value does agree with the true value.

In the examples and problems in this book we usually give numerical values with three significant figures, so your answers should usually have no more than three significant figures. (Many numbers in the real world have even less accuracy. The speedometer in a car, for example, usually gives only two significant figures.) Even if you do the arithmetic with a

calculator that displays ten digits, a ten-digit answer would misrepresent the accuracy of the results. Always round your final answer to keep only the correct number of significant figures or, in doubtful cases, one more at most. In Example 1.1 it would have been wrong to state the answer as 341.01861 m/s. Note that when you reduce such an answer to the appropriate number of significant figures, you must *round*, not *truncate*. Your calculator will tell you that the ratio of 525 m to 311 m is 1.688102894; to three significant figures, this is 1.69, not 1.68.

Here's a special note about calculations that involve multiple steps: As you work, it's helpful to keep extra significant figures in your calculations. Once you have your final answer, round it to the correct number of significant figures. This will give you the most accurate results.

When we work with very large or very small numbers, we can show significant figures much more easily by using **scientific notation**, sometimes called **powers-of-10 notation**. The distance from the earth to the moon is about 384,000,000 m, but writing the number in this form doesn't indicate the number of significant figures. Instead, we move the decimal point eight places to the left (corresponding to dividing by 10^8) and multiply by 10^8 ; that is,

$$384,000,000 \text{ m} = 3.84 \times 10^8 \text{ m}$$

In this form, it is clear that we have three significant figures. The number 4.00×10^{-7} also has three significant figures, even though two of them are zeros. Note that in scientific notation the usual practice is to express the quantity as a number between 1 and 10 multiplied by the appropriate power of 10.

When an integer or a fraction occurs in an algebraic equation, we treat that number as having no uncertainty at all. For example, in the equation $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$, which is Eq. (2.13) in Chapter 2, the coefficient 2 is *exactly* 2. We can consider this coefficient as having an infinite number of significant figures (2.000000 . . .). The same is true of the exponent 2 in v_x^2 and v_{0x}^2 .

Finally, let's note that **precision** is not the same as *accuracy*. A cheap digital watch that gives the time as 10:35:17 a.m. is very *precise* (the time is given to the second), but if the watch runs several minutes slow, then this value isn't very *accurate*. On the other hand, a grandfather clock might be very accurate (that is, display the correct time), but if the clock has no second hand, it isn't very precise. A high-quality measurement is both precise *and* accurate.

EXAMPLE 1.3 Significant figures in multiplication

The rest energy E of an object with rest mass m is given by Albert Einstein's famous equation $E = mc^2$, where c is the speed of light in vacuum. Find E for an electron for which (to three significant figures) $m = 9.11 \times 10^{-31} \text{ kg}$. The SI unit for E is the joule (J); $1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$.

IDENTIFY and SET UP Our target variable is the energy E . We are given the value of the mass m ; from Section 1.3 (or Appendix F) the speed of light is $c = 2.99792458 \times 10^8 \text{ m/s}$.

EXECUTE Substituting the values of m and c into Einstein's equation, we find

$$\begin{aligned} E &= (9.11 \times 10^{-31} \text{ kg})(2.99792458 \times 10^8 \text{ m/s})^2 \\ &= (9.11)(2.99792458)^2(10^{-31})(10^8)^2 \text{ kg} \cdot \text{m}^2/\text{s}^2 \\ &= (81.87659678)(10^{[-31+(2 \times 8)]}) \text{ kg} \cdot \text{m}^2/\text{s}^2 \\ &= 8.187659678 \times 10^{-14} \text{ kg} \cdot \text{m}^2/\text{s}^2 \end{aligned}$$

Since the value of m was given to only three significant figures, we must round this to

$$E = 8.19 \times 10^{-14} \text{ kg} \cdot \text{m}^2/\text{s}^2 = 8.19 \times 10^{-14} \text{ J}$$

EVALUATE While the rest energy contained in an electron may seem ridiculously small, on the atomic scale it is tremendous. Compare our answer to 10^{-19} J , the energy gained or lost by a single atom during a typical chemical reaction. The rest energy of an electron is about 1,000,000 times larger! (We'll discuss the significance of rest energy in Chapter 37.)

KEYCONCEPT When you are multiplying (or dividing) quantities, the result can have no more significant figures than the quantity with the fewest significant figures.

TEST YOUR UNDERSTANDING OF SECTION 1.5 The density of a material is equal to its mass divided by its volume. What is the density (in kg/m^3) of a rock of mass 1.80 kg and volume $6.0 \times 10^{-4} \text{ m}^3$? (i) $3 \times 10^3 \text{ kg}/\text{m}^3$; (ii) $3.0 \times 10^3 \text{ kg}/\text{m}^3$; (iii) $3.00 \times 10^3 \text{ kg}/\text{m}^3$; (iv) $3.000 \times 10^3 \text{ kg}/\text{m}^3$; (v) any of these—all of these answers are mathematically equivalent.

ANSWER

number with the fewest significant figures controls the number of significant figures in the result. (ii) Density = $(1.80 \text{ kg})/(6.0 \times 10^{-4} \text{ m}^3) = 3.0 \times 10^3 \text{ kg}/\text{m}^3$. When we multiply or divide, the

1.6 ESTIMATES AND ORDERS OF MAGNITUDE

We have stressed the importance of knowing the accuracy of numbers that represent physical quantities. But even a very crude estimate of a quantity often gives us useful information. Sometimes we know how to calculate a certain quantity, but we have to guess at the data we need for the calculation. Or the calculation might be too complicated to carry out exactly, so we make rough approximations. In either case our result is also a guess, but such a guess can be useful even if it is uncertain by a factor of two, ten, or more. Such calculations are called **order-of-magnitude estimates**. The great Italian-American nuclear physicist Enrico Fermi (1901–1954) called them “back-of-the-envelope calculations.”

Exercises 1.15 through 1.20 at the end of this chapter are of the estimating, or order-of-magnitude, variety. Most require guesswork for the needed input data. Don’t try to look up a lot of data; make the best guesses you can. Even when they are off by a factor of ten, the results can be useful and interesting.

EXAMPLE 1.4 An order-of-magnitude estimate

You are writing an adventure novel in which the hero escapes with a billion dollars’ worth of gold in his suitcase. Could anyone carry that much gold? Would it fit in a suitcase?

IDENTIFY, SET UP, and EXECUTE Gold sells for about \$1400 an ounce, or about \$100 for $\frac{1}{14}$ ounce. (The price per ounce has varied between \$200 and \$1900 over the past twenty years or so.) An ounce is about 30 grams, so \$100 worth of gold has a mass of about $\frac{1}{14}$ of 30 grams, or roughly 2 grams. A billion (10^9) dollars’ worth of gold has a mass 10^7 times greater, about 2×10^7 (20 million) grams or 2×10^4 (20,000) kilograms. A thousand kilograms has a weight in British units of about a ton, so the suitcase weighs roughly 20 tons! No human could lift it.

Roughly what is the *volume* of this gold? The density of water is 10^3 kg/m^3 ; if gold, which is much denser than water, has a density 10 times greater, then 10^4 kg of gold fits into a volume of 1 m^3 . So 10^9 dollars’ worth of gold has a volume of 2 m^3 , many times the volume of a suitcase.

EVALUATE Clearly your novel needs rewriting. Try the calculation again with a suitcase full of five-carat (1-gram) diamonds, each worth \$500,000. Would this work?

KEYCONCEPT To decide whether the numerical value of a quantity is reasonable, assess the quantity in terms of other quantities that you can estimate, even if only roughly.

TEST YOUR UNDERSTANDING OF SECTION 1.6 Can you estimate the total number of teeth in the mouths of all the students on your campus? (*Hint:* How many teeth are in your mouth? Count them!)

ANSWER

The answer depends on how many students are enrolled at your campus.

APPLICATION Scalar

Temperature, Vector Wind The comfort level on a wintry day depends on the temperature, a scalar quantity that can be positive or negative (say, $+5^\circ\text{C}$ or -20°C) but has no direction. It also depends on the wind velocity, a vector quantity with both magnitude and direction (for example, 15 km/h from the west).



1.7 VECTORS AND VECTOR ADDITION

Some physical quantities, such as time, temperature, mass, and density, can be described completely by a single number with a unit. But many other important quantities in physics have a *direction* associated with them and cannot be described by a single number. A simple example is the motion of an airplane: We must say not only how fast the plane is moving but also in what direction. The speed of the airplane combined with its direction of motion constitute a quantity called *velocity*. Another example is *force*, which in physics means a push or pull exerted on an object. Giving a complete description of a force means describing both how hard the force pushes or pulls on the object and the direction of the push or pull.

When a physical quantity is described by a single number, we call it a **scalar quantity**. In contrast, a **vector quantity** has both a **magnitude** (the “how much” or “how big” part) and a direction in space. Calculations that combine scalar quantities use the operations of ordinary arithmetic. For example, $6 \text{ kg} + 3 \text{ kg} = 9 \text{ kg}$, or $4 \times 2 \text{ s} = 8 \text{ s}$. However, combining vectors requires a different set of operations.

To understand more about vectors and how they combine, we start with the simplest vector quantity, **displacement**. Displacement is a change in the position of an object.

Displacement is a vector quantity because we must state not only how far the object moves but also in what direction. Walking 3 km north from your front door doesn't get you to the same place as walking 3 km southeast; these two displacements have the same magnitude but different directions.

We usually represent a vector quantity such as displacement by a single letter, such as $\vec{A}$ in Fig. 1.9a. In this book we always print vector symbols in **boldface italic type with an arrow above them**. We do this to remind you that vector quantities have different properties from scalar quantities; the arrow is a reminder that vectors have direction. When you handwrite a symbol for a vector, *always* write it with an arrow on top. If you don't distinguish between scalar and vector quantities in your notation, you probably won't make the distinction in your thinking either, and confusion will result.

We always *draw* a vector as a line with an arrowhead at its tip. The length of the line shows the vector's magnitude, and the direction of the arrowhead shows the vector's direction. Displacement is always a straight-line segment directed from the starting point to the ending point, even though the object's actual path may be curved (Fig. 1.9b). Note that displacement is not related directly to the total *distance* traveled. If the object were to continue past P_2 and then return to P_1 , the displacement for the entire trip would be *zero* (Fig. 1.9c).

If two vectors have the same direction, they are **parallel**. If they have the same magnitude *and* the same direction, they are *equal*, no matter where they are located in space. The vector $\vec{A}'$ from point P_3 to point P_4 in Fig. 1.10 has the same length and direction as the vector $\vec{A}$ from P_1 to P_2 . These two displacements are equal, even though they start at different points. We write this as $\vec{A}' = \vec{A}$ in Fig. 1.10; the boldface equals sign emphasizes that equality of two vector quantities is not the same relationship as equality of two scalar quantities. Two vector quantities are equal only when they have the same magnitude *and* the same direction.

Vector $\vec{B}$ in Fig. 1.10, however, is not equal to $\vec{A}$ because its direction is *opposite* that of $\vec{A}$. We define the **negative of a vector** as a vector having the same magnitude as the original vector but the *opposite* direction. The negative of vector quantity $\vec{A}$ is denoted as $-\vec{A}$, and we use a boldface minus sign to emphasize the vector nature of the quantities. If $\vec{A}$ is 87 m south, then $-\vec{A}$ is 87 m north. Thus we can write the relationship between $\vec{A}$ and $\vec{B}$ in Fig. 1.10 as $\vec{A} = -\vec{B}$ or $\vec{B} = -\vec{A}$. When two vectors $\vec{A}$ and $\vec{B}$ have opposite directions, whether their magnitudes are the same or not, we say that they are **antiparallel**.

We usually represent the *magnitude* of a vector quantity by the same letter used for the vector, but in *lightface italic type* with *no* arrow on top. For example, if displacement vector $\vec{A}$ is 87 m south, then $A = 87$ m. An alternative notation is the vector symbol with vertical bars on both sides:

$$(\text{Magnitude of } \vec{A}) = A = |\vec{A}| \quad (1.1)$$

The magnitude of a vector quantity is a scalar quantity (a number) and is *always positive*. Note that a vector can never be equal to a scalar because they are different kinds of quantities. The expression " $\vec{A} = 6$ m" is just as wrong as "2 oranges = 3 apples"!

When we draw diagrams with vectors, it's best to use a scale similar to those used for maps. For example, a displacement of 5 km might be represented in a diagram by a vector 1 cm long, and a displacement of 10 km by a vector 2 cm long.

Vector Addition and Subtraction

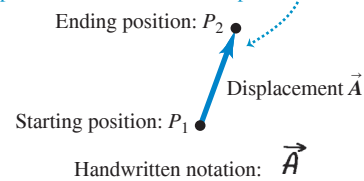
Suppose a particle undergoes a displacement $\vec{A}$ followed by a second displacement $\vec{B}$. The final result is the same as if the particle had started at the same initial point and undergone a single displacement $\vec{C}$ (Fig. 1.11a, next page). We call displacement $\vec{C}$ the **vector sum**, or **resultant**, of displacements $\vec{A}$ and $\vec{B}$. We express this relationship symbolically as

$$\vec{C} = \vec{A} + \vec{B} \quad (1.2)$$

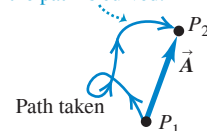
The boldface plus sign emphasizes that adding two vector quantities requires a geometrical process and is not the same operation as adding two scalar quantities such as $2 + 3 = 5$. In vector addition we usually place the *tail* of the *second* vector at the *head*, or tip, of the *first* vector (Fig. 1.11a).

Figure 1.9 Displacement as a vector quantity.

(a) We represent a displacement by an arrow that points in the direction of displacement.



(b) A displacement is always a straight arrow directed from the starting position to the ending position. It does not depend on the path taken, even if the path is curved.



(c) Total displacement for a round trip is 0, regardless of the path taken or distance traveled.

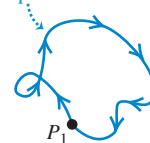


Figure 1.10 The meaning of vectors that have the same magnitude and the same or opposite direction.

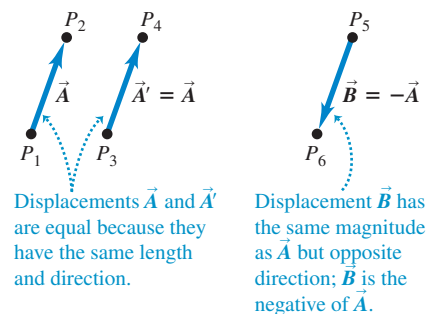
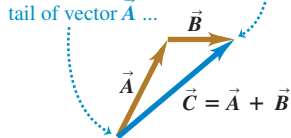


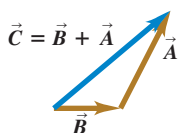
Figure 1.11 Three ways to add two vectors.

(a) We can add two vectors by placing them head to tail.

The vector sum $\vec{C}$ extends from the tail of vector $\vec{A}$... to the head of vector $\vec{B}$.



(b) Adding them in reverse order gives the same result: $\vec{A} + \vec{B} = \vec{B} + \vec{A}$. The order doesn't matter in vector addition.



(c) We can also add two vectors by placing them tail to tail and constructing a parallelogram.

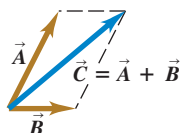
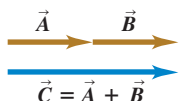
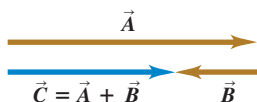


Figure 1.12 Adding vectors that are (a) parallel and (b) antiparallel.

(a) Only when vectors $\vec{A}$ and $\vec{B}$ are parallel does the magnitude of their vector sum $\vec{C}$ equal the sum of their magnitudes: $C = A + B$.



(b) When $\vec{A}$ and $\vec{B}$ are antiparallel, the magnitude of their vector sum $\vec{C}$ equals the difference of their magnitudes: $C = |A - B|$.



If we make the displacements $\vec{A}$ and $\vec{B}$ in reverse order, with $\vec{B}$ first and $\vec{A}$ second, the result is the same (Fig. 1.11b). Thus

$$\vec{C} = \vec{B} + \vec{A} \quad \text{and} \quad \vec{A} + \vec{B} = \vec{B} + \vec{A} \quad (1.3)$$

This shows that the order of terms in a vector sum doesn't matter. In other words, vector addition obeys the *commutative* law.

Figure 1.11c shows another way to represent the vector sum: If we draw vectors $\vec{A}$ and $\vec{B}$ with their tails at the same point, vector $\vec{C}$ is the diagonal of a parallelogram constructed with $\vec{A}$ and $\vec{B}$ as two adjacent sides.

CAUTION **Magnitudes in vector addition** It's a common error to conclude that if $\vec{C} = \vec{A} + \vec{B}$, then magnitude C equals magnitude A plus magnitude B . In general, this conclusion is *wrong*; for the vectors shown in Fig. 1.11, $C < A + B$. The magnitude of $\vec{A} + \vec{B}$ depends on the magnitudes of $\vec{A}$ and $\vec{B}$ and on the angle between $\vec{A}$ and $\vec{B}$. Only in the special case in which $\vec{A}$ and $\vec{B}$ are *parallel* is the magnitude of $\vec{C} = \vec{A} + \vec{B}$ equal to the sum of the magnitudes of $\vec{A}$ and $\vec{B}$ (Fig. 1.12a). When the vectors are *antiparallel* (Fig. 1.12b), the magnitude of $\vec{C}$ equals the *difference* of the magnitudes of $\vec{A}$ and $\vec{B}$. Be careful to distinguish between scalar and vector quantities, and you'll avoid making errors about the magnitude of a vector sum. ■

Figure 1.13a shows *three* vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$. To find the vector sum of all three, in Fig. 1.13b we first add $\vec{A}$ and $\vec{B}$ to give a vector sum $\vec{D}$; we then add vectors $\vec{C}$ and $\vec{D}$ by the same process to obtain the vector sum $\vec{R}$:

$$\vec{R} = (\vec{A} + \vec{B}) + \vec{C} = \vec{D} + \vec{C}$$

Alternatively, we can first add $\vec{B}$ and $\vec{C}$ to obtain vector $\vec{E}$ (Fig. 1.13c), and then add $\vec{A}$ and $\vec{E}$ to obtain $\vec{R}$:

$$\vec{R} = \vec{A} + (\vec{B} + \vec{C}) = \vec{A} + \vec{E}$$

We don't even need to draw vectors $\vec{D}$ and $\vec{E}$; all we need to do is draw $\vec{A}$, $\vec{B}$, and $\vec{C}$ in succession, with the tail of each at the head of the one preceding it. The sum vector $\vec{R}$ extends from the tail of the first vector to the head of the last vector (Fig. 1.13d). The order makes no difference; Fig. 1.13e shows a different order, and you should try others. Vector addition obeys the *associative* law.

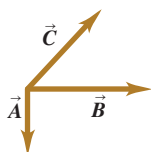
We can *subtract* vectors as well as add them. To see how, recall that vector $-\vec{A}$ has the same magnitude as $\vec{A}$ but the opposite direction. We define the difference $\vec{A} - \vec{B}$ of two vectors $\vec{A}$ and $\vec{B}$ to be the vector sum of $\vec{A}$ and $-\vec{B}$:

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) \quad (1.4)$$

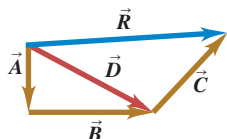
Figure 1.14 shows an example of vector subtraction.

Figure 1.13 Several constructions for finding the vector sum $\vec{A} + \vec{B} + \vec{C}$.

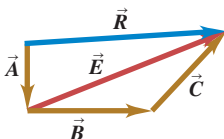
(a) To find the sum of these three vectors ...



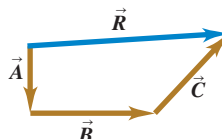
(b) ... add $\vec{A}$ and $\vec{B}$ to get $\vec{D}$ and then add $\vec{C}$ to $\vec{D}$ to get the final sum (resultant) $\vec{R}$...



(c) ... or add $\vec{B}$ and $\vec{C}$ to get $\vec{E}$ and then add $\vec{E}$ to $\vec{A}$ to get $\vec{R}$...



(d) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ to get $\vec{R}$ directly ...



(e) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ in any other order and still get $\vec{R}$.

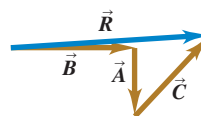
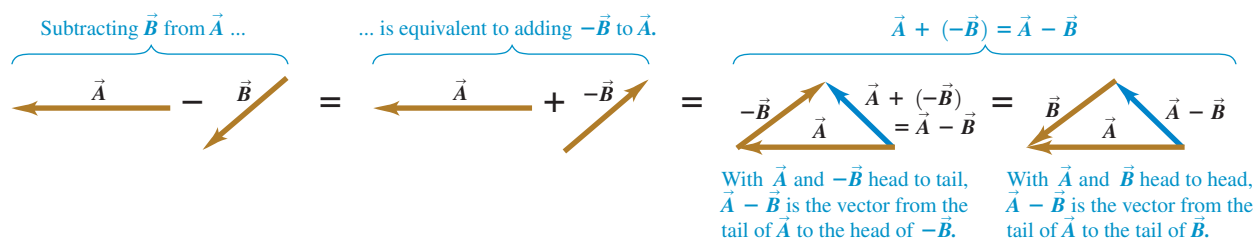


Figure 1.14 To construct the vector difference $\vec{A} - \vec{B}$, you can either place the tail of $-\vec{B}$ at the head of $\vec{A}$ or place the two vectors $\vec{A}$ and $\vec{B}$ head to head.

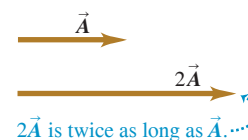


A vector quantity such as a displacement can be multiplied by a scalar quantity (an ordinary number). The displacement $2\vec{A}$ is a displacement (vector quantity) in the same direction as vector $\vec{A}$ but twice as long; this is the same as adding $\vec{A}$ to itself (Fig. 1.15a). In general, when we multiply a vector $\vec{A}$ by a scalar c , the result $c\vec{A}$ has magnitude $|c|A$ (the absolute value of c multiplied by the magnitude of vector $\vec{A}$). If c is positive, $c\vec{A}$ is in the same direction as $\vec{A}$; if c is negative, $c\vec{A}$ is in the direction opposite to $\vec{A}$. Thus $3\vec{A}$ is parallel to $\vec{A}$, while $-3\vec{A}$ is antiparallel to $\vec{A}$ (Fig. 1.15b).

A scalar used to multiply a vector can also be a physical quantity. For example, you may be familiar with the relationship $\vec{F} = m\vec{a}$; the net force $\vec{F}$ (a vector quantity) that acts on an object is equal to the product of the object's mass m (a scalar quantity) and its acceleration $\vec{a}$ (a vector quantity). The direction of $\vec{F}$ is the same as that of $\vec{a}$ because m is positive, and the magnitude of $\vec{F}$ is equal to the mass m multiplied by the magnitude of $\vec{a}$. The unit of force is the unit of mass multiplied by the unit of acceleration.

Figure 1.15 Multiplying a vector by a scalar.

(a) Multiplying a vector by a positive scalar changes the magnitude (length) of the vector but not its direction.



(b) Multiplying a vector by a negative scalar changes its magnitude and reverses its direction.



EXAMPLE 1.5 Adding two vectors at right angles

A cross-country skier skis 1.00 km north and then 2.00 km east on a horizontal snowfield. How far and in what direction is she from the starting point?

IDENTIFY and SET UP The problem involves combining two displacements at right angles to each other. This vector addition amounts to solving a right triangle, so we can use the Pythagorean theorem and trigonometry. The target variables are the skier's straight-line distance and direction from her starting point. **Figure 1.16** is a scale diagram of the two displacements and the resultant net displacement. We denote the direction from the starting point by the angle ϕ (the Greek letter phi). The displacement appears to be a bit more than 2 km. Measuring the angle with a protractor indicates that ϕ is about 63° .

EXECUTE The distance from the starting point to the ending point is equal to the length of the hypotenuse:

$$\sqrt{(1.00 \text{ km})^2 + (2.00 \text{ km})^2} = 2.24 \text{ km}$$

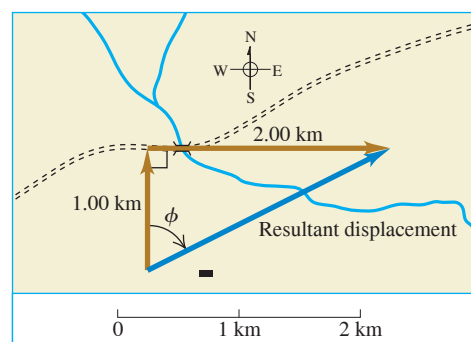
A little trigonometry (from Appendix B) allows us to find angle ϕ :

$$\tan \phi = \frac{\text{Opposite side}}{\text{Adjacent side}} = \frac{2.00 \text{ km}}{1.00 \text{ km}} = 2.00$$

$$\phi = \arctan 2.00 = 63.4^\circ$$

We can describe the direction as 63.4° east of north or $90^\circ - 63.4^\circ = 26.6^\circ$ north of east.

Figure 1.16 The vector diagram, drawn to scale, for a ski trip.



EVALUATE Our answers (2.24 km and $\phi = 63.4^\circ$) are close to our predictions. In Section 1.8 we'll learn how to easily add two vectors *not* at right angles to each other.

KEYCONCEPT In every problem involving vector addition, draw the two vectors being added as well as the vector sum. The head-to-tail arrangement shown in Figs. 1.11a and 1.11b is easiest. This will help you to visualize the vectors and understand the direction of the vector sum. Drawing the vectors is equally important for problems involving vector subtraction (see Fig. 1.14).

TEST YOUR UNDERSTANDING OF SECTION 1.7 Two displacement vectors, $\vec{S}$ and $\vec{T}$, have magnitudes $S = 3$ m and $T = 4$ m. Which of the following could be the magnitude of the difference vector $\vec{S} - \vec{T}$? (There may be more than one correct answer.) (i) 9 m; (ii) 7 m; (iii) 5 m; (iv) 1 m; (v) 0 m; (vi) -1 m.

ANSWER

(i), (ii), and (iii) Vector $\vec{S} - \vec{T}$ has the same magnitude as vector $\vec{T}$, so $\vec{S} - \vec{T} = \vec{T}$ and $|\vec{S} - \vec{T}| = |\vec{T}| = 4$ m. This sum has magnitude 7 m. The magnitude of $\vec{S} - \vec{T}$ is 1 m if $\vec{S}$ and $\vec{T}$ are antiparallel. The magnitude of $\vec{S} - \vec{T}$ is 5 m if $\vec{S}$ and $\vec{T}$ are perpendicular, when vectors $\vec{S}$, $\vec{T}$, and $\vec{S} - \vec{T}$ form a 3-4-5 right triangle. Answer (i) is impossible because the magnitude of the sum of two vectors cannot be greater than the sum of the magnitudes; answer (v) is impossible because the sum of two vectors can be zero only if the two vectors are antiparallel and have the same magnitude; and answer (vi) is impossible because the magnitude of a vector cannot be negative.

1.8 COMPONENTS OF VECTORS

In Section 1.7 we added vectors by using a scale diagram and properties of right triangles. But calculations with right triangles work only when the two vectors are perpendicular. So we need a simple but general method for adding vectors. This is called the method of *components*.

To define what we mean by the components of a vector $\vec{A}$, we begin with a rectangular (Cartesian) coordinate system of axes (**Fig. 1.17**). If we think of $\vec{A}$ as a displacement vector, we can regard $\vec{A}$ as the sum of a displacement parallel to the x -axis and a displacement parallel to the y -axis. We use the numbers A_x and A_y to tell us how much displacement there is parallel to the x -axis and how much there is parallel to the y -axis, respectively. For example, if the $+x$ -axis points east and the $+y$ -axis points north, $\vec{A}$ in **Fig. 1.17** could be the sum of a 2.00 m displacement to the east and a 1.00 m displacement to the north. Then $A_x = +2.00$ m and $A_y = +1.00$ m. We can use the same idea for any vectors, not just displacement vectors. The two numbers A_x and A_y are called the **components** of $\vec{A}$.

CAUTION Components are not vectors The components A_x and A_y of a vector $\vec{A}$ are numbers; they are *not* vectors themselves. This is why we print the symbols for components in lightface italic type with *no* arrow on top instead of in boldface italic with an arrow, which is reserved for vectors. ■

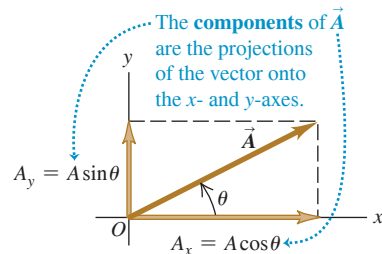
We can calculate the components of vector $\vec{A}$ if we know its magnitude A and its direction. We'll describe the direction of a vector by its angle relative to some reference direction. In **Fig. 1.17** this reference direction is the positive x -axis, and the angle between vector $\vec{A}$ and the positive x -axis is θ (the Greek letter theta). Imagine that vector $\vec{A}$ originally lies along the $+x$ -axis and that you then rotate it to its true direction, as indicated by the arrow in **Fig. 1.17** on the arc for angle θ . If this rotation is from the $+x$ -axis toward the $+y$ -axis, as is the case in **Fig. 1.17**, then θ is *positive*; if the rotation is from the $+x$ -axis toward the $-y$ -axis, then θ is *negative*. Thus the $+y$ -axis is at an angle of 90° , the $-x$ -axis at 180° , and the $-y$ -axis at 270° (or -90°). If θ is measured in this way, then from the definition of the trigonometric functions,

$$\begin{aligned} \frac{A_x}{A} &= \cos \theta & \text{and} & & \frac{A_y}{A} &= \sin \theta \\ A_x &= A \cos \theta & \text{and} & & A_y &= A \sin \theta \end{aligned} \quad (1.5)$$

(θ measured from the $+x$ -axis, rotating toward the $+y$ -axis)

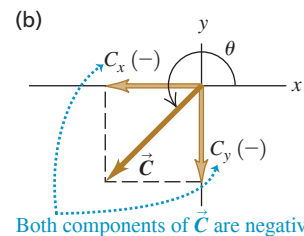
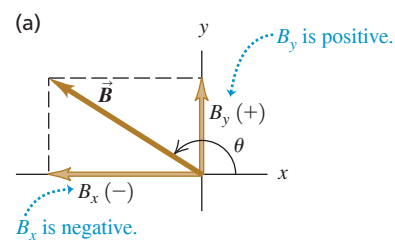
In **Fig. 1.17** A_x and A_y are positive. This is consistent with Eqs. (1.5); θ is in the first quadrant (between 0° and 90°), and both the cosine and the sine of an angle in this quadrant are positive. But in **Fig. 1.18a** the component B_x is negative and the component B_y is positive. (If the $+x$ -axis points east and the $+y$ -axis points north, $\vec{B}$ could represent a displacement of 2.00 m west and 1.00 m north. Since west is in the $-x$ -direction and north is in the $+y$ -direction, $B_x = -2.00$ m is negative and $B_y = +1.00$ m is positive.) Again,

Figure 1.17 Representing a vector $\vec{A}$ in terms of its components A_x and A_y .



In this case, both A_x and A_y are positive.

Figure 1.18 The components of a vector may be positive or negative numbers.



this is consistent with Eqs. (1.5); now θ is in the second quadrant, so $\cos \theta$ is negative and $\sin \theta$ is positive. In Fig. 1.18b both C_x and C_y are negative (both $\cos \theta$ and $\sin \theta$ are negative in the third quadrant).

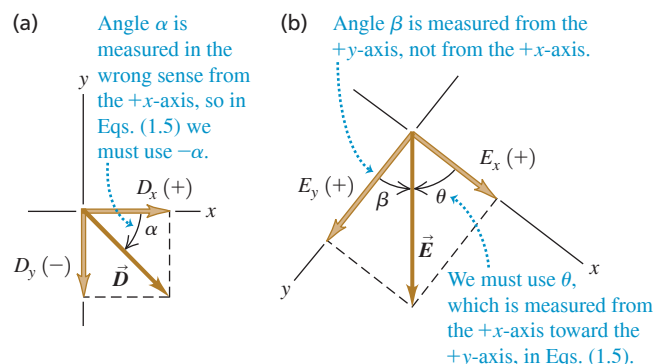
CAUTION **Relating a vector's magnitude and direction to its components** Equations (1.5) are correct *only* when the angle θ is measured from the positive x -axis. If the angle of the vector is given from a different reference direction or you use a different rotation direction, the relationships are different! Example 1.6 illustrates this point. ■

EXAMPLE 1.6 Finding components

(a) What are the x - and y -components of vector $\vec{D}$ in Fig. 1.19a? The magnitude of the vector is $D = 3.00$ m, and angle $\alpha = 45^\circ$. (b) What are the x - and y -components of vector $\vec{E}$ in Fig. 1.19b? The magnitude of the vector is $E = 4.50$ m, and angle $\beta = 37.0^\circ$.

IDENTIFY and SET UP We can use Eqs. (1.5) to find the components of these vectors, but we must be careful: Neither angle α nor β in Fig. 1.19 is measured from the $+x$ -axis toward the $+y$ -axis. We estimate from the figure that the lengths of both components in part (a) are roughly 2 m, and that those in part (b) are 3 m and 4 m. The figure indicates the signs of the components.

Figure 1.19 Calculating the x - and y -components of vectors.



EXECUTE (a) The angle α (the Greek letter alpha) between the positive x -axis and $\vec{D}$ is measured toward the *negative* y -axis. The angle we must use in Eqs. (1.5) is $\theta = -\alpha = -45^\circ$. We then find

$$D_x = D \cos \theta = (3.00 \text{ m})(\cos(-45^\circ)) = +2.1 \text{ m}$$

$$D_y = D \sin \theta = (3.00 \text{ m})(\sin(-45^\circ)) = -2.1 \text{ m}$$

Had we carelessly substituted $+45^\circ$ for θ in Eqs. (1.5), our result for D_y would have had the wrong sign.

(b) The x - and y -axes in Fig. 1.19b are at right angles, so it doesn't matter that they aren't horizontal and vertical, respectively. But we can't use the angle β (the Greek letter beta) in Eqs. (1.5), because β is measured from the $+y$ -axis. Instead, we must use the angle $\theta = 90.0^\circ - \beta = 90.0^\circ - 37.0^\circ = 53.0^\circ$. Then we find

$$E_x = E \cos 53.0^\circ = (4.50 \text{ m})(\cos 53.0^\circ) = +2.71 \text{ m}$$

$$E_y = E \sin 53.0^\circ = (4.50 \text{ m})(\sin 53.0^\circ) = +3.59 \text{ m}$$

EVALUATE Our answers to both parts are close to our predictions. But why do the answers in part (a) correctly have only two significant figures?

KEYCONCEPT When you are finding the components of a vector, always use a diagram of the vector and the coordinate axes to guide your calculations.

Using Components to Do Vector Calculations

Using components makes it relatively easy to do various calculations involving vectors. Let's look at three important examples: finding a vector's magnitude and direction, multiplying a vector by a scalar, and calculating the vector sum of two or more vectors.

- 1. Finding a vector's magnitude and direction from its components.** We can describe a vector completely by giving either its magnitude and direction or its x - and y -components. Equations (1.5) show how to find the components if we know the magnitude and direction. We can also reverse the process: We can find the magnitude and direction if we know the components. By applying the Pythagorean theorem to Fig. 1.17, we find that the magnitude of vector $\vec{A}$ is

$$A = \sqrt{A_x^2 + A_y^2} \quad (1.6)$$

(We always take the positive root.) Equation (1.6) is valid for any choice of x -axis and y -axis, as long as they are mutually perpendicular. The expression for the vector direction comes from the definition of the tangent of an angle. If θ is measured from

Figure 1.20 Drawing a sketch of a vector reveals the signs of its x - and y -components.

Suppose that $\tan \theta = \frac{A_y}{A_x} = +1$. What is θ ?

Two angles have tangents of $+1$: 45° and 225° .
The diagram shows that θ must be 225° .

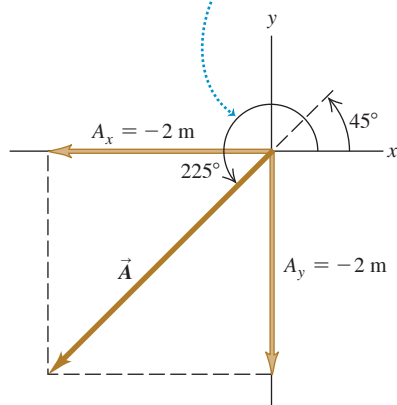


Figure 1.21 Finding the vector sum (resultant) of $\vec{A}$ and $\vec{B}$ using components.

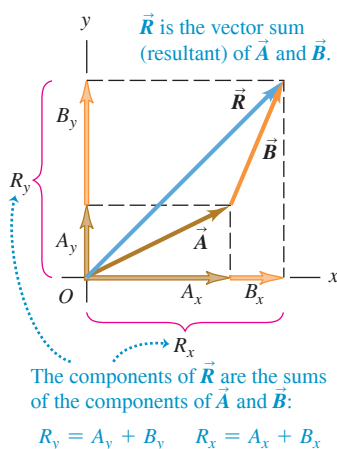
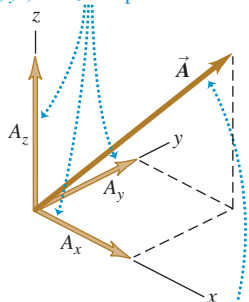


Figure 1.22 A vector in three dimensions.

In three dimensions, a vector has x -, y -, and z -components.



The magnitude of vector $\vec{A}$ is $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$.

the positive x -axis, and a positive angle is measured toward the positive y -axis (as in Fig. 1.17), then

$$\tan \theta = \frac{A_y}{A_x} \quad \text{and} \quad \theta = \arctan \frac{A_y}{A_x} \quad (1.7)$$

We'll always use the notation $\arctan$ for the inverse tangent function (see Example 1.5 in Section 1.7). The notation $\tan^{-1}$ is also commonly used, and your calculator may have an INV or 2ND button to be used with the TAN button.

CAUTION Finding the direction of a vector from its components There's one complication in using Eqs. (1.7) to find θ : Any two angles that differ by 180° have the same tangent. For example, in Fig. 1.20 the tangent of the angle θ is $\tan \theta = A_y/A_x = +1$. A calculator will tell you that $\theta = \tan^{-1}(+1) = 45^\circ$. But the tangent of $180^\circ + 45^\circ = 225^\circ$ is also equal to $+1$, so θ could also be 225° (which is actually the case in Fig. 1.20). Always draw a sketch like Fig. 1.20 to determine which of the two possibilities is correct.

2. **Multiplying a vector by a scalar.** If we multiply a vector $\vec{A}$ by a scalar c , each component of the product $\vec{D} = c\vec{A}$ is the product of c and the corresponding component of $\vec{A}$:

$$D_x = cA_x, \quad D_y = cA_y \quad (\text{components of } \vec{D} = c\vec{A}) \quad (1.8)$$

For example, Eqs. (1.8) say that each component of the vector $2\vec{A}$ is twice as great as the corresponding component of $\vec{A}$, so $2\vec{A}$ is in the same direction as $\vec{A}$ but has twice the magnitude. Each component of the vector $-3\vec{A}$ is three times as great as the corresponding component of $\vec{A}$ but has the opposite sign, so $-3\vec{A}$ is in the opposite direction from $\vec{A}$ and has three times the magnitude. Hence Eqs. (1.8) are consistent with our discussion in Section 1.7 of multiplying a vector by a scalar (see Fig. 1.15).

3. **Using components to calculate the vector sum (resultant) of two or more vectors.** Figure 1.21 shows two vectors $\vec{A}$ and $\vec{B}$ and their vector sum $\vec{R}$, along with the x - and y -components of all three vectors. The x -component R_x of the vector sum is simply the sum ($A_x + B_x$) of the x -components of the vectors being added. The same is true for the y -components. In symbols,

$$\begin{aligned} \text{Each component of } \vec{R} = \vec{A} + \vec{B} \dots \\ R_x = A_x + B_x, \quad R_y = A_y + B_y \quad (1.9) \\ \dots \text{ is the sum of the corresponding components of } \vec{A} \text{ and } \vec{B}. \end{aligned}$$

Figure 1.21 shows this result for the case in which the components A_x , A_y , B_x , and B_y are all positive. Draw additional diagrams to verify for yourself that Eqs. (1.9) are valid for *any* signs of the components of $\vec{A}$ and $\vec{B}$.

If we know the components of any two vectors $\vec{A}$ and $\vec{B}$, perhaps by using Eqs. (1.5), we can compute the components of the vector sum $\vec{R}$. Then if we need the magnitude and direction of $\vec{R}$, we can obtain them from Eqs. (1.6) and (1.7) with the A 's replaced by R 's.

We can use the same procedure to find the sum of any number of vectors. If $\vec{R}$ is the vector sum of $\vec{A}$, $\vec{B}$, $\vec{C}$, $\vec{D}$, $\vec{E}$, $\dots$, the components of $\vec{R}$ are

$$\begin{aligned} R_x &= A_x + B_x + C_x + D_x + E_x + \dots \\ R_y &= A_y + B_y + C_y + D_y + E_y + \dots \end{aligned} \quad (1.10)$$

We have talked about vectors that lie in the xy -plane only, but the component method works just as well for vectors having any direction in space. We can introduce a z -axis perpendicular to the xy -plane; then in general a vector $\vec{A}$ has components A_x , A_y , and A_z in the three coordinate directions. Its magnitude A is

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (1.11)$$

Again, we always take the positive root (Fig. 1.22). Also, Eqs. (1.10) for the vector sum $\vec{R}$ have a third component:

$$R_z = A_z + B_z + C_z + D_z + E_z + \dots$$

We've focused on adding *displacement* vectors, but the method is applicable to all vector quantities. When we study the concept of force in Chapter 4, we'll find that forces are vectors that obey the same rules of vector addition.

PROBLEM-SOLVING STRATEGY 1.3 Vector Addition

IDENTIFY *the relevant concepts:* Decide what the target variable is. It may be the magnitude of the vector sum, the direction, or both.

SET UP *the problem:* Sketch the vectors being added, along with suitable coordinate axes. Place the tail of the first vector at the origin of the coordinates, place the tail of the second vector at the head of the first vector, and so on. Draw the vector sum $\vec{R}$ from the tail of the first vector (at the origin) to the head of the last vector. Use your sketch to estimate the magnitude and direction of $\vec{R}$. Select the equations you'll need: Eqs. (1.5) to obtain the components of the vectors given, if necessary, Eqs. (1.10) to obtain the components of the vector sum, Eq. (1.11) to obtain its magnitude, and Eqs. (1.7) to obtain its direction.

EXECUTE *the solution* as follows:

1. Find the x - and y -components of each individual vector and record your results in a table, as in Example 1.7 below. If a vector is described by a magnitude A and an angle θ , measured from the $+x$ -axis toward the $+y$ -axis, then its components are given by Eqs. (1.5):

$$A_x = A \cos \theta \quad A_y = A \sin \theta$$

If the angles of the vectors are given in some other way, perhaps using a different reference direction, convert them to angles measured from the $+x$ -axis as in Example 1.6.

2. Add the individual x -components algebraically (including signs) to find R_x , the x -component of the vector sum. Do the same for the y -components to find R_y . See Example 1.7.
3. Calculate the magnitude R and direction θ of the vector sum by using Eqs. (1.6) and (1.7):

$$R = \sqrt{R_x^2 + R_y^2} \quad \theta = \arctan \frac{R_y}{R_x}$$

EVALUATE *your answer:* Confirm that your results for the magnitude and direction of the vector sum agree with the estimates you made from your sketch. The value of θ that you find with a calculator may be off by 180° ; your drawing will indicate the correct value. (See Example 1.7 below for an illustration of this.)

EXAMPLE 1.7 Using components to add vectors

WITH VARIATION PROBLEMS

Three players on a reality TV show are brought to the center of a large, flat field. Each is given a meter stick, a compass, a calculator, a shovel, and (in a different order for each contestant) the following three displacements:

- $\vec{A}$: 72.4 m, 32.0° east of north
 $\vec{B}$: 57.3 m, 36.0° south of west
 $\vec{C}$: 17.8 m due south

The three displacements lead to the point in the field where the keys to a new Porsche are buried. Two players start measuring immediately, but the winner first *calculates* where to go. What does she calculate?

IDENTIFY and SET UP The goal is to find the sum (resultant) of the three displacements, so this is a problem in vector addition. See Fig. 1.23. We have chosen the $+x$ -axis as east and the $+y$ -axis as north. We estimate from the diagram that the vector sum $\vec{R}$ is about 10 m, 40° west of north (so θ is about 90° plus 40° , or about 130°).

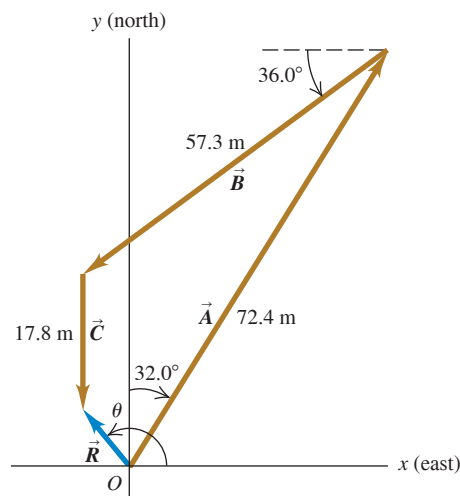
EXECUTE The angles of the vectors, measured from the $+x$ -axis toward the $+y$ -axis, are $(90.0^\circ - 32.0^\circ) = 58.0^\circ$, $(180.0^\circ + 36.0^\circ) = 216.0^\circ$, and 270.0° , respectively. We may now use Eqs. (1.5) to find the components of $\vec{A}$:

$$A_x = A \cos \theta_A = (72.4 \text{ m})(\cos 58.0^\circ) = 38.37 \text{ m}$$

$$A_y = A \sin \theta_A = (72.4 \text{ m})(\sin 58.0^\circ) = 61.40 \text{ m}$$

We've kept an extra significant figure in the components; we'll round to the correct number of significant figures at the end of our calculation. The table at right shows the components of all the displacements, the addition of the components, and the other calculations from Eqs. (1.6) and (1.7).

Figure 1.23 Three successive displacements $\vec{A}$, $\vec{B}$, and $\vec{C}$ and the resultant (vector sum) displacement $\vec{R} = \vec{A} + \vec{B} + \vec{C}$.



Distance	Angle	x -component	y -component
$A = 72.4 \text{ m}$	58.0°	38.37 m	61.40 m
$B = 57.3 \text{ m}$	216.0°	-46.36 m	-33.68 m
$C = 17.8 \text{ m}$	270.0°	0.00 m	-17.80 m
		$R_x = -7.99 \text{ m}$	$R_y = 9.92 \text{ m}$

$$R = \sqrt{(-7.99 \text{ m})^2 + (9.92 \text{ m})^2} = 12.7 \text{ m}$$

$$\theta = \arctan \frac{9.92 \text{ m}}{-7.99 \text{ m}} = -51^\circ$$

Continued

Comparing to angle θ in Fig. 1.23 shows that the calculated angle is clearly off by 180° . The correct value is $\theta = 180^\circ + (-51^\circ) = 129^\circ$, or 39° west of north.

EVALUATE Our calculated answers for R and θ agree with our estimates. Notice how drawing the diagram in Fig. 1.23 made it easy to avoid a 180° error in the direction of the vector sum.

KEYCONCEPT When you are adding vectors, the x -component of the vector sum is equal to the sum of the x -components of the vectors being added, and likewise for the y -component. Always use a diagram to help determine the direction of the vector sum.

TEST YOUR UNDERSTANDING OF SECTION 1.8 Two vectors $\vec{A}$ and $\vec{B}$ lie in the xy -plane.

(a) Can $\vec{A}$ have the same magnitude as $\vec{B}$ but different components? (b) Can $\vec{A}$ have the same components as $\vec{B}$ but a different magnitude?

ANSWER

(a) **yes**, (b) **no** Vectors $\vec{A}$ and $\vec{B}$ can have the same magnitude but different components if they point in different directions. If they have the same components, however, they are the same vector and so must have the same magnitude. ($\vec{A} = \vec{B}$)

1.9 UNIT VECTORS

A **unit vector** is a vector that has a magnitude of 1, with no units. Its only purpose is to *point*—that is, to describe a direction in space. Unit vectors provide a convenient notation for many expressions involving components of vectors. We'll always include a caret, or "hat" ($\wedge$), in the symbol for a unit vector to distinguish it from ordinary vectors whose magnitude may or may not be equal to 1.

In an xy -coordinate system we can define a unit vector $\hat{i}$ that points in the direction of the positive x -axis and a unit vector $\hat{j}$ that points in the direction of the positive y -axis (Fig. 1.24a). Then we can write a vector $\vec{A}$ in terms of its components as

$$\vec{A} = A_x \hat{i} + A_y \hat{j} \quad (1.12)$$

Equation (1.12) is a vector equation; each term, such as $A_x \hat{i}$, is a vector quantity (Fig. 1.24b).

Using unit vectors, we can express the vector sum $\vec{R}$ of two vectors $\vec{A}$ and $\vec{B}$ as follows:

$$\begin{aligned} \vec{A} &= A_x \hat{i} + A_y \hat{j} \\ \vec{B} &= B_x \hat{i} + B_y \hat{j} \\ \vec{R} &= \vec{A} + \vec{B} \\ &= (A_x \hat{i} + A_y \hat{j}) + (B_x \hat{i} + B_y \hat{j}) \\ &= (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j} \\ &= R_x \hat{i} + R_y \hat{j} \end{aligned} \quad (1.13)$$

Equation (1.13) restates the content of Eqs. (1.9) in the form of a single vector equation rather than two component equations.

If not all of the vectors lie in the xy -plane, then we need a third component. We introduce a third unit vector $\hat{k}$ that points in the direction of the positive z -axis (Fig. 1.25). Then Eqs. (1.12) and (1.13) become

Any vector can be expressed in terms of its x -, y -, and z -components ...

$$\begin{aligned} \vec{A} &= A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \\ \vec{B} &= B_x \hat{i} + B_y \hat{j} + B_z \hat{k} \\ &\dots \text{and unit vectors } \hat{i}, \hat{j}, \text{ and } \hat{k}. \end{aligned} \quad (1.14)$$

$$\begin{aligned} \vec{R} &= (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j} + (A_z + B_z) \hat{k} \\ &= R_x \hat{i} + R_y \hat{j} + R_z \hat{k} \end{aligned} \quad (1.15)$$

Figure 1.24 (a) The unit vectors $\hat{i}$ and $\hat{j}$. (b) Expressing a vector $\vec{A}$ in terms of its components.

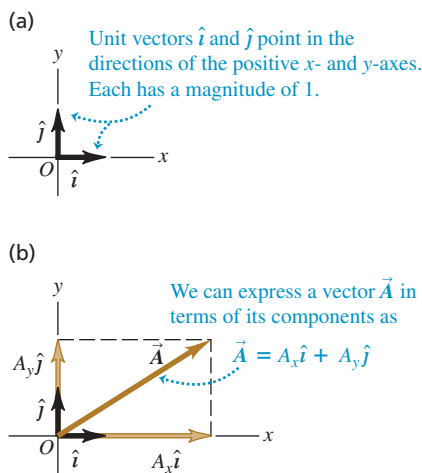
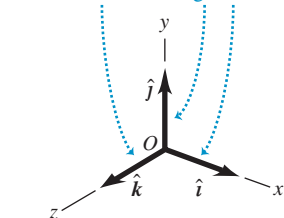


Figure 1.25 The unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$.

Unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$ point in the directions of the positive x -, y -, and z -axes. Each has a magnitude of 1.



EXAMPLE 1.8 Using unit vectors

Given the two displacements

$$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) \text{ m} \quad \text{and} \\ \vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) \text{ m}$$

find the magnitude of the displacement $2\vec{D} - \vec{E}$.

IDENTIFY and SET UP We are to multiply vector $\vec{D}$ by 2 (a scalar) and subtract vector $\vec{E}$ from the result, so as to obtain the vector $\vec{F} = 2\vec{D} - \vec{E}$. Equation (1.8) says that to multiply $\vec{D}$ by 2, we multiply each of its components by 2. We can use Eq. (1.15) to do the subtraction; recall from Section 1.7 that subtracting a vector is the same as adding the negative of that vector.

EXECUTE We have

$$\begin{aligned} \vec{F} &= 2(6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) \text{ m} - (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) \text{ m} \\ &= [(12.00 - 4.00)\hat{i} + (6.00 + 5.00)\hat{j} + (-2.00 - 8.00)\hat{k}] \text{ m} \\ &= (8.00\hat{i} + 11.00\hat{j} - 10.00\hat{k}) \text{ m} \end{aligned}$$

From Eq. (1.11) the magnitude of $\vec{F}$ is

$$\begin{aligned} F &= \sqrt{F_x^2 + F_y^2 + F_z^2} \\ &= \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} \\ &= 16.9 \text{ m} \end{aligned}$$

EVALUATE Our answer is of the same order of magnitude as the larger components that appear in the sum. We wouldn't expect our answer to be much larger than this, but it could be much smaller.

KEYCONCEPT By using unit vectors, you can write a single equation for vector addition that incorporates the x -, y -, and z -components.

TEST YOUR UNDERSTANDING OF SECTION 1.9 Arrange the following vectors in order of their magnitude, with the vector of largest magnitude first. (i) $\vec{A} = (3\hat{i} + 5\hat{j} - 2\hat{k}) \text{ m}$; (ii) $\vec{B} = (-3\hat{i} + 5\hat{j} - 2\hat{k}) \text{ m}$; (iii) $\vec{C} = (3\hat{i} - 5\hat{j} - 2\hat{k}) \text{ m}$; (iv) $\vec{D} = (3\hat{i} + 5\hat{j} + 2\hat{k}) \text{ m}$.

ANSWER $w 2.9 = z w 83 \sqrt{=} = z w 4 + z w 25 + z w 6 \sqrt{=} =$
 $z(w 2 \mp) + z(w 5 \mp) + z(w 3 \mp) \sqrt{=} = D = C = B = A$

same magnitude:

I All have the same magnitude. Vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, and $\vec{D}$ point in different directions but have the

1.10 PRODUCTS OF VECTORS

We saw how vector addition develops naturally from the problem of combining displacements. It will prove useful for calculations with many other vector quantities. We can also express many physical relationships by using *products* of vectors. Vectors are not ordinary numbers, so we can't directly apply ordinary multiplication to vectors. We'll define two different kinds of products of vectors. The first, called the *scalar product*, yields a result that is a scalar quantity. The second, the *vector product*, yields another vector.

Scalar Product

We denote the **scalar product** of two vectors $\vec{A}$ and $\vec{B}$ by $\vec{A} \cdot \vec{B}$. Because of this notation, the scalar product is also called the **dot product**. Although $\vec{A}$ and $\vec{B}$ are vectors, the quantity $\vec{A} \cdot \vec{B}$ is a scalar.

To define the scalar product $\vec{A} \cdot \vec{B}$ we draw the two vectors $\vec{A}$ and $\vec{B}$ with their tails at the same point (**Fig. 1.26a**). The angle ϕ (the Greek letter phi) between their directions ranges from 0° to 180° . Figure 1.26b shows the projection of vector $\vec{B}$ onto the direction of $\vec{A}$; this projection is the component of $\vec{B}$ in the direction of $\vec{A}$ and is equal to $B \cos \phi$. (We can take components along *any* direction that's convenient, not just the x - and y -axes.) We define $\vec{A} \cdot \vec{B}$ to be the magnitude of $\vec{A}$ multiplied by the component of $\vec{B}$ in the direction of $\vec{A}$, or

$$\begin{aligned} \text{Scalar (dot) product of vectors } \vec{A} \text{ and } \vec{B} \quad & \text{Magnitudes of } \vec{A} \text{ and } \vec{B} \\ \vec{A} \cdot \vec{B} = AB \cos \phi = |\vec{A}| |\vec{B}| \cos \phi \quad & \text{Angle between } \vec{A} \text{ and } \vec{B} \text{ when placed tail to tail} \end{aligned} \quad (1.16)$$

Figure 1.26 Calculating the scalar product of two vectors, $\vec{A} \cdot \vec{B} = AB \cos \phi$.

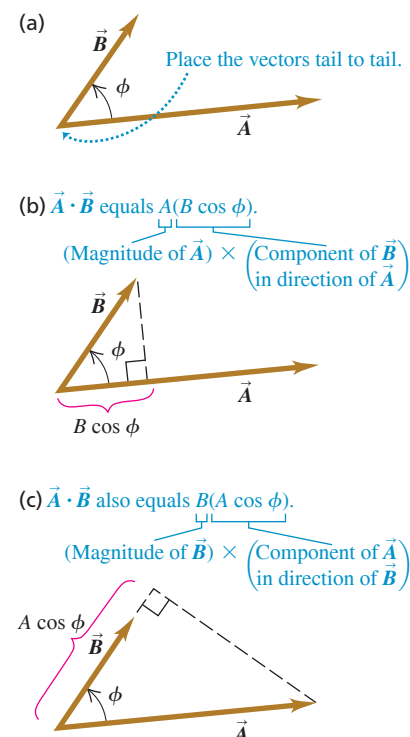
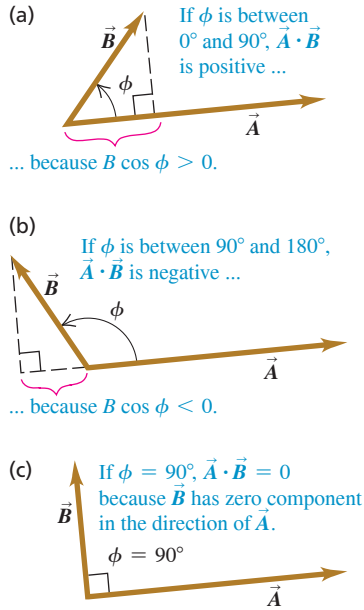


Figure 1.27 The scalar product $\vec{A} \cdot \vec{B} = AB \cos \phi$ can be positive, negative, or zero, depending on the angle between $\vec{A}$ and $\vec{B}$.



Alternatively, we can define $\vec{A} \cdot \vec{B}$ to be the magnitude of $\vec{B}$ multiplied by the component of $\vec{A}$ in the direction of $\vec{B}$, as in Fig. 1.26c. Hence $\vec{A} \cdot \vec{B} = B(A \cos \phi) = AB \cos \phi$, which is the same as Eq. (1.16).

The scalar product is a scalar quantity, not a vector, and it may be positive, negative, or zero. When ϕ is between 0° and 90° , $\cos \phi > 0$ and the scalar product is positive (Fig. 1.27a). When ϕ is between 90° and 180° so $\cos \phi < 0$, the component of $\vec{B}$ in the direction of $\vec{A}$ is negative, and $\vec{A} \cdot \vec{B}$ is negative (Fig. 1.27b). Finally, when $\phi = 90^\circ$, $\vec{A} \cdot \vec{B} = 0$ (Fig. 1.27c). *The scalar product of two perpendicular vectors is always zero.*

For any two vectors $\vec{A}$ and $\vec{B}$, $AB \cos \phi = BA \cos \phi$. This means that $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$. The scalar product obeys the commutative law of multiplication; the order of the two vectors does not matter.

We'll use the scalar product in Chapter 6 to describe work done by a force. In later chapters we'll use the scalar product for a variety of purposes, from calculating electric potential to determining the effects that varying magnetic fields have on electric circuits.

Using Components to Calculate the Scalar Product

We can calculate the scalar product $\vec{A} \cdot \vec{B}$ directly if we know the x -, y -, and z -components of $\vec{A}$ and $\vec{B}$. To see how this is done, let's first work out the scalar products of the unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$. All unit vectors have magnitude 1 and are perpendicular to each other. Using Eq. (1.16), we find

$$\begin{aligned}\hat{i} \cdot \hat{i} &= \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = (1)(1) \cos 0^\circ = 1 \\ \hat{i} \cdot \hat{j} &= \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = (1)(1) \cos 90^\circ = 0\end{aligned}\quad (1.17)$$

Now we express $\vec{A}$ and $\vec{B}$ in terms of their components, expand the product, and use these products of unit vectors:

$$\begin{aligned}\vec{A} \cdot \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= A_x \hat{i} \cdot B_x \hat{i} + A_x \hat{i} \cdot B_y \hat{j} + A_x \hat{i} \cdot B_z \hat{k} \\ &\quad + A_y \hat{j} \cdot B_x \hat{i} + A_y \hat{j} \cdot B_y \hat{j} + A_y \hat{j} \cdot B_z \hat{k} \\ &\quad + A_z \hat{k} \cdot B_x \hat{i} + A_z \hat{k} \cdot B_y \hat{j} + A_z \hat{k} \cdot B_z \hat{k} \\ &= A_x B_x \hat{i} \cdot \hat{i} + A_x B_y \hat{i} \cdot \hat{j} + A_x B_z \hat{i} \cdot \hat{k} \\ &\quad + A_y B_x \hat{j} \cdot \hat{i} + A_y B_y \hat{j} \cdot \hat{j} + A_y B_z \hat{j} \cdot \hat{k} \\ &\quad + A_z B_x \hat{k} \cdot \hat{i} + A_z B_y \hat{k} \cdot \hat{j} + A_z B_z \hat{k} \cdot \hat{k}\end{aligned}\quad (1.18)$$

From Eqs. (1.17) you can see that six of these nine terms are zero. The three that survive give

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z \quad (1.19)$$

Scalar (dot) product of vectors $\vec{A}$ and $\vec{B}$ Components of $\vec{A}$ Components of $\vec{B}$

Thus *the scalar product of two vectors is the sum of the products of their respective components.*

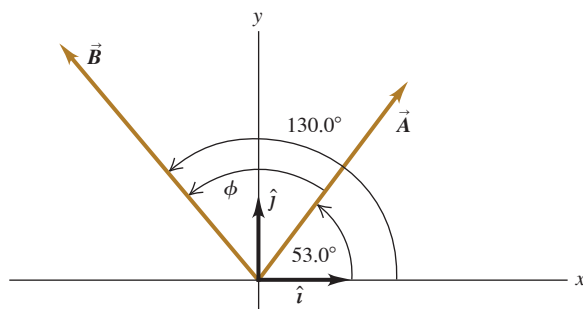
The scalar product gives a straightforward way to find the angle ϕ between any two vectors $\vec{A}$ and $\vec{B}$ whose components are known. In this case we can use Eq. (1.19) to find the scalar product of $\vec{A}$ and $\vec{B}$. Example 1.10 shows how to do this.

EXAMPLE 1.9 Calculating a scalar product**WITH VARIATION PROBLEMS**

Find the scalar product $\vec{A} \cdot \vec{B}$ of the two vectors in **Fig. 1.28**. The magnitudes of the vectors are $A = 4.00$ and $B = 5.00$.

IDENTIFY and SET UP We can calculate the scalar product in two ways: using the magnitudes of the vectors and the angle between them (Eq. 1.16) and using the components of the vectors (Eq. 1.19). We'll do it both ways, and the results will check each other.

Figure 1.28 Two vectors $\vec{A}$ and $\vec{B}$ in two dimensions.



EXECUTE The angle between the two vectors $\vec{A}$ and $\vec{B}$ is $\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ$, so Eq. (1.16) gives us

$$\vec{A} \cdot \vec{B} = AB \cos \phi = (4.00)(5.00) \cos 77.0^\circ = 4.50$$

To use Eq. (1.19), we must first find the components of the vectors. The angles of $\vec{A}$ and $\vec{B}$ are given with respect to the $+x$ -axis and are measured in the sense from the $+x$ -axis to the $+y$ -axis, so we can use Eqs. (1.5):

$$A_x = (4.00) \cos 53.0^\circ = 2.407$$

$$A_y = (4.00) \sin 53.0^\circ = 3.195$$

$$B_x = (5.00) \cos 130.0^\circ = -3.214$$

$$B_y = (5.00) \sin 130.0^\circ = 3.830$$

As in Example 1.7, we keep an extra significant figure in the components and round at the end. Equation (1.19) now gives us

$$\begin{aligned} \vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.407)(-3.214) + (3.195)(3.830) + (0)(0) = 4.50 \end{aligned}$$

EVALUATE Both methods give the same result, as they should.

KEYCONCEPT The scalar product $\vec{A} \cdot \vec{B}$ is a scalar (a number) that equals the sum of the products of the x -components, y -components, and z -components of $\vec{A}$ and $\vec{B}$.

EXAMPLE 1.10 Finding an angle with the scalar product**WITH VARIATION PROBLEMS**

Find the angle between the vectors

$$\vec{A} = 2.00\hat{i} + 3.00\hat{j} + 1.00\hat{k}$$

and

$$\vec{B} = -4.00\hat{i} + 2.00\hat{j} - 1.00\hat{k}$$

IDENTIFY and SET UP We're given the x -, y -, and z -components of two vectors. Our target variable is the angle ϕ between them (**Fig. 1.29**). To find this, we'll solve Eq. (1.16), $\vec{A} \cdot \vec{B} = AB \cos \phi$, for ϕ in terms of the scalar product $\vec{A} \cdot \vec{B}$ and the magnitudes A and B . We can use Eq. (1.19) to evaluate the scalar product, $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$, and we can use Eq. (1.6) to find A and B .

EXECUTE We solve Eq. (1.16) for $\cos \phi$ and use Eq. (1.19) to write $\vec{A} \cdot \vec{B}$:

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{A_x B_x + A_y B_y + A_z B_z}{AB}$$

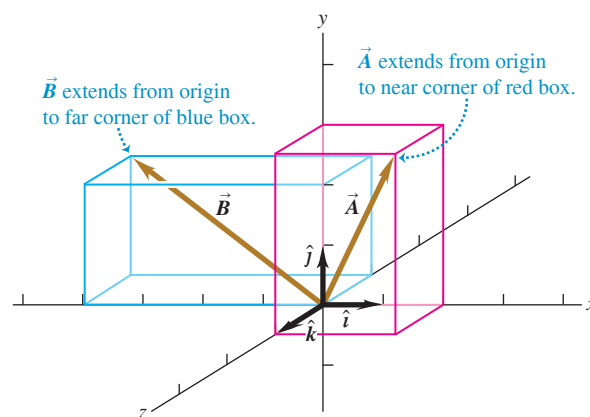
We can use this formula to find the angle between *any* two vectors $\vec{A}$ and $\vec{B}$. Here we have $A_x = 2.00$, $A_y = 3.00$, and $A_z = 1.00$, and $B_x = -4.00$, $B_y = 2.00$, and $B_z = -1.00$. Thus

$$\begin{aligned} \vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y + A_z B_z \\ &= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00) \\ &= -3.00 \end{aligned}$$

$$\begin{aligned} A &= \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2} \\ &= \sqrt{14.00} \end{aligned}$$

$$\begin{aligned} B &= \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2} \\ &= \sqrt{21.00} \end{aligned}$$

Figure 1.29 Two vectors in three dimensions.



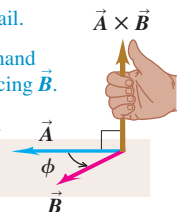
$$\begin{aligned} \cos \phi &= \frac{A_x B_x + A_y B_y + A_z B_z}{AB} = \frac{-3.00}{\sqrt{14.00} \sqrt{21.00}} = -0.175 \\ \phi &= 100^\circ \end{aligned}$$

EVALUATE As a check on this result, note that the scalar product $\vec{A} \cdot \vec{B}$ is negative. This means that ϕ is between 90° and 180° (see Fig. 1.27), which agrees with our answer.

KEYCONCEPT You can find the angle ϕ between two vectors $\vec{A}$ and $\vec{B}$ whose components are known by first finding their scalar product, then using the equation $\vec{A} \cdot \vec{B} = AB \cos \phi$.

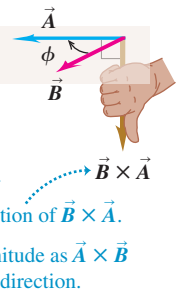
Figure 1.30 The vector product of (a) $\vec{A} \times \vec{B}$ and (b) $\vec{B} \times \vec{A}$.

(a) Using the right-hand rule to find the direction of $\vec{A} \times \vec{B}$

- ① Place $\vec{A}$ and $\vec{B}$ tail to tail.
 - ② Point fingers of right hand along $\vec{A}$, with palm facing $\vec{B}$.
 - ③ Curl fingers toward $\vec{B}$.
 - ④ Thumb points in direction of $\vec{A} \times \vec{B}$.
- 

(b) Using the right-hand rule to find the direction of $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$

(vector product is anticommutative)

- ① Place $\vec{B}$ and $\vec{A}$ tail to tail.
 - ② Point fingers of right hand along $\vec{B}$, with palm facing $\vec{A}$.
 - ③ Curl fingers toward $\vec{A}$.
 - ④ Thumb points in direction of $\vec{B} \times \vec{A}$.
 - ⑤ $\vec{B} \times \vec{A}$ has same magnitude as $\vec{A} \times \vec{B}$ but points in opposite direction.
- 

Vector Product

We denote the **vector product** of two vectors $\vec{A}$ and $\vec{B}$, also called the **cross product**, by $\vec{A} \times \vec{B}$. As the name suggests, the vector product is itself a vector. We'll use this product in Chapter 10 to describe torque and angular momentum; in Chapters 27 and 28 we'll use it to describe magnetic fields and forces.

To define the vector product $\vec{A} \times \vec{B}$, we again draw the two vectors $\vec{A}$ and $\vec{B}$ with their tails at the same point (Fig. 1.30a). The two vectors then lie in a plane. We define the vector product to be a vector quantity with a direction perpendicular to this plane (that is, perpendicular to both $\vec{A}$ and $\vec{B}$) and a magnitude equal to $AB \sin \phi$. That is, if $\vec{C} = \vec{A} \times \vec{B}$, then

$$C = AB \sin \phi \quad (1.20)$$

Magnitude of vector (cross) product of vectors $\vec{B}$ and $\vec{A}$

Magnitudes of $\vec{A}$ and $\vec{B}$ Angle between $\vec{A}$ and $\vec{B}$ when placed tail to tail

We measure the angle ϕ from $\vec{A}$ toward $\vec{B}$ and take it to be the smaller of the two possible angles, so ϕ ranges from 0° to 180° . Then $\sin \phi \geq 0$ and C in Eq. (1.20) is never negative, as must be the case for a vector magnitude. Note that when $\vec{A}$ and $\vec{B}$ are parallel or antiparallel, $\phi = 0^\circ$ or 180° and $C = 0$. That is, *the vector product of two parallel or antiparallel vectors is always zero*. In particular, *the vector product of any vector with itself is zero*.

CAUTION Vector product vs. scalar product Don't confuse the expression $AB \sin \phi$ for the magnitude of the vector product $\vec{A} \times \vec{B}$ with the similar expression $AB \cos \phi$ for the scalar product $\vec{A} \cdot \vec{B}$. To see the difference between these two expressions, imagine that we vary the angle between $\vec{A}$ and $\vec{B}$ while keeping their magnitudes constant. When $\vec{A}$ and $\vec{B}$ are parallel, the magnitude of the vector product will be zero and the scalar product will be maximum. When $\vec{A}$ and $\vec{B}$ are perpendicular, the magnitude of the vector product will be maximum and the scalar product will be zero.

There are always *two* directions perpendicular to a given plane, one on each side of the plane. We choose which of these is the direction of $\vec{A} \times \vec{B}$ as follows. Imagine rotating vector $\vec{A}$ about the perpendicular line until $\vec{A}$ is aligned with $\vec{B}$, choosing the smaller of the two possible angles between $\vec{A}$ and $\vec{B}$. Curl the fingers of your right hand around the perpendicular line so that your fingertips point in the direction of rotation; your thumb will then point in the direction of $\vec{A} \times \vec{B}$. Figure 1.30a shows this **right-hand rule** and describes a second way to think about this rule.

Similarly, we determine the direction of $\vec{B} \times \vec{A}$ by rotating $\vec{B}$ into $\vec{A}$ as in Fig. 1.30b. The result is a vector that is *opposite* to the vector $\vec{A} \times \vec{B}$. The vector product is *not* commutative but instead is *anticommutative*: For any two vectors $\vec{A}$ and $\vec{B}$,

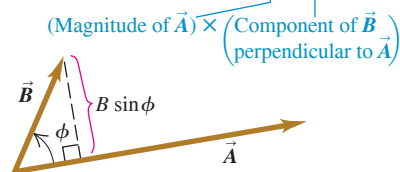
$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A} \quad (1.21)$$

Just as we did for the scalar product, we can give a geometrical interpretation of the magnitude of the vector product. In Fig. 1.31a, $B \sin \phi$ is the component of vector $\vec{B}$ that is *perpendicular* to the direction of vector $\vec{A}$. From Eq. (1.20) the magnitude of $\vec{A} \times \vec{B}$ equals the magnitude of $\vec{A}$ multiplied by the component of $\vec{B}$ that is perpendicular to $\vec{A}$. Figure 1.31b shows that the magnitude of $\vec{A} \times \vec{B}$ also equals the magnitude of $\vec{B}$ multiplied by the component of $\vec{A}$ that is perpendicular to $\vec{B}$. Note that Fig. 1.31 shows the case in which ϕ is between 0° and 90° ; draw a similar diagram for ϕ between 90° and 180° to show that the same geometrical interpretation of the magnitude of $\vec{A} \times \vec{B}$ applies.

Figure 1.31 Calculating the magnitude $AB \sin \phi$ of the vector product of two vectors, $\vec{A} \times \vec{B}$.

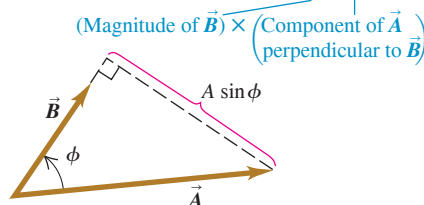
(a)

(Magnitude of $\vec{A} \times \vec{B}$) equals $A(B \sin \phi)$.



(b)

(Magnitude of $\vec{A} \times \vec{B}$) also equals $B(A \sin \phi)$.



Using Components to Calculate the Vector Product

If we know the components of $\vec{A}$ and $\vec{B}$, we can calculate the components of the vector product by using a procedure similar to that for the scalar product. First we work out the multiplication table for unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$, all three of which are perpendicular to each other (Fig. 1.32a). The vector product of any vector with itself is zero, so

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \mathbf{0}$$

The boldface zero is a reminder that each product is a zero *vector*—that is, one with all components equal to zero and an undefined direction. Using Eqs. (1.20) and (1.21) and the right-hand rule, we find

$$\begin{aligned}\hat{i} \times \hat{j} &= -\hat{j} \times \hat{i} = \hat{k} \\ \hat{j} \times \hat{k} &= -\hat{k} \times \hat{j} = \hat{i} \\ \hat{k} \times \hat{i} &= -\hat{i} \times \hat{k} = \hat{j}\end{aligned}\quad (1.22)$$

You can verify these equations by referring to Fig. 1.32a.

Next we express $\vec{A}$ and $\vec{B}$ in terms of their components and the corresponding unit vectors, and we expand the expression for the vector product:

$$\begin{aligned}\vec{A} \times \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= A_x \hat{i} \times B_x \hat{i} + A_x \hat{i} \times B_y \hat{j} + A_x \hat{i} \times B_z \hat{k} \\ &\quad + A_y \hat{j} \times B_x \hat{i} + A_y \hat{j} \times B_y \hat{j} + A_y \hat{j} \times B_z \hat{k} \\ &\quad + A_z \hat{k} \times B_x \hat{i} + A_z \hat{k} \times B_y \hat{j} + A_z \hat{k} \times B_z \hat{k}\end{aligned}\quad (1.23)$$

We can also rewrite the individual terms in Eq. (1.23) as $A_x \hat{i} \times B_y \hat{j} = (A_x B_y) \hat{i} \times \hat{j}$, and so on. Evaluating these by using the multiplication table for the unit vectors in Eqs. (1.22) and then grouping the terms, we get

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k} \quad (1.24)$$

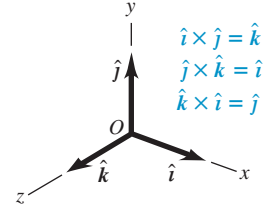
If you compare Eq. (1.24) with Eq. (1.14), you'll see that the components of $\vec{C} = \vec{A} \times \vec{B}$ are

$$\begin{aligned}C_x &= A_y B_z - A_z B_y & C_y &= A_z B_x - A_x B_z & C_z &= A_x B_y - A_y B_x \\ A_x, A_y, A_z &= \text{components of } \vec{A} & B_x, B_y, B_z &= \text{components of } \vec{B}\end{aligned}\quad (1.25)$$

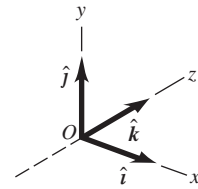
With the axis system of Fig. 1.32a, if we reverse the direction of the z -axis, we get the system shown in Fig. 1.32b. Then, as you may verify, the definition of the vector product gives $\hat{i} \times \hat{j} = -\hat{k}$ instead of $\hat{i} \times \hat{j} = \hat{k}$. In fact, all vector products of unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$ would have signs opposite to those in Eqs. (1.22). So there are two kinds of coordinate systems, which differ in the signs of the vector products of unit vectors. An axis system in which $\hat{i} \times \hat{j} = \hat{k}$, as in Fig. 1.32a, is called a **right-handed system**. The usual practice is to use *only* right-handed systems, and we'll follow that practice throughout this book.

Figure 1.32 (a) We'll always use a right-handed coordinate system, like this one. (b) We'll never use a left-handed coordinate system (in which $\hat{i} \times \hat{j} = -\hat{k}$, and so on).

(a) A right-handed coordinate system



(b) A left-handed coordinate system; we will not use these.

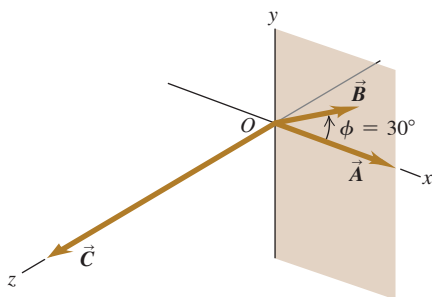


EXAMPLE 1.11 Calculating a vector product

Vector $\vec{A}$ has magnitude 6 units and is in the direction of the $+x$ -axis. Vector $\vec{B}$ has magnitude 4 units and lies in the xy -plane, making an angle of 30° with the $+x$ -axis (**Fig. 1.33**). Find the vector product $\vec{C} = \vec{A} \times \vec{B}$.

IDENTIFY and SET UP We'll find the vector product in two ways, which will provide a check of our calculations. First we'll use Eq. (1.20) and the right-hand rule; then we'll use Eqs. (1.25) to find the vector product by using components.

Figure 1.33 Vectors $\vec{A}$ and $\vec{B}$ and their vector product $\vec{C} = \vec{A} \times \vec{B}$. Vector $\vec{B}$ lies in the xy -plane.



EXECUTE From Eq. (1.20) the magnitude of the vector product is

$$AB \sin \phi = (6)(4)(\sin 30^\circ) = 12$$

By the right-hand rule, the direction of $\vec{A} \times \vec{B}$ is along the $+z$ -axis (the direction of the unit vector $\hat{k}$), so $\vec{C} = \vec{A} \times \vec{B} = 12\hat{k}$.

To use Eqs. (1.25), we first determine the components of $\vec{A}$ and $\vec{B}$. Note that $\vec{A}$ points along the x -axis, so its only nonzero component is A_x . For $\vec{B}$, Fig. 1.33 shows that $\phi = 30^\circ$ is measured from the $+x$ -axis toward the $+y$ -axis, so we can use Eqs. (1.5):

$$A_x = 6 \qquad A_y = 0 \qquad A_z = 0$$

$$B_x = 4 \cos 30^\circ = 2\sqrt{3} \qquad B_y = 4 \sin 30^\circ = 2 \qquad B_z = 0$$

Then Eqs. (1.25) yield

$$C_x = (0)(0) - (0)(2) = 0$$

$$C_y = (0)(2\sqrt{3}) - (6)(0) = 0$$

$$C_z = (6)(2) - (0)(2\sqrt{3}) = 12$$

Thus again we have $\vec{C} = 12\hat{k}$.

EVALUATE Both methods give the same result. Depending on the situation, one or the other of the two approaches may be the more convenient one to use.

KEYCONCEPT The vector product $\vec{A} \times \vec{B}$ of two vectors is a third vector that is perpendicular to both $\vec{A}$ and $\vec{B}$. You can find the vector product either from the magnitudes of the two vectors, the angle between them, and the right-hand rule, or from the components of the two vectors.

TEST YOUR UNDERSTANDING OF SECTION 1.10 Vector $\vec{A}$ has magnitude 2 and vector $\vec{B}$ has magnitude 3. The angle ϕ between $\vec{A}$ and $\vec{B}$ is (i) 0° , (ii) 90° , or (iii) 180° . For each of the following situations, state what the value of ϕ must be. (In each situation there may be more than one correct answer.) (a) $\vec{A} \cdot \vec{B} = 0$; (b) $\vec{A} \times \vec{B} = \mathbf{0}$; (c) $\vec{A} \cdot \vec{B} = 6$; (d) $\vec{A} \cdot \vec{B} = -6$; (e) (magnitude of $\vec{A} \times \vec{B}$) = 6.

ANSWER

(a) $\phi = 90^\circ$; (b) $\phi = 0^\circ$ or 180° ; (c) $\phi = 0^\circ$; (d) $\phi = 180^\circ$; (e) (magnitude of $\vec{A} \times \vec{B}$) = 6. The magnitude of the vector product is equal to the product of the magnitudes of $\vec{A}$ and $\vec{B}$ only if $\vec{A}$ and $\vec{B}$ are perpendicular. (c) The scalar product is equal to the product of the magnitudes of $\vec{A}$ and $\vec{B}$ only if $\vec{A}$ and $\vec{B}$ are parallel. (d) The scalar product is equal to the negative of the product of the magnitudes of $\vec{A}$ and $\vec{B}$ only if $\vec{A}$ and $\vec{B}$ are antiparallel. (e) The magnitude of the vector product is equal to the product of the magnitudes of $\vec{A}$ and $\vec{B}$ only if $\vec{A}$ and $\vec{B}$ are perpendicular.

CHAPTER 1 SUMMARY

Physical quantities and units: Three fundamental physical quantities are mass, length, and time. The corresponding fundamental SI units are the kilogram, the meter, and the second. Derived units for other physical quantities are products or quotients of the basic units. Equations must be dimensionally consistent; two terms can be added only when they have the same units. (See Examples 1.1 and 1.2.)

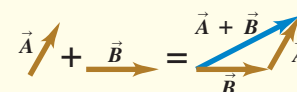
Significant figures: The accuracy of a measurement can be indicated by the number of significant figures or by a stated uncertainty. The significant figures in the result of a calculation are determined by the rules summarized in Table 1.2. When only crude estimates are available for input data, we can often make useful order-of-magnitude estimates. (See Examples 1.3 and 1.4.)

Significant figures in magenta

$$\pi = \frac{C}{2r} = \frac{0.424 \text{ m}}{2(0.06750 \text{ m})} = 3.14$$

$$123.62 + 8.9 = 132.5$$

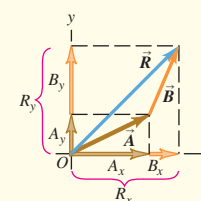
Scalars, vectors, and vector addition: Scalar quantities are numbers and combine according to the usual rules of arithmetic. Vector quantities have direction as well as magnitude and combine according to the rules of vector addition. The negative of a vector has the same magnitude but points in the opposite direction. (See Example 1.5.)



Vector components and vector addition: Vectors can be added by using components of vectors. The x-component of $\vec{R} = \vec{A} + \vec{B}$ is the sum of the x-components of $\vec{A}$ and $\vec{B}$, and likewise for the y- and z-components. (See Examples 1.6 and 1.7.)

$$\begin{aligned} R_x &= A_x + B_x \\ R_y &= A_y + B_y \\ R_z &= A_z + B_z \end{aligned}$$

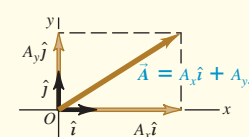
(1.9)



Unit vectors: Unit vectors describe directions in space. A unit vector has a magnitude of 1, with no units. The unit vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$, aligned with the x-, y-, and z-axes of a rectangular coordinate system, are especially useful. (See Example 1.8.)

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

(1.14)

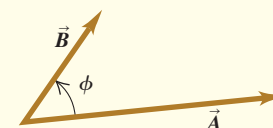


Scalar product: The scalar product $C = \vec{A} \cdot \vec{B}$ of two vectors $\vec{A}$ and $\vec{B}$ is a scalar quantity. It can be expressed in terms of the magnitudes of $\vec{A}$ and $\vec{B}$ and the angle ϕ between the two vectors, or in terms of the components of $\vec{A}$ and $\vec{B}$. The scalar product is commutative; $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$. The scalar product of two perpendicular vectors is zero. (See Examples 1.9 and 1.10.)

$$\vec{A} \cdot \vec{B} = AB \cos \phi = |\vec{A}| |\vec{B}| \cos \phi \quad (1.16)$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z \quad (1.19)$$

Scalar product $\vec{A} \cdot \vec{B} = AB \cos \phi$



Vector product: The vector product $\vec{C} = \vec{A} \times \vec{B}$ of two vectors $\vec{A}$ and $\vec{B}$ is a third vector $\vec{C}$. The magnitude of $\vec{A} \times \vec{B}$ depends on the magnitudes of $\vec{A}$ and $\vec{B}$ and the angle ϕ between the two vectors. The direction of $\vec{A} \times \vec{B}$ is perpendicular to the plane of the two vectors being multiplied, as given by the right-hand rule. The components of $\vec{C} = \vec{A} \times \vec{B}$ can be expressed in terms of the components of $\vec{A}$ and $\vec{B}$. The vector product is not commutative; $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$. The vector product of two parallel or antiparallel vectors is zero. (See Example 1.11.)

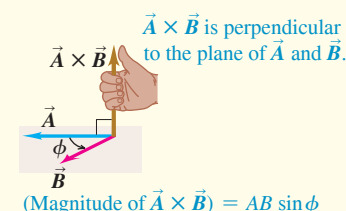
$$C = AB \sin \phi \quad (1.20)$$

$$C_x = A_y B_z - A_z B_y$$

$$C_y = A_z B_x - A_x B_z$$

$$C_z = A_x B_y - A_y B_x$$

(1.25)





GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLE 1.7** (Section 1.8) before attempting these problems.

VP1.7.1 Consider the three vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ in Example 1.7. If a fourth vector $\vec{D}$ is added to $\vec{A} + \vec{B} + \vec{C}$, the result is zero: $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$. Find the magnitude and direction of $\vec{D}$. State the direction of $\vec{D}$ in terms of an angle measured counterclockwise from the positive x -axis, and state in which quadrant this angle lies.

VP1.7.2 Consider the three vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ in Example 1.7. Calculate the magnitude and direction of the vector $\vec{S} = \vec{A} - \vec{B} + \vec{C}$. State the direction of $\vec{S}$ in terms of an angle measured counterclockwise from the positive x -axis, and state in which quadrant this angle lies. (*Hint:* The components of $-\vec{B}$ are just the negatives of the components of $\vec{B}$.)

VP1.7.3 Consider the three vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ in Example 1.7. (a) Find the components of the vector $\vec{T} = \vec{A} + \vec{B} + 2\vec{C}$. (b) Find the magnitude and direction of $\vec{T}$. State the direction of $\vec{T}$ in terms of an angle measured counterclockwise from the positive x -axis, and state in which quadrant this angle lies.

VP1.7.4 A hiker undergoes the displacement $\vec{A}$ shown in Example 1.7. The hiker then undergoes a second displacement such that she ends up 38.0 m from her starting point, in a direction from her starting point that is 37.0° west of north. Find the magnitude and direction of this second displacement. State the direction in terms of an angle measured

counterclockwise from the positive x -axis, and state in which quadrant this angle lies.

Be sure to review **EXAMPLES 1.9 and 1.10** (Section 1.10) before attempting these problems.

VP1.10.1 Vector $\vec{A}$ has magnitude 5.00 and is at an angle of 36.9° south of east. Vector $\vec{B}$ has magnitude 6.40 and is at an angle of 20.0° west of north. (a) Choose the positive x -direction to the east and the positive y -direction to the north. Find the components of $\vec{A}$ and $\vec{B}$. (b) Calculate the scalar product $\vec{A} \cdot \vec{B}$.

VP1.10.2 Vector $\vec{C}$ has magnitude 6.50 and is at an angle of 55.0° measured counterclockwise from the $+x$ -axis toward the $+y$ -axis. Vector $\vec{D}$ has components $D_x = +4.80$ and $D_y = -8.40$. (a) Calculate the scalar product $\vec{C} \cdot \vec{D}$. (b) Find the angle ϕ between the vectors $\vec{C}$ and $\vec{D}$.

VP1.10.3 Vector $\vec{A}$ has components $A_x = -5.00$, $A_y = 3.00$, and $A_z = 0$. Vector $\vec{B}$ has components $B_x = 2.50$, $B_y = 4.00$, and $B_z = -1.50$. Find the angle between the two vectors.

VP1.10.4 If a force $\vec{F}$ acts on an object as that object moves through a displacement $\vec{s}$, the *work* done by that force equals the scalar product of $\vec{F}$ and $\vec{s}$: $W = \vec{F} \cdot \vec{s}$. A certain object moves through displacement $\vec{s} = (4.00 \text{ m})\hat{i} + (5.00 \text{ m})\hat{j}$. As it moves it is acted on by force $\vec{F}$, which has x -component $F_x = -12.0 \text{ N}$ (1 N = 1 newton is the SI unit of force). The work done by this force is $26.0 \text{ N} \cdot \text{m} = 26.0 \text{ J}$ (1 J = 1 joule = 1 newton-meter is the SI unit of work). (a) Find the y -component of $\vec{F}$. (b) Find the angle between $\vec{F}$ and $\vec{s}$.

BRIDGING PROBLEM Vectors on the Roof

An air-conditioning unit is fastened to a roof that slopes at an angle of 35° above the horizontal (**Fig. 1.34**). Its weight is a force $\vec{F}$ on the air conditioner that is directed vertically downward. In order that the unit not crush the roof tiles, the component of the unit's weight perpendicular to the roof cannot exceed 425 N. (One newton, or 1 N, is the SI unit of force. It is equal to 0.2248 lb.) (a) What is the maximum allowed weight of the unit? (b) If the fasteners fail, the unit slides 1.50 m along the roof before it comes to a halt against a ledge. How much work does the weight force do on the unit during its slide if the unit has the weight calculated in part (a)? The work done by a force $\vec{F}$ on an object that undergoes a displacement $\vec{s}$ is $W = \vec{F} \cdot \vec{s}$.

SOLUTION GUIDE

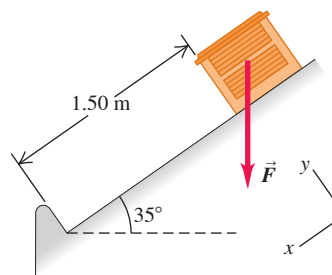
IDENTIFY and SET UP

1. This problem involves vectors and components. What are the known quantities? Which aspect(s) of the weight vector (magnitude, direction, and/or particular components) represent the target variable for part (a)? Which aspect(s) must you know to solve part (b)?
2. Make a sketch based on Fig. 1.34. Draw the x - and y -axes, choosing the positive direction for each. Your axes don't have to be horizontal and vertical, but they do have to be mutually perpendicular. Figure 1.34 shows a convenient choice of axes: The x -axis is parallel to the slope of the roof.
3. Choose the equations you'll use to determine the target variables.

EXECUTE

4. Use the relationship between the magnitude and direction of a vector and its components to solve for the target variable in

Figure 1.34 An air-conditioning unit on a slanted roof.



part (a). Be careful: Is 35° the correct angle to use in the equation? (*Hint:* Check your sketch.)

5. Make sure your answer has the correct number of significant figures.
6. Use the definition of the scalar product to solve for the target variable in part (b). Again, use the correct number of significant figures.

EVALUATE

7. Did your answer to part (a) include a vector component whose absolute value is greater than the magnitude of the vector? Is that possible?
8. There are two ways to find the scalar product of two vectors, one of which you used to solve part (b). Check your answer by repeating the calculation, using the other way. Do you get the same answer?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

- Q1.1** How many correct experiments do we need to disprove a theory? How many do we need to prove a theory? Explain.
- Q1.2** Suppose you are asked to compute the tangent of 5.00 meters. Is this possible? Why or why not?
- Q1.3** What is your height in centimeters? What is your weight in newtons?
- Q1.4** The U.S. National Institute of Standards and Technology (NIST) maintains several accurate copies of the international standard kilogram. Even after careful cleaning, these national standard kilograms are gaining mass at an average rate of about $1 \mu\text{g}/\text{y}$ ($\text{y} = \text{year}$) when compared every 10 years or so to the standard international kilogram. Does this apparent increase have any importance? Explain.
- Q1.5** What physical phenomena (other than a pendulum or cesium clock) could you use to define a time standard?
- Q1.6** Describe how you could measure the thickness of a sheet of paper with an ordinary ruler.
- Q1.7** The quantity $\pi = 3.14159 \dots$ is a number with no dimensions, since it is a ratio of two lengths. Describe two or three other geometrical or physical quantities that are dimensionless.
- Q1.8** What are the units of volume? Suppose another student tells you that a cylinder of radius r and height h has volume given by $\pi r^3 h$. Explain why this cannot be right.
- Q1.9** Three archers each fire four arrows at a target. Joe's four arrows hit points that are spread around in a region that goes 10 cm above, 10 cm below, 10 cm to the left, and 10 cm to the right of the center of the target. All four of Moe's arrows hit within 1 cm of a point 20 cm from the center, and Flo's four arrows hit within 1 cm of the center. The contest judge says that one of the archers is precise but not accurate, another archer is accurate but not precise, and the third archer is both accurate and precise. Which description applies to which archer? Explain.
- Q1.10** Is the vector $(\hat{i} + \hat{j} + \hat{k})$ a unit vector? Is the vector $(3.0\hat{i} - 2.0\hat{j})$ a unit vector? Justify your answers.
- Q1.11** A circular racetrack has a radius of 500 m. What is the displacement of a bicyclist when she travels around the track from the north side to the south side? When she makes one complete circle around the track? Explain.
- Q1.12** Can you find two vectors with different lengths that have a vector sum of zero? What length restrictions are required for three vectors to have a vector sum of zero? Explain.
- Q1.13** The "direction of time" is said to proceed from past to future. Does this mean that time is a vector quantity? Explain.
- Q1.14** Air traffic controllers give instructions called "vectors" to tell airline pilots in which direction they are to fly. If these are the only instructions given, is the name "vector" used correctly? Why or why not?
- Q1.15** Can you find a vector quantity that has a magnitude of zero but components that are not zero? Explain. Can the magnitude of a vector be less than the magnitude of any of its components? Explain.
- Q1.16** (a) Does it make sense to say that a vector is *negative*? Why? (b) Does it make sense to say that one vector is the negative of another? Why? Does your answer here contradict what you said in part (a)?
- Q1.17** If $\vec{C} = \vec{A} + \vec{B}$, what must be true about the directions and magnitudes of $\vec{A}$ and $\vec{B}$ if $C = A + B$? What must be true about the directions and magnitudes of $\vec{A}$ and $\vec{B}$ if $C = 0$?

- Q1.18** If $\vec{A}$ and $\vec{B}$ are nonzero vectors, is it possible for both $\vec{A} \cdot \vec{B}$ and $\vec{A} \times \vec{B}$ to be zero? Explain.
- Q1.19** What does $\vec{A} \cdot \vec{A}$, the scalar product of a vector with itself, give? What about $\vec{A} \times \vec{A}$, the vector product of a vector with itself?
- Q1.20** Let $\vec{A}$ represent any nonzero vector. Why is $\vec{A}/A$ a unit vector, and what is its direction? If θ is the angle that $\vec{A}$ makes with the $+x$ -axis, explain why $(\vec{A}/A) \cdot \hat{i}$ is called the *direction cosine* for that axis.
- Q1.21** Figure 1.7 shows the result of an unacceptable error in the stopping position of a train. If a train travels 890 km from Berlin to Paris and then overshoots the end of the track by 10.0 m, what is the percent error in the total distance covered? Is it correct to write the total distance covered by the train as 890,010 m? Explain.
- Q1.22** Which of the following are legitimate mathematical operations: (a) $\vec{A} \cdot (\vec{B} - \vec{C})$; (b) $(\vec{A} - \vec{B}) \times \vec{C}$; (c) $\vec{A} \cdot (\vec{B} \times \vec{C})$; (d) $\vec{A} \times (\vec{B} \times \vec{C})$; (e) $\vec{A} \times (\vec{B} \cdot \vec{C})$? In each case, give the reason for your answer.
- Q1.23** Consider the vector products $\vec{A} \times (\vec{B} \times \vec{C})$ and $(\vec{A} \times \vec{B}) \times \vec{C}$. Give an example that illustrates the general rule that these two vector products do not have the same magnitude or direction. Can you choose vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ such that these two vector products are equal? If so, give an example.
- Q1.24** Show that, no matter what $\vec{A}$ and $\vec{B}$ are, $\vec{A} \cdot (\vec{A} \times \vec{B}) = 0$. (*Hint*: Do not look for an elaborate mathematical proof. Consider the definition of the direction of the cross product.)
- Q1.25** (a) If $\vec{A} \cdot \vec{B} = 0$, does it necessarily follow that $A = 0$ or $B = 0$? Explain. (b) If $\vec{A} \times \vec{B} = \vec{0}$, does it necessarily follow that $A = 0$ or $B = 0$? Explain.
- Q1.26** If $\vec{A} = \vec{0}$ for a vector in the xy -plane, does it follow that $A_x = -A_y$? What can you say about A_x and A_y ?

EXERCISES

Section 1.3 Standards and Units

Section 1.4 Using and Converting Units

- 1.1** • Starting with the definition $1 \text{ in.} = 2.54 \text{ cm}$, find the number of (a) kilometers in 1.00 mile and (b) feet in 1.00 km.
- 1.2** •• According to the label on a bottle of salad dressing, the volume of the contents is 0.473 liter (L). Using only the conversions $1 \text{ L} = 1000 \text{ cm}^3$ and $1 \text{ in.} = 2.54 \text{ cm}$, express this volume in cubic inches.
- 1.3** •• How many nanoseconds does it take light to travel 1.00 ft in vacuum? (This result is a useful quantity to remember.)
- 1.4** •• The density of gold is 19.3 g/cm^3 . What is this value in kilograms per cubic meter?
- 1.5** • How many years older will you be 1.00 gigasecond from now? (Assume a 365-day year.)
- 1.6** • The following conversions occur frequently in physics and are very useful. (a) Use $1 \text{ mi} = 5280 \text{ ft}$ and $1 \text{ h} = 3600 \text{ s}$ to convert 60 mph to units of ft/s. (b) The acceleration of a freely falling object is 32 ft/s^2 . Use $1 \text{ ft} = 30.48 \text{ cm}$ to express this acceleration in units of m/s^2 . (c) The density of water is 1.0 g/cm^3 . Convert this density to units of kg/m^3 .

1.7 • A certain fuel-efficient hybrid car gets gasoline mileage of 55.0 mpg (miles per gallon). (a) If you are driving this car in Europe and want to compare its mileage with that of other European cars, express this mileage in km/L (L = liter). Use the conversion factors in Appendix E. (b) If this car's gas tank holds 45 L, how many tanks of gas will you use to drive 1500 km?

1.8 • BIO (a) The recommended daily allowance (RDA) of the trace metal magnesium is 410 mg/day for males. Express this quantity in $\mu\text{g/day}$. (b) For adults, the RDA of the amino acid lysine is 12 mg per kg of body weight. How many grams per day should a 75 kg adult receive? (c) A typical multivitamin tablet can contain 2.0 mg of vitamin B₂ (riboflavin), and the RDA is 0.0030 g/day. How many such tablets should a person take each day to get the proper amount of this vitamin, if he gets none from other sources? (d) The RDA for the trace element selenium is 0.000070 g/day. Express this dose in mg/day.

1.9 •• Neptunium. In the fall of 2002, scientists at Los Alamos National Laboratory determined that the critical mass of neptunium-237 is about 60 kg. The critical mass of a fissionable material is the minimum amount that must be brought together to start a nuclear chain reaction. Neptunium-237 has a density of 19.5 g/cm^3 . What would be the radius of a sphere of this material that has a critical mass?

1.10 •• BIO Bacteria. Bacteria vary in size, but a diameter of $2.0 \mu\text{m}$ is not unusual. What are the volume (in cubic centimeters) and surface area (in square millimeters) of a spherical bacterium of that size? (Consult Appendix B for relevant formulas.)

Section 1.5 Uncertainty and Significant Figures

1.11 • With a wooden ruler, you measure the length of a rectangular piece of sheet metal to be 12 mm. With micrometer calipers, you measure the width of the rectangle to be 5.98 mm. Use the correct number of significant figures: What are (a) the area of the rectangle; (b) the ratio of the rectangle's width to its length; (c) the perimeter of the rectangle; (d) the difference between the length and the width; and (e) the ratio of the length to the width?

1.12 • The volume of a solid cylinder is given by $V = \pi r^2 h$, where r is the radius and h is the height. You measure the radius and height of a thin cylindrical wire and obtain the results $r = 0.036 \text{ cm}$ and $h = 12.1 \text{ cm}$. What do your measurements give for the volume of the wire in mm^3 ? Use the correct number of significant figures in your answer.

1.13 •• A useful and easy-to-remember approximate value for the number of seconds in a year is $\pi \times 10^7$. Determine the percent error in this approximate value. (There are 365.24 days in one year.)

1.14 • Express each approximation of π to six significant figures: (a) $22/7$ and (b) $355/113$. (c) Are these approximations accurate to that precision?

Section 1.6 Estimates and Orders of Magnitude

1.15 •• BIO A rather ordinary middle-aged man is in the hospital for a routine checkup. The nurse writes "200" on the patient's medical chart but forgets to include the units. Which of these quantities could the 200 plausibly represent? The patient's (a) mass in kilograms; (b) height in meters; (c) height in centimeters; (d) height in millimeters; (e) age in months.

1.16 • How many gallons of gasoline are used in the United States in one day? Assume that there are two cars for every three people, that each car is driven an average of 10,000 miles per year, and that the average car gets 20 miles per gallon.

1.17 • In Wagner's opera *Das Rheingold*, the goddess Freia is ransomed for a pile of gold just tall enough and wide enough to hide her from sight. Estimate the monetary value of this pile. The density of gold is 19.3 g/cm^3 , and take its value to be about \$40 per gram.

1.18 • BIO Four astronauts are in a spherical space station. (a) If, as is typical, each of them breathes about 500 cm^3 of air with each breath, approximately what volume of air (in cubic meters) do these astronauts breathe in a year? (b) What would the diameter (in meters) of the space station have to be to contain all this air?

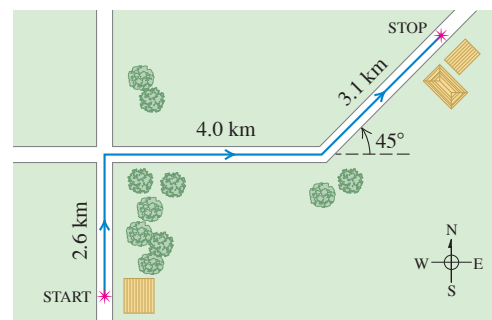
1.19 • You are using water to dilute small amounts of chemicals in the laboratory, drop by drop. How many drops of water are in a 1.0 L bottle? (Hint: Start by estimating the diameter of a drop of water.)

1.20 • BIO How many times does a human heart beat during a person's lifetime? How many gallons of blood does it pump? (Estimate that the heart pumps 50 cm^3 of blood with each beat.)

Section 1.7 Vectors and Vector Addition

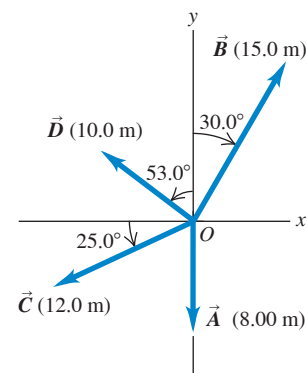
1.21 •• A postal employee drives a delivery truck along the route shown in Fig. E1.21. Determine the magnitude and direction of the resultant displacement by drawing a scale diagram. (See also Exercise 1.28 for a different approach.)

Figure E1.21



1.22 •• For the vectors $\vec{A}$ and $\vec{B}$ in Fig. E1.22, use a scale drawing to find the magnitude and direction of (a) the vector sum $\vec{A} + \vec{B}$ and (b) the vector difference $\vec{A} - \vec{B}$. Use your answers to find the magnitude and direction of (c) $-\vec{A} - \vec{B}$ and (d) $\vec{B} - \vec{A}$. (See also Exercise 1.29 for a different approach.)

Figure E1.22



1.23 •• A spelunker is surveying a cave. She follows a passage 180 m straight west, then 210 m in a direction 45° east of south, and then 280 m at 30° east of north. After a fourth displacement, she finds herself back where she started. Use a scale drawing to determine the magnitude and direction of the fourth displacement. (See also Problem 1.57 for a different approach.)

Section 1.8 Components of Vectors

1.24 •• Let θ be the angle that the vector $\vec{A}$ makes with the $+x$ -axis, measured counterclockwise from that axis. Find angle θ for a vector that has these components: (a) $A_x = 2.00 \text{ m}$, $A_y = -1.00 \text{ m}$; (b) $A_x = 2.00 \text{ m}$, $A_y = 1.00 \text{ m}$; (c) $A_x = -2.00 \text{ m}$, $A_y = 1.00 \text{ m}$; (d) $A_x = -2.00 \text{ m}$, $A_y = -1.00 \text{ m}$.

1.25 • Compute the x - and y -components of the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, and $\vec{D}$ in Fig. E1.22.

1.26 • Vector $\vec{A}$ is in the direction 34.0° clockwise from the $-y$ -axis. The x -component of $\vec{A}$ is $A_x = -16.0$ m. (a) What is the y -component of $\vec{A}$? (b) What is the magnitude of $\vec{A}$?

1.27 • Vector $\vec{A}$ has y -component $A_y = +9.60$ m. $\vec{A}$ makes an angle of 32.0° counterclockwise from the $+y$ -axis. (a) What is the x -component of $\vec{A}$? (b) What is the magnitude of $\vec{A}$?

1.28 •• A postal employee drives a delivery truck over the route shown in Fig. E1.21. Use the method of components to determine the magnitude and direction of her resultant displacement. In a vector-addition diagram (roughly to scale), show that the resultant displacement found from your diagram is in qualitative agreement with the result you obtained by using the method of components.

1.29 • For the vectors $\vec{A}$ and $\vec{B}$ in Fig. E1.22, use the method of components to find the magnitude and direction of (a) the vector sum $\vec{A} + \vec{B}$; (b) the vector sum $\vec{B} + \vec{A}$; (c) the vector difference $\vec{A} - \vec{B}$; (d) the vector difference $\vec{B} - \vec{A}$.

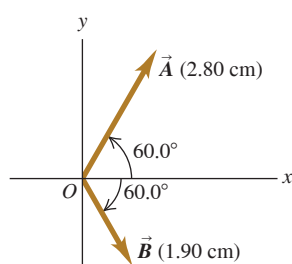
1.30 • Find the magnitude and direction of the vector represented by the following pairs of components: (a) $A_x = -8.60$ cm, $A_y = 5.20$ cm; (b) $A_x = -9.70$ m, $A_y = -2.45$ m; (c) $A_x = 7.75$ km, $A_y = -2.70$ km.

1.31 •• A disoriented physics professor drives 3.25 km north, then 2.20 km west, and then 1.50 km south. Find the magnitude and direction of the resultant displacement, using the method of components. In a vector-addition diagram (roughly to scale), show that the resultant displacement found from your diagram is in qualitative agreement with the result you obtained by using the method of components.

1.32 •• Vector $\vec{A}$ has magnitude 8.00 m and is in the xy -plane at an angle of 127° counterclockwise from the $+x$ -axis (37° past the $+y$ -axis). What are the magnitude and direction of vector $\vec{B}$ if the sum $\vec{A} + \vec{B}$ is in the $-y$ -direction and has magnitude 12.0 m?

1.33 •• Vector $\vec{A}$ is 2.80 cm long and is 60.0° above the x -axis in the first quadrant. Vector $\vec{B}$ is 1.90 cm long and is 60.0° below the x -axis in the fourth quadrant (Fig. E1.33). Use components to find the magnitude and direction of (a) $\vec{A} + \vec{B}$; (b) $\vec{A} - \vec{B}$; (c) $\vec{B} - \vec{A}$. In each case, sketch the vector addition or subtraction and show that your numerical answers are in qualitative agreement with your sketch.

Figure E1.33



Section 1.9 Unit Vectors

1.34 • In each case, find the x - and y -components of vector $\vec{A}$: (a) $\vec{A} = 5.0\hat{i} - 6.3\hat{j}$; (b) $\vec{A} = 11.2\hat{j} - 9.91\hat{i}$; (c) $\vec{A} = -15.0\hat{i} + 22.4\hat{j}$; (d) $\vec{A} = 5.0\vec{B}$, where $\vec{B} = 4\hat{i} - 6\hat{j}$.

1.35 •• Write each vector in Fig. E1.22 in terms of the unit vectors $\hat{i}$ and $\hat{j}$.

1.36 •• Given two vectors $\vec{A} = 4.00\hat{i} + 7.00\hat{j}$ and $\vec{B} = 5.00\hat{i} - 2.00\hat{j}$, (a) find the magnitude of each vector; (b) use unit vectors to write an expression for the vector difference $\vec{A} - \vec{B}$; and (c) find the magnitude and direction of the vector difference $\vec{A} - \vec{B}$. (d) In a vector diagram show $\vec{A}$, $\vec{B}$, and $\vec{A} - \vec{B}$, and show that your diagram agrees qualitatively with your answer to part (c).

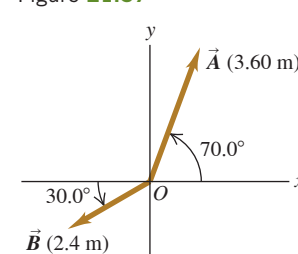
1.37 •• (a) Write each vector in Fig. E1.37 in terms of the unit vectors $\hat{i}$ and $\hat{j}$. (b) Use unit vectors to express vector $\vec{C}$, where $\vec{C} = 3.00\vec{A} - 4.00\vec{B}$. (c) Find the magnitude and direction of $\vec{C}$.

1.38 • You are given two vectors $\vec{A} = -3.00\hat{i} + 6.00\hat{j}$ and $\vec{B} = 7.00\hat{i} + 2.00\hat{j}$. Let counterclockwise angles be positive.

(a) What angle does $\vec{A}$ make with the $+x$ -axis? (b) What angle does $\vec{B}$ make with the $+x$ -axis? (c) Vector $\vec{C}$ is the sum of $\vec{A}$ and $\vec{B}$, so $\vec{C} = \vec{A} + \vec{B}$. What angle does $\vec{C}$ make with the $+x$ -axis?

1.39 • Given two vectors $\vec{A} = -2.00\hat{i} + 3.00\hat{j} + 4.00\hat{k}$ and $\vec{B} = 3.00\hat{i} + 1.00\hat{j} - 3.00\hat{k}$, (a) find the magnitude of each vector; (b) use unit vectors to write an expression for the vector difference $\vec{A} - \vec{B}$; and (c) find the magnitude of the vector difference $\vec{A} - \vec{B}$. Is this the same as the magnitude of $\vec{B} - \vec{A}$? Explain.

Figure E1.37



Section 1.10 Products of Vectors

1.40 •• (a) Find the scalar product of the vectors $\vec{A}$ and $\vec{B}$ given in Exercise 1.36. (b) Find the angle between these two vectors.

1.41 • For the vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ in Fig. E1.22, find the scalar products (a) $\vec{A} \cdot \vec{B}$; (b) $\vec{B} \cdot \vec{C}$; (c) $\vec{A} \cdot \vec{C}$.

1.42 •• Find the vector product $\vec{A} \times \vec{B}$ (expressed in unit vectors) of the two vectors given in Exercise 1.36. What is the magnitude of the vector product?

1.43 •• Find the angle between each of these pairs of vectors:

(a) $\vec{A} = -2.00\hat{i} + 6.00\hat{j}$ and $\vec{B} = 2.00\hat{i} - 3.00\hat{j}$

(b) $\vec{A} = 3.00\hat{i} + 5.00\hat{j}$ and $\vec{B} = 10.00\hat{i} + 6.00\hat{j}$

(c) $\vec{A} = -4.00\hat{i} + 2.00\hat{j}$ and $\vec{B} = 7.00\hat{i} + 14.00\hat{j}$

1.44 • For the two vectors in Fig. E1.33, find the magnitude and direction of (a) the vector product $\vec{A} \times \vec{B}$; (b) the vector product $\vec{B} \times \vec{A}$.

1.45 • For the two vectors $\vec{A}$ and $\vec{D}$ in Fig. E1.22, find the magnitude and direction of (a) the vector product $\vec{A} \times \vec{D}$; (b) the vector product $\vec{D} \times \vec{A}$.

1.46 • For the two vectors $\vec{A}$ and $\vec{B}$ in Fig. E1.37, find (a) the scalar product $\vec{A} \cdot \vec{B}$; (b) the magnitude and direction of the vector product $\vec{A} \times \vec{B}$.

1.47 •• The vector product of vectors $\vec{A}$ and $\vec{B}$ has magnitude 16.0 m² and is in the $+z$ -direction. If vector $\vec{A}$ has magnitude 8.0 m and is in the $-x$ -direction, what are the magnitude and direction of vector $\vec{B}$ if it has no x -component?

1.48 • The angle between two vectors is θ . (a) If $\theta = 30.0^\circ$, which has the greater magnitude: the scalar product or the vector product of the two vectors? (b) For what value (or values) of θ are the magnitudes of the scalar product and the vector product equal?

PROBLEMS

1.49 •• White Dwarfs and Neutron Stars. Recall that density is mass divided by volume, and consult Appendix B as needed. (a) Calculate the average density of the earth in g/cm³, assuming our planet is a perfect sphere. (b) In about 5 billion years, at the end of its lifetime, our sun will end up as a white dwarf that has about the same mass as it does now but is reduced to about 15,000 km in diameter. What will be its density at that stage? (c) A neutron star is the remnant of certain supernovae (explosions of giant stars). Typically, neutron stars are about 20 km in diameter and have about the same mass as our sun. What is a typical neutron star density in g/cm³?

1.50 •• The Hydrogen Maser. A maser is a laser-type device that produces electromagnetic waves with frequencies in the microwave and radio-wave bands of the electromagnetic spectrum. You can use the radio waves generated by a hydrogen maser as a standard of frequency. The frequency of these waves is 1,420,405,751.786 hertz. (A hertz is another name for one cycle per second.) A clock controlled by a hydrogen maser is off by only 1 s in 100,000 years. For the following questions, use only three significant figures. (The large number of significant figures given for the frequency simply illustrates the remarkable accuracy to which it has been measured.) (a) What is the time for one cycle of the radio wave? (b) How many cycles occur in 1 h? (c) How many cycles would have occurred during the age of the earth, which is estimated to be 4.6×10^9 years? (d) By how many seconds would a hydrogen maser clock be off after a time interval equal to the age of the earth?

1.51 •• An Earthlike Planet. In January 2006 astronomers reported the discovery of a planet, comparable in size to the earth, orbiting another star and having a mass about 5.5 times the earth's mass. It is believed to consist of a mixture of rock and ice, similar to Neptune. If this planet has the same density as Neptune (1.76 g/cm^3), what is its radius expressed (a) in kilometers and (b) as a multiple of earth's radius? Consult Appendix F for astronomical data.

1.52 ••• A rectangular piece of aluminum is $7.60 \pm 0.01 \text{ cm}$ long and $1.90 \pm 0.01 \text{ cm}$ wide. (a) Find the area of the rectangle and the uncertainty in the area. (b) Verify that the fractional uncertainty in the area is equal to the sum of the fractional uncertainties in the length and in the width. (This is a general result.)

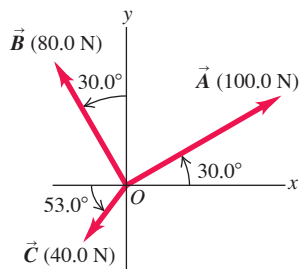
1.53 • BIO Estimate the number of atoms in your body. (*Hint:* Based on what you know about biology and chemistry, what are the most common types of atom in your body? What is the mass of each type of atom? Appendix D gives the atomic masses of different elements, measured in atomic mass units; you can find the value of an atomic mass unit, or 1 u, in Appendix E.)

1.54 • BIO Biological tissues are typically made up of 98% water. Given that the density of water is $1.0 \times 10^3 \text{ kg/m}^3$, estimate the mass of (a) the heart of an adult human; (b) a cell with a diameter of $0.5 \text{ }\mu\text{m}$; (c) a honeybee.

1.55 •• Vector $\vec{A} = 3.0\hat{i} - 4.0\hat{k}$. (a) Construct a unit vector that is parallel to $\vec{A}$. (b) Construct a unit vector that is antiparallel to $\vec{A}$. (c) Construct two unit vectors that are perpendicular to $\vec{A}$ and that have no y-component.

1.56 •• Three horizontal ropes pull on a large stone stuck in the ground, producing the vector forces $\vec{A}$, $\vec{B}$, and $\vec{C}$ shown in **Fig. P1.56**. Find the magnitude and direction of a fourth force on the stone that will make the vector sum of the four forces zero.

Figure P1.56



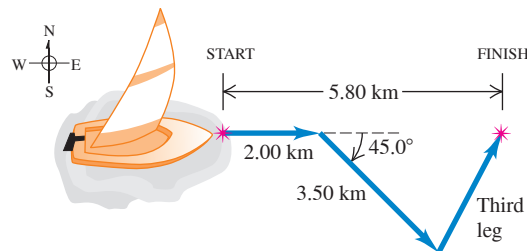
1.57 ••• As noted in Exercise 1.23, a spelunker is surveying a cave. She follows a passage 180 m straight west, then 210 m in a direction 45° east of south, and then 280 m at 30° east of north. After a fourth displacement, she finds herself back where she started. Use the method of components to determine the magnitude and direction of the fourth displacement. Draw the vector-addition diagram and show that it is in qualitative agreement with your numerical solution.

1.58 ••• Emergency Landing. A plane leaves the airport in Galisteo and flies 170 km at 68.0° east of north; then it changes direction to fly 230 km at 36.0° south of east, after which it makes an immediate emergency landing in a pasture. When the airport sends out a rescue crew, in which direction and how far should this crew fly to go directly to this plane?

1.59 •• A charged object with electric charge q produces an electric field. The SI unit for electric field is N/C, where N is the SI unit for force and C is the SI unit for charge. If at point P there are electric fields from two or more charged objects, then the resultant field is the vector sum of the fields from each object. At point P the electric field $\vec{E}_1$ from charge q_1 is 450 N/C in the $+y$ -direction, and the electric field $\vec{E}_2$ from charge q_2 is 600 N/C in the direction 36.9° from the $-y$ -axis toward the $-x$ -axis. What are the magnitude and direction of the resultant field $\vec{E} = \vec{E}_1 + \vec{E}_2$ at point P due to these two charges?

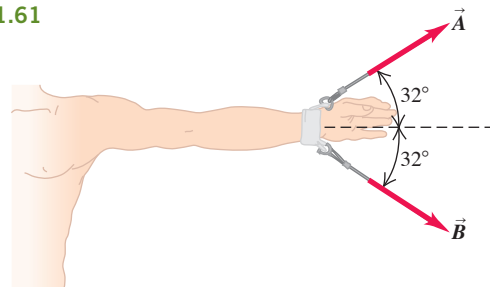
1.60 •• A sailor in a small sailboat encounters shifting winds. She sails 2.00 km east, next 3.50 km southeast, and then an additional distance in an unknown direction. Her final position is 5.80 km directly east of the starting point (**Fig. P1.60**). Find the magnitude and direction of the third leg of the journey. Draw the vector-addition diagram and show that it is in qualitative agreement with your numerical solution.

Figure P1.60



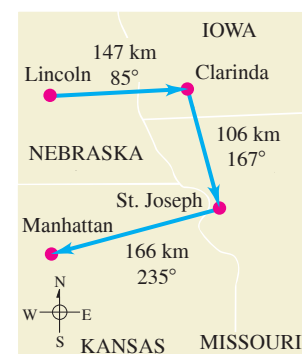
1.61 ••• BIO Dislocated Shoulder. A patient with a dislocated shoulder is put into a traction apparatus as shown in **Fig. P1.61**. The pulls $\vec{A}$ and $\vec{B}$ have equal magnitudes and must combine to produce an outward traction force of 12.8 N on the patient's arm. How large should these pulls be?

Figure P1.61



1.62 ••• On a training flight, a student pilot flies from Lincoln, Nebraska, to Clarinda, Iowa, next to St. Joseph, Missouri, and then to Manhattan, Kansas (**Fig. P1.62**). The directions are shown relative to north: 0° is north, 90° is east, 180° is south, and 270° is west. Use the method of components to find (a) the distance she has to fly from Manhattan to get back to Lincoln, and (b) the direction (relative to north) she must fly to get there. Illustrate your solutions with a vector diagram.

Figure P1.62



1.63 •• You leave the airport in College Station and fly 23.0 km in a direction 34.0° south of east. You then fly 46.0 km due north. How far and in what direction must you then fly to reach a private landing strip that is 32.0 km due west of the College Station airport?

1.64 ••• Getting Back. An explorer in Antarctica leaves his shelter during a whiteout. He takes 40 steps northeast, next 80 steps at 60° north of west, and then 50 steps due south. Assume all of his steps are equal in length. (a) Sketch, roughly to scale, the three vectors and their resultant. (b) Save the explorer from becoming hopelessly lost by giving him the displacement, calculated by using the method of components, that will return him to his shelter.

1.65 •• As a test of orienteering skills, your physics class holds a contest in a large, open field. Each contestant is told to travel 20.8 m due north from the starting point, then 38.0 m due east, and finally 18.0 m in the direction 33.0° west of south. After the specified displacements, a contestant will find a silver dollar hidden under a rock. The winner is the person who takes the shortest time to reach the location of the silver dollar. Remembering what you learned in class, you run on a straight line from the starting point to the hidden coin. How far and in what direction do you run?

1.66 • You are standing on a street corner with your friend. You then travel 14.0 m due west across the street and into your apartment building. You travel in the elevator 22.0 m upward to your floor, walk 12.0 m north to the door of your apartment, and then walk 6.0 m due east to your balcony that overlooks the street. Your friend is standing where you left her. Now how far are you from your friend?

1.67 •• You are lost at night in a large, open field. Your GPS tells you that you are 122.0 m from your truck, in a direction 58.0° east of south. You walk 72.0 m due west along a ditch. How much farther, and in what direction, must you walk to reach your truck?

1.68 ••• You live in a town where the streets are straight but are in a variety of directions. On Saturday you go from your apartment to the grocery store by driving 0.60 km due north and then 1.40 km in the direction 60.0° west of north. On Sunday you again travel from your apartment to the same store but this time by driving 0.80 km in the direction 50.0° north of west and then in a straight line to the store. (a) How far is the store from your apartment? (b) On which day do you travel the greater distance, and how much farther do you travel? Or, do you travel the same distance on each route to the store?

1.69 •• While following a treasure map, you start at an old oak tree. You first walk 825 m directly south, then turn and walk 1.25 km at 30.0° west of north, and finally walk 1.00 km at 32.0° north of east, where you find the treasure: a biography of Isaac Newton! (a) To return to the old oak tree, in what direction should you head and how far will you walk? Use components to solve this problem. (b) To see whether your calculation in part (a) is reasonable, compare it with a graphical solution drawn roughly to scale.

1.70 •• A fence post is 52.0 m from where you are standing, in a direction 37.0° north of east. A second fence post is due south from you. How far are you from the second post if the distance between the two posts is 68.0 m?

1.71 •• A dog in an open field runs 12.0 m east and then 28.0 m in a direction 50.0° west of north. In what direction and how far must the dog then run to end up 10.0 m south of her original starting point?

1.72 ••• Ricardo and Jane are standing under a tree in the middle of a pasture. An argument ensues, and they walk away in different directions. Ricardo walks 26.0 m in a direction 60.0° west of north. Jane walks 16.0 m in a direction 30.0° south of west. They then stop and turn to face each other. (a) What is the distance between them? (b) In what direction should Ricardo walk to go directly toward Jane?

1.73 ••• You are camping with Joe and Karl. Since all three of you like your privacy, you don't pitch your tents close together. Joe's tent is 21.0 m from yours, in the direction 23.0° south of east. Karl's tent is 32.0 m from yours, in the direction 37.0° north of east. What is the distance between Karl's tent and Joe's tent?

1.74 •• Bond Angle in Methane. In the methane molecule, CH_4 , each hydrogen atom is at a corner of a regular tetrahedron with the carbon atom at the center. In coordinates for which one of the C—H bonds is in the direction of $\hat{i} + \hat{j} + \hat{k}$, an adjacent C—H bond is in the $\hat{i} - \hat{j} - \hat{k}$ direction. Calculate the angle between these two bonds.

1.75 •• The work W done by a constant force $\vec{F}$ on an object that undergoes displacement $\vec{s}$ from point 1 to point 2 is $W = \vec{F} \cdot \vec{s}$. For F in newtons (N) and s in meters (m), W is in joules (J). If, during a displacement of the object, $\vec{F}$ has constant direction 60.0° above the $-x$ -axis and constant magnitude 5.00 N and if the displacement is 0.800 m in the $+x$ -direction, what is the work done by the force $\vec{F}$?

1.76 •• Magnetic fields are produced by moving charges and exert forces on moving charges. When a particle with charge q is moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$, the force $\vec{F}$ that the field exerts on the particle is given by $\vec{F} = q\vec{v} \times \vec{B}$. The SI units are as follows: For charge it is the coulomb (C), for magnetic field it is tesla (T), for force it is newton (N), and for velocity it is m/s. If $q = -8.00 \times 10^{-6}$ C, $\vec{v}$ is 3.00×10^4 m/s in the $+x$ -direction, and $\vec{B}$ is 5.00 T in the $-y$ -direction, what are the magnitude and direction of the force that the magnetic field exerts on the charged particle?

1.77 •• Vectors $\vec{A}$ and $\vec{B}$ have scalar product -6.00 , and their vector product has magnitude $+9.00$. What is the angle between these two vectors?

1.78 •• Torque is a vector quantity that specifies the effectiveness of a force in causing the rotation of an object. The torque that a force $\vec{F}$ exerts on a rigid object depends on the point where the force acts and on the location of the axis of rotation. If $\vec{r}$ is the length vector from the axis to the point of application of the force, then the torque is $\vec{r} \times \vec{F}$. If $\vec{F}$ is 22.0 N in the $-y$ -direction and if $\vec{r}$ is in the xy -plane at an angle of 36° from the $+y$ -axis toward the $-x$ -axis and has magnitude 4.0 m, what are the magnitude and direction of the torque exerted by $\vec{F}$?

1.79 •• Vector $\vec{A} = a\hat{i} - b\hat{k}$ and vector $\vec{B} = -c\hat{j} + d\hat{k}$. (a) In terms of the positive scalar quantities a , b , c , and d , what are $\vec{A} \cdot \vec{B}$ and $\vec{A} \times \vec{B}$? (b) If $c = 0$, what is the magnitude of $\vec{A} \cdot \vec{B}$ and what are the magnitude and direction of $\vec{A} \times \vec{B}$? Does your result for the direction for $\vec{A} \times \vec{B}$ agree with the result you get if you sketch $\vec{A}$ and $\vec{B}$ in the xz -plane and apply the right-hand rule? The scalar product can be described as the magnitude of $\vec{B}$ times the component of $\vec{A}$ that is parallel to $\vec{B}$. Does this agree with your result? The magnitude of the vector product can be described as the magnitude of $\vec{B}$ times the component of $\vec{A}$ that is perpendicular to $\vec{B}$. Does this agree with your result?

1.80 •• Vectors $\vec{A}$ and $\vec{B}$ are in the xy -plane. Vector $\vec{A}$ is in the $+x$ -direction, and the direction of vector $\vec{B}$ is at an angle θ from the $+x$ -axis measured toward the $+y$ -axis. (a) If θ is in the range $0^\circ \leq \theta \leq 180^\circ$, for what two values of θ does the scalar product $\vec{A} \cdot \vec{B}$ have its maximum magnitude? For each of these values of θ , what is the magnitude of the vector product $\vec{A} \times \vec{B}$? (b) If θ is in the range $0^\circ \leq \theta \leq 180^\circ$, for what value of θ does the vector product $\vec{A} \times \vec{B}$ have its maximum value? For this value of θ , what is the magnitude of the scalar product $\vec{A} \cdot \vec{B}$? (c) What is the angle θ in the range $0^\circ \leq \theta \leq 180^\circ$ for which $\vec{A} \cdot \vec{B}$ is twice $|\vec{A} \times \vec{B}|$?

1.81 •• Vector $\vec{A}$ has magnitude 12.0 m, and vector $\vec{B}$ has magnitude 16.0 m. The scalar product $\vec{A} \cdot \vec{B}$ is 112.0 m^2 . What is the magnitude of the vector product between these two vectors?

1.82 •• Vector $\vec{A}$ has magnitude 5.00 m and lies in the xy -plane in a direction 53.0° from the $+x$ -axis axis measured toward the $+y$ -axis. Vector $\vec{B}$ has magnitude 8.00 m and a direction you can adjust. (a) You want the vector product $\vec{A} \times \vec{B}$ to have a positive z -component of the largest possible magnitude. What direction should you select for vector $\vec{B}$? (b) What is the direction of $\vec{B}$ for which $\vec{A} \times \vec{B}$ has the most negative z -component? (c) What are the two directions of $\vec{B}$ for which $\vec{A} \times \vec{B}$ is zero?

1.83 •• The scalar product of vectors $\vec{A}$ and $\vec{B}$ is $+48.0 \text{ m}^2$. Vector $\vec{A}$ has magnitude 9.00 m and direction 28.0° west of south. If vector $\vec{B}$ has direction 39.0° south of east, what is the magnitude of $\vec{B}$?

1.84 ••• Obtain a *unit vector* perpendicular to the two vectors given in Exercise 1.39.

1.85 •• You are given vectors $\vec{A} = 5.0\hat{i} - 6.5\hat{j}$ and $\vec{B} = 3.5\hat{i} - 7.0\hat{j}$. A third vector, $\vec{C}$, lies in the xy -plane. Vector $\vec{C}$ is perpendicular to vector $\vec{A}$, and the scalar product of $\vec{C}$ with $\vec{B}$ is 15.0. From this information, find the components of vector $\vec{C}$.

1.86 •• Two vectors $\vec{A}$ and $\vec{B}$ have magnitudes $A = 3.00$ and $B = 3.00$. Their vector product is $\vec{A} \times \vec{B} = -5.00\hat{k} + 2.00\hat{i}$. What is the angle between $\vec{A}$ and $\vec{B}$?

1.87 ••• DATA You are a team leader at a pharmaceutical company. Several technicians are preparing samples, and you want to compare the densities of the samples (density = mass/volume) by using the mass and volume values they have reported. Unfortunately, you did not specify what units to use. The technicians used a variety of units in reporting their values, as shown in the following table.

Sample ID	Mass	Volume
A	8.00 g	$1.67 \times 10^{-6} \text{ m}^3$
B	$6.00 \mu\text{g}$	$9.38 \times 10^6 \mu\text{m}^3$
C	8.00 mg	$2.50 \times 10^{-3} \text{ cm}^3$
D	$9.00 \times 10^{-4} \text{ kg}$	$2.81 \times 10^3 \text{ mm}^3$
E	$9.00 \times 10^4 \text{ ng}$	$1.41 \times 10^{-2} \text{ mm}^3$
F	$6.00 \times 10^{-2} \text{ mg}$	$1.25 \times 10^8 \mu\text{m}^3$

List the sample IDs in order of increasing density of the sample.

1.88 ••• DATA You are a mechanical engineer working for a manufacturing company. Two forces, $\vec{F}_1$ and $\vec{F}_2$, act on a component part of a piece of equipment. Your boss asked you to find the magnitude of the larger of these two forces. You can vary the angle between $\vec{F}_1$ and $\vec{F}_2$ from 0° to 90° while the magnitude of each force stays constant. And, you can measure the magnitude of the resultant force they produce (their vector sum), but you cannot directly measure the magnitude of each separate force. You measure the magnitude of the resultant force for four angles θ between the directions of the two forces as follows:

θ	Resultant force (N)
0.0°	8.00
45.0°	7.43
60.0°	7.00
90.0°	5.83

(a) What is the magnitude of the larger of the two forces? (b) When the equipment is used on the production line, the angle between the two forces is 30.0° . What is the magnitude of the resultant force in this case?

1.89 ••• DATA Navigating in the Solar System. The *Mars Polar Lander* spacecraft was launched on January 3, 1999. On December 3, 1999, the day *Mars Polar Lander* impacted the Martian surface at high velocity and probably disintegrated, the positions of the earth and Mars were given by these coordinates:

	x	y	z
Earth	0.3182 AU	0.9329 AU	-0.0000 AU
Mars	1.3087 AU	-0.4423 AU	-0.0414 AU

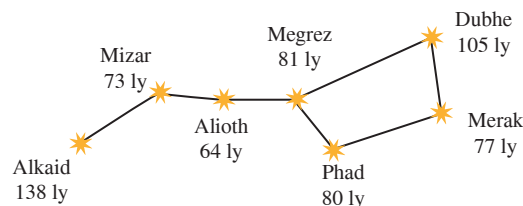
With these coordinates, the sun is at the origin and the earth's orbit is in the xy -plane. The earth passes through the $+x$ -axis once a year on the autumnal equinox, the first day of autumn in the northern hemisphere (on or about September 22). One AU, or *astronomical unit*, is equal to $1.496 \times 10^8 \text{ km}$, the average distance from the earth to the sun. (a) Draw the positions of the sun, the earth, and Mars on December 3, 1999. (b) Find these distances in AU on December 3, 1999: from (i) the sun to the earth; (ii) the sun to Mars; (iii) the earth to Mars. (c) As seen from the earth, what was the angle between the direction to the sun and the direction to Mars on December 3, 1999? (d) Explain whether Mars was visible from your current location at midnight on December 3, 1999. (When it is midnight, the sun is on the opposite side of the earth from you.)

CHALLENGE PROBLEMS

1.90 ••• Completed Pass. The football team at Enormous State University (ESU) uses vector displacements to record its plays, with the origin taken to be the position of the ball before the play starts. In a certain pass play, the receiver starts at $+1.0\hat{i} - 5.0\hat{j}$, where the units are yards, $\hat{i}$ is to the right, and $\hat{j}$ is downfield. Subsequent displacements of the receiver are $+9.0\hat{i}$ (he is in motion before the snap), $+11.0\hat{j}$ (breaks downfield), $-6.0\hat{i} + 4.0\hat{j}$ (zigs), and $+12.0\hat{i} + 18.0\hat{j}$ (zags). Meanwhile, the quarterback has dropped straight back to a position $-7.0\hat{j}$. How far and in which direction must the quarterback throw the ball? (Like the coach, you'll be well advised to diagram the situation before solving this numerically.)

1.91 ••• Navigating in the Big Dipper. All of the stars of the Big Dipper (part of the constellation Ursa Major) may appear to be the same distance from the earth, but in fact they are very far from each other. **Figure P1.91** shows the distances from the earth to each of these stars. The distances are given in light-years (ly), the distance that light travels in one year. One light-year equals $9.461 \times 10^{15} \text{ m}$. (a) Alkaid and Merak are 25.6° apart in the earth's sky. In a diagram, show the relative positions of Alkaid, Merak, and our sun. Find the distance in light-years from Alkaid to Merak. (b) To an inhabitant of a planet orbiting Merak, how many degrees apart in the sky would Alkaid and our sun be?

Figure P1.91



MCAT-STYLE PASSAGE PROBLEMS

BIO Calculating Lung Volume in Humans. In humans, oxygen and carbon dioxide are exchanged in the blood within many small sacs called alveoli in the lungs. Alveoli provide a large surface area for gas exchange. Recent careful measurements show that the total number of alveoli in a typical pair of lungs is about 480×10^6 and that the average volume of a single alveolus is $4.2 \times 10^6 \mu\text{m}^3$. (The volume of a sphere is $V = \frac{4}{3}\pi r^3$, and the area of a sphere is $A = 4\pi r^2$.)

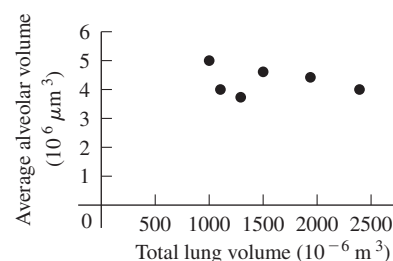
1.92 What is total volume of the gas-exchanging region of the lungs? (a) $2000 \mu\text{m}^3$; (b) 2 m^3 ; (c) 2.0 L ; (d) 120 L .

1.93 If we assume that alveoli are spherical, what is the diameter of a typical alveolus? (a) 0.20 mm ; (b) 2 mm ; (c) 20 mm ; (d) 200 mm .

1.94 Individuals vary considerably in total lung volume. **Figure P1.94** shows the results of measuring the total lung volume and average alveolar volume of six individuals. From these data, what can you infer about the relationship among alveolar size, total lung volume, and number of alveoli per individual? As the total volume of the lungs increases,

(a) the number and volume of individual alveoli increase; (b) the number of alveoli increases and the volume of individual alveoli decreases; (c) the volume of the individual alveoli remains constant and the number of alveoli increases; (d) both the number of alveoli and the volume of individual alveoli remain constant.

Figure P1.94



ANSWERS

Chapter Opening Question ?

(iii) Take the $+x$ -axis to point east and the $+y$ -axis to point north. Then we need to find the y -component of the velocity vector, which has magnitude $v = 15 \text{ km/h}$ and is at an angle $\theta = 37^\circ$ measured from the $+x$ -axis toward the $+y$ -axis. From Eqs. (1.5) we have $v_y = v \sin \theta = (15 \text{ km/h}) \sin 37^\circ = 9.0 \text{ km/h}$. So the thunderstorm moves 9.0 km north in 1 h and 18 km north in 2 h .

Key Example **✓** VARIATION Problems

VP1.7.1 $D = 12.7 \text{ m}$, $\theta = -51^\circ = 309^\circ$ (fourth quadrant)

VP1.7.2 $S = 115 \text{ m}$, $\theta = 42^\circ$ (first quadrant)

VP1.7.3 (a) $T_x = -7.99 \text{ m}$, $T_y = -7.88 \text{ m}$ (b) $T = 11.2 \text{ m}$, $\theta = 225^\circ$ (third quadrant)

VP1.7.4 68.7 m , $\theta = 207^\circ$ (third quadrant)

VP1.10.1 (a) $A_x = 4.00$, $A_y = -3.00$, $B_x = -2.19$, $B_y = 6.01$

(b) $\vec{A} \cdot \vec{B} = -26.8$

VP1.10.2 (a) $\vec{C} \cdot \vec{D} = -26.8$ (b) $\phi = 115^\circ$

VP1.10.3 $\phi = 91^\circ$

VP1.10.4 (a) 14.8 N (b) 77.7°

Bridging Problem

(a) $5.2 \times 10^2 \text{ N}$

(b) $4.5 \times 10^2 \text{ N} \cdot \text{m}$